

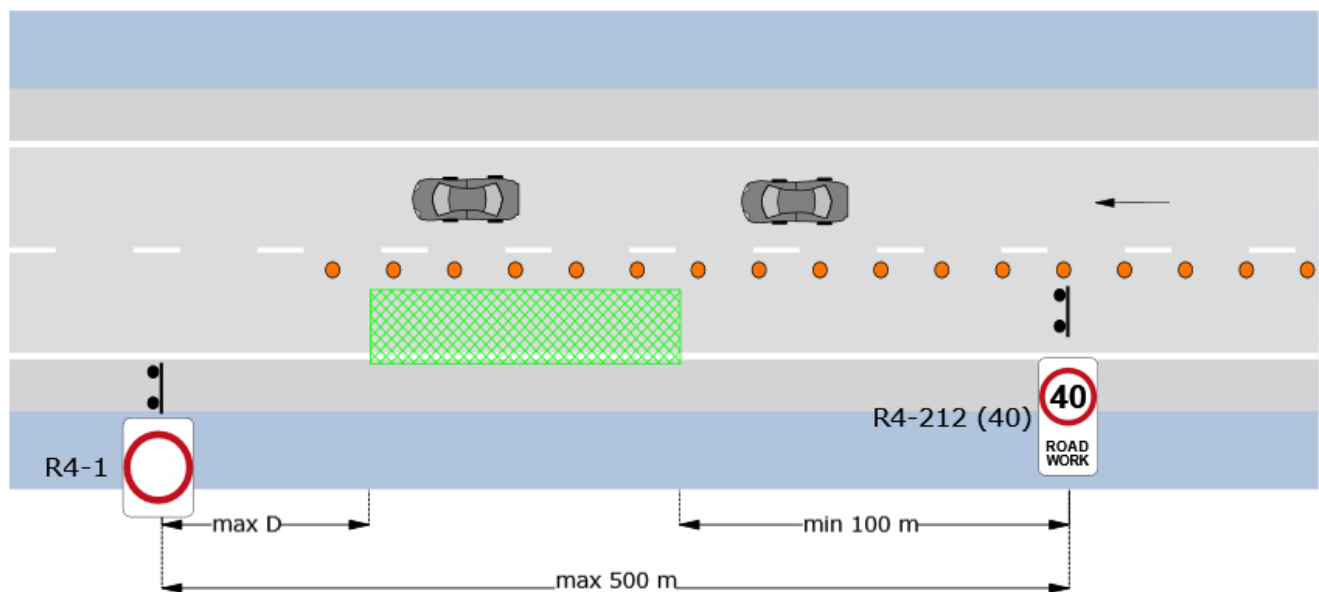


Figure 4-3. Minimum length zones

Note to Figure 4-3: In urban areas the location of the roadwork speed zone may be adjusted so that signs can be erected clear of parked vehicles, other signs or obstructions.

Roadworks speed zones may be offset so that there is a different speed restriction for opposing directions of traffic. An offset speed zone may be used:

- To reduce the length of speed restriction leaving the work site;
- Where a speed zone is extended in one direction to protect the end of a queue;
- On a divided road where roadworks affect traffic conditions on one side of the median only;
- Where a roadworks speed zone is used in a transition zone to reduce traffic speeds on approach to a changed traffic condition, but is not needed for traffic flow in the opposite direction; or
- Where work is underway on the shoulder which does not affect traffic in the opposing traffic lane.

4.5.6 Minimum length zones

Temporary speed zones at work sites are significantly more effective if they appear reasonable to drivers (see also [Section 4.7 Credibility of work sites](#)). The lengths of roadworks speed zones must be in accordance with [Table 4-10](#).

Table 4-10. Length of roadworks speed zones

Roadwork Speed Zone	Minimum length	Maximum length
less than 35 km/h	100 m	200 m
40 km/h	150 m	500 m
60 km/h	150 m	Not specified*
70 km/h transition zone	200 m	Not specified*
80 km/h	500 m	Not specified*
80 km/h transition zones	300 m	Not specified*

Note* to Table 4-10: For traffic safety, there is no maximum length specified however for longer lengths, additional controls to manage traffic speed should be considered, for example use of pilot vehicles.

It must be noted that increasing the length of reduced temporary speed zone without reason may cause adverse road user behaviours, as drivers may return to the normal speed limit if they see no reason to slow down.

4.5.7 Strategies for achieving speed zone compliance

In addition to a reduced speed limit, the following devices or techniques should be considered to assist with driver speed compliance:

- PTCs;
- Pilot vehicles;
- Additional or oversized signs;
- Duplication of signs;
- Variable message signs;
- Electronic speed display signs (such as a variable speed limit sign (VSLS) or radar assisted speed sign (RASS));
- Narrowing of lanes using bollards or cones;
- Chicanes;
- Temporary portable rumble strips (TPRS);
- Flashing lights on signs;
- Thermoplastic tape or similar laid either longitudinally in zig zags or transversely to act as gentle 'speed humps'; or
- VMS advance warning vehicle.

4.5.8 Approval of roadwork speed zones

Authorisation must be provided by an appropriately delegated officer under item 1.30 of the Transport Regulatory Delegations, Part 2 Road Network and Operations Delegations, before a speed limit sign is installed, displayed, altered or removed on any State road.

For work undertaken on State roads, councils and private contractors must seek the authorisation of Transport to erect roadwork speed limit (R4-212n) signs.

For work undertaken on regional roads, before a speed limit sign is installed, displayed, altered or removed, authorisation must be carried out in accordance with Item 13 of Transport's Delegation to Councils – Regulation of Traffic (dated 31 October 2011), which states the following:

When the installation period for a 'Roadwork Speed Limit' (R4-212n) sign is to be for six working days or less:

- *Authorisation of the use of the 'Roadwork Speed Limit' (R4-212n) sign must be carried out by council or a sub-delegate holding a current Prepare Work Zone Traffic Management Plan accreditation issued by the Authority*
- *The nearest Transport office is to be notified in writing of council's intention to implement a roadwork speed limit prior to works commencing.*
- *The nearest police station is to be notified in writing of council's intention to implement a roadwork speed limit prior to works commencing."*

When the installation period for a 'Roadwork Speed Limit' (R4-212n) sign is to be for more than six working days:

- *Authorisation of the use of the 'Roadwork Speed Limit' (R4-212n) sign must be carried out by council or a sub-delegate holding a current Prepare Work Zone Traffic Management Plan accreditation issued by the Authority*
- *The nearest Transport office is to be notified in writing of council's intention to implement a roadwork speed limit seven days prior to works commencing*
- *The nearest police station is to be notified in writing of council's intention to implement a roadwork speed limit seven days prior to works commencing.*

4.5.9 Speed enforcement at roadwork sites

4.5.9.1 General

Compliance with roadwork speed limits is critical to effectively managing the safety of road workers and road users at roadwork sites. Police speed enforcement is a method that may be used to improve compliance.

Early communications and planning with police is crucial to providing an environment conducive to safe and effective police speed enforcement operations. As such, provision of suitable areas for police operations should be considered during the development and construction phases of projects and when planning maintenance activities.

During the development, delivery and maintenance stages of a project, the project team must consider the use of police speed enforcement as part of their risk management approach.

The planning of police speed enforcement is typically split in to three phases:

- *Phase 1* – early site investigation;
- *Phase 2* – planning for police speed enforcement; and
- *Phase 3* – operational engagement of police.

4.5.9.2 Phase 1: Early site investigation

The risks presented by speeding drivers to the safety of road workers and road users on the roadwork site must be assessed for all roadwork projects during TMP development.

The traffic speed and crash history data obtained as part of the development of the TTM strategy and plan must be reviewed to determine if there are any existing speed related issues within the project area. This data forms a basis for early police engagement and highlight high risk areas to be dealt through appropriate TTM planning.

Potential projects requiring police involvement include:

- Long-term work located on high volume and/or high speed traffic routes;
- High-risk short-term work such as:
 - Where traffic barriers cannot be used to separate workers from adjacent traffic and work site safety depends on a lateral separation distance and reduced traffic speed; or
 - Where there is limited sight distance, complex traffic arrangements, or other factors requiring a high level of speed compliance.

When it has been identified that there is the potential for non-compliance with roadworks speed zones, which presents an unacceptable risk to the safety of road workers and road users, the project representative must begin planning for police speed enforcement.

4.5.9.3 Phase 2: Planning for police speed enforcement

The objectives of Phase 2 are to:

- Raise awareness of the project and the potential need for police assistance;
- Ensure the work site can accommodate police enforcement operations in a safe and effective manner; and
- Ensure that the work site speed zones are lawful and enforceable.

This phase should normally occur during the development stage of a project and involves initial engagement with police, review ongoing data collection to monitor speed compliance and the identification of the work site requirements to support police enforcement.

Initial police engagement

The project representative must engage the police to alert them of the safety risk of road workers and road users as identified in the early site investigation. The appropriate contact details to be used for police engagement should be determined by contacting the Manager, Regional Operations (in the respective region).

Police are most likely to provide speed enforcement operations if they are confident it will be effective, the speed zone is appropriately sign posted and lawfully enforceable and the operation can be carried out within a safe operating environment that complies with their standard operating procedures.

Engaging the police early in the project is important as it creates an opportunity to:

- Describe the project and any existing conditions that affect speed compliance and safety;
- Identify and plan the work site to accommodate police speed enforcement;
- Identify data collection parameters to support a request for police involvement;
- Identify site conditions that can impact on their operations; and
- Alert them to the potential need for police speed enforcement.

Police may not commit to speed enforcement at this stage; however, this consultation can be used to obtain an in-principle agreement with the police for speed enforcement during the project delivery when the need arises, provided the work site can provide sufficient supporting data.

The type and quantity of supporting data to be collected should be confirmed with police during the initial engagement process.

Work site requirements

To enable police to undertake speed enforcement on roadwork sites, project teams must ensure that:

- Roadwork speed zones are implemented in accordance with this document;
- Speed Zone Authorisation (SZA) has been issued for the speed zone in operation and the documentation is available to police;
- Location reference points of start and end of roadwork speed zones are documented; and
- Roadwork speed zones are long enough for practical enforcement.

The following considerations must be implemented in each stage of the traffic management plan in order to create an environment for safe and effective police speed enforcement:

- Appropriately sized areas for positioning police vehicles, for example, wide shoulders or breakdown bays or pull-over bays for OSOM vehicles;
- Appropriately located areas for positioning police vehicles with unobscured line of sight to approaching traffic which allow safe departure and acceleration when conducting mobile intercept operations;
- Ability for police to pull over offending vehicles with sufficient length to accommodate the longest vehicle type (access under Notice) that police may pull over. This may include provision of:
 - Sufficient shoulder width; or
 - Regular (every 500 m) breakdown bays.
- Suitable shoulder and breakdown bay surfaces.

These considerations along with any other work site requirements should be confirmed with police during the initial engagement process.

4.5.9.4 Phase 3: Operational engagement of police

Police engagement for speed enforcement

The appropriate contact details to be used for police engagement should be determined by contacting the Manager, Regional Operations (in the respective region).

At the site meeting between the project representative and police, the project representative must:

- Provide an overview of the project;
- Provide a copy of the SZA for the speed zone;
- Provide details of speed zone locations, dates and start and end times of speed zones;
- Provide evidence of speed non-compliance and any other supporting data requested during the initial engagement;
- Request confirmation of the type and quantity of data to be collected to support enforcement activities;
- Demonstrate that any work site requirements for police speed enforcement have been met;
- Provide information about activities that can impact on police operations, for example, construction vehicle movements; and

- Maintain a record of the meeting.

The project representative must then request for speed enforcement operations to be undertaken within the work site on an agreed schedule. In certain situations a 'user pays' arrangement in accordance with NSW Police Force User Pays Policing Services guidelines may need to be agreed upon.

As police speed enforcement operations are not always available for the entire time road workers are on-site, police enforcement then becomes an additional control to complement existing control measures determined through risk assessment and the application of the hierarchy of controls framework.

Ongoing data collection

Police speed enforcement data must be continuously collected throughout the duration of the project to monitor the effectiveness of the traffic management plan and the police speed enforcement when in place. This data allows police to focus their enforcement operation around the particular times of day/s of the week when the risk to road workers and road users is most significant.

This data may be used to monitor speed compliance and can be used as evidence when deciding if ongoing police speed enforcement will be requested in future discussions with police.

Ongoing communications with police

Regular communication (such as a recurring meeting) between project representatives and police must be scheduled throughout the delivery and/or maintenance stage of the project to keep police informed. The meetings must:

- Discuss site conditions that impact police operations;
- Provide a project schedule of works to police informing them of planned changes to the road configuration that affect their speed enforcement operations;
- Discuss user pays arrangements where applicable;
- Discuss the outcomes from the previous police speed enforcement operations;
- Discuss the data that has been collected and determine if police speed enforcement is still warranted; and
- Agree on a future police speed enforcement schedule.

Responsibilities

While Transport project representatives work with police to design site conditions conducive to police speed enforcement/presence, police are responsible for their own management of risks associated with their activities. The method of police speed enforcement is to be determined by the police.

4.5.10 Records and inspections

Records must be maintained of all roadwork speed zones. Records must include:

- Written authorisation of the installation from Transport, or council or its sub-delegate. The sub-delegate's Prepare a Work Zone Traffic Management Plan certificate number must be shown;
- Reference points or location of the roadwork speed zone;
- The installation time and date;
- The speed displayed;
- Location of repeater signs;

- Location of advanced warning signs; and
- The removal time and date.

This information must be kept for at least seven years as it can be required as evidence for litigation purposes including speeding prosecutions or accident compensation claims.

Application and completion for roadwork speed limit authorisation and occupancy licences must be lodged via Transports [Road Occupancy Licence website](#).

For long term projects, copies of the approved forms can be forwarded to the police station nearest to the appropriate sections of road by the project manager and a record of this should be kept on site.

Roadwork speed zones must be regularly inspected as part of daily inspections carried out by an Implement Traffic Control Plans or Prepare Work Zone TMP qualified person who was not involved in the physical implementation of the signage. These inspections should be carried out before work starts, during the works and pre-closedown of the site and completed in accordance with [Section 8.1 Work site inspections, reviews and audits](#).

In accordance with [Section 8.2 Record keeping of TTM documentation](#), roadwork speed zones must be inspected and associated documentation examined on a regular basis. The ITCP qualified person must ensure that speed restriction signs are properly erected, conflicting signs are covered and advance signs are in place, when inspecting the traffic control on the site. Consideration should be given to keeping photographic records of signs and their locations. Any issues or breaches should be recorded in daily diaries.

4.6 End-of-queue management

4.6.1 General

End-of-queue management strategies must be provided when:

- Traffic is planned to be stopped;
- Queues are expected to be generated; or
- The TMP or risk assessment identifies there is a potential for end-of-queue collisions.

Wherever traffic is required to slow significantly or stop, particularly at active traffic control positions, long queues can form, depending on traffic volumes and the length of delay. This can result in an increased risk of rear end collisions occurring.

The risk of end-of-queue collisions is increased:

- Where sight distance to the end of the queue for approaching traffic is likely to be less than 2D (open road areas) or 1.5D (built-up areas);
- Where queues are unexpected, for instance on rural highways, and can be due to the lack of attention of road users or fatigue;
- Where queue lengths are long and can vary considerably, for instance in built-up areas, and can be due to queue ends extending beyond advance warning signs and devices;
- On high speed roads such as motorways, particularly when lane closures are introduced; and
- At night or in adverse weather conditions.

4.6.2 Assessment of expected queue length

Anticipated queue lengths must be estimated and documented prior to stopping traffic. Factors that influence queue lengths include:

- Expected hourly traffic volumes at the time of the work;
- Expected delays at each phase of the work;
- Number of traffic lanes closed or affected;
- Traffic composition, such as percentage of heavy vehicles;
- Predicted distances or gaps between stopped vehicles – drivers in rural areas tend to leave greater gaps than those in urban environments;
- Known special events;
- Terrain at the site; and
- Road alignment (horizontal and vertical).

Traffic queues must be monitored at all times to ensure that queue lengths do not extend back beyond the limits of the advance warning signs.

At work sites where longer than normal delays can be expected, such as the laying of bridge beams, additional planning may be required which is beyond the scope of this Technical Manual. When required, this additional planning must be included in the preparation of the TMP and site-specific or activity-specific risk assessment to identify the risks associated with such long delays, and be completed in consultation with relevant divisions of Transport, emergency services and other relevant stakeholders.

4.6.3 Procedure for reducing end-of-queue collisions

As required by Section 5.4 Traffic control, when traffic is being controlled, a PREPARE TO STOP (T1-18) sign must be installed and the traffic controller operating the PTC or STOP/SLOW bat or an observer with the ability to communicate to the traffic controller, must be able to maintain visibility to the end of the queue.

Additional advance warning or other mitigation measures should be implemented, depending on the speed of traffic and sight distance to the end of the queue, to alert road users of the changed road conditions and to avoid end-of-queue collisions. Where applicable, this may include extending the length of a sign posted roadwork speed zone, where the queue length extends beyond the originally established zone, and provided the zone is self-regulating.

The use of such devices and techniques must be documented in the TMP or risk assessment and included on the approved TGS.

Where significant queues are expected to form, active monitoring of the queue must be undertaken as the primary control. In addition, at least one the following end-of-queue protection strategies must be used:

- Where the maximum queue length can be predicted in advance, the primary PREPARE TO STOP sign must be located such that the distance from this sign to the end of the queue is not less than D, see Figure 4-4. The B size PREPARE TO STOP sign should be used in this application.

The distance may need to be adjusted if the queue length proves to be underestimated. If the primary PREPARE TO STOP sign needs to be placed more than 4D, approximately 15 seconds of travel time from the control point, repeater PREPARE TO STOP signs at intervals of not more than 4D should be provided between that point and the control point to provide for conditions after the queue has dispersed.

In any relocation of the primary PREPARE TO STOP sign, the distance D to the roadwork ahead sign must be maintained.

- Under the immediate direction of an ITCP qualified person, a traffic controller may be employed to move the PREPARE TO STOP (T1-18) sign and the ROADWORK AHEAD (T1-1) sign as necessary to maintain their minimum required distance in advance of the end-of-queue.
- A portable traffic signal (flashing amber) may also be added, at each location of the PREPARE TO STOP sign to alert road users of the approaching work site.

Provided the conditions of [Section 5.4.2 Traffic control types](#) are met and a temporary speed zone of less than 65 km/h is implemented, a traffic controller displaying the SLOW sign at each location of the PREPARE TO STOP sign, may be substituted for a portable traffic signal.

All other advance and position signs required for the work must be located at the distances shown in [Figure 4-4](#) from the start of the work area.

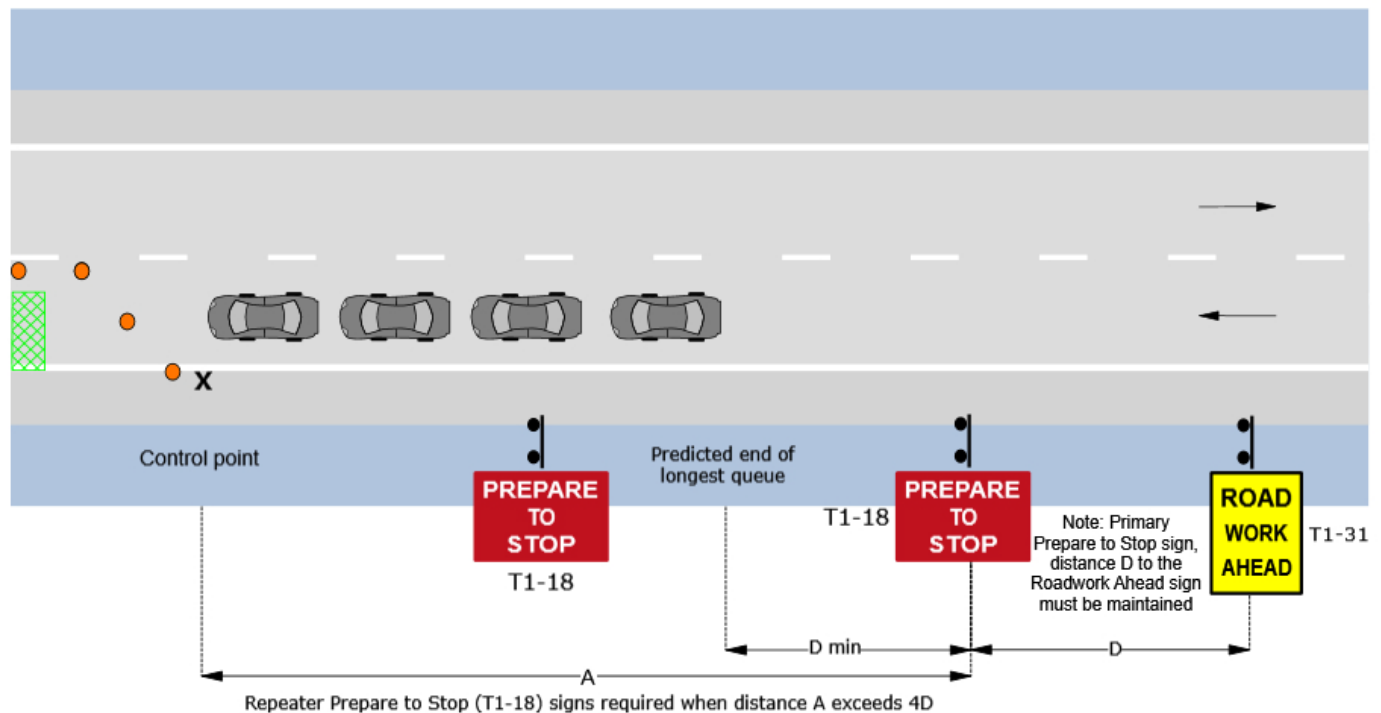


Figure 4-4. Avoiding end-of-queue collisions example

4.7 Credibility of work sites

4.7.1 General

Research has shown that drivers do not perceive roadwork zones as hazardous unless they can see active workers or machinery. In a survey undertaken by the Centre for Accident Research and Road Safety, Queensland, road users stated that the credibility of work zones is undermined by the lack of activity drivers see, particularly when a speed reduction is in place, or having to slow for a speed reduction that is perceived to be unrealistic. It was reported that not only may this lead to speed limit non-compliance, driver frustration may result in other unsafe behaviour, such as tailgating, which generates risk for the road workers and road users.

It has also been demonstrated that drivers become complacent to roadwork signage when it is misused, left out or not appropriate for the works. This includes irrelevant roadwork signage left out during aftercare or the use of PREPARE TO STOP (T1-18) or traffic controller symbolic (T1-34) sign while the traffic control is not present. Not only does this affect the credibility of the work site, it can lessen the effectiveness of the signs when next encountered by the drivers.

4.7.2 Enhancing work site credibility

The study reported that controls most likely to affect driving speed through a work zone were visibility of police, presence of workers on the road and protection of workers by barriers. To ensure credibility of Transport roadwork sites, it is important to consider the following in the TTM planning phase:

- Only reduce speed limits where required to ensure worker or road user safety;
- Reinstate speed limits when work is not occurring;
- Minimise delays to road users;
- Ensure temporary signage and road marking are clear and do not cause confusion;
- Remove or cover temporary signs that are not relevant;
- Provide ongoing road user information and education programs; and
- Engage with police to develop an enforcement strategy through the site.

4.7.3 Improving public awareness

The following may be used to raise public awareness of work sites:

- Transport for NSW website;
- Approved social media;
- Community consultation sessions;
- Conducting campaigns with the police to reinforce responsible behaviour on the roads especially at work sites;
- Additional VMS / static signs near work sites;
- Presenting road safety visuals and posters in public Transport offices, particularly Service NSW centres;
- Advertising the times and duration of planned works on various mediums e.g. such as radio or newspaper;
- Explaining detour options;
- Signs that thank drivers for their cooperation; and
- Communication about the road user or community safety benefits of the work being completed and contact details for enquiries or feedback.

5 Providing for works

5.1 General

When developing a TMP, it is important to consider the impact of the works and its management within the road corridor. This includes the operation of traffic control and works vehicles and activity specific considerations that need to be made.

5.2 Providing safe movements for works traffic

5.2.1 General

In addition to the requirements outlined in, Section 4 Inputs to a Traffic Management Plan planning must also be undertaken to determine the management of works traffic within and around the work site. In particular, the vehicle entry to and exit from the site, and their movements on the road network surrounding the work site, to ensure the safety of workers and road users while maintaining the service provided by the road network.

For the purposes of this document, works traffic is any light or heavy vehicle associated with the undertaking of the work being performed, and includes, but is not limited to:

- Work vehicles carrying crews, signage or equipment;
- Haulage trucks delivering or receiving plant, materials or goods; and
- Any light vehicle attending site that is associated with the work.

When planning and designing for works traffic the following core principles must be considered and applied:

- **Safety** – providing safe movements within and around site for construction works vehicles while not increasing the risk to the general traffic.
- **Efficiency** – ensuring access to construction sites does not impede traffic flows and is designed and managed to minimise impact on traffic efficiency.
- **Consistency** – ensuring consistent and uniform access treatments are applied, so that the message conveyed to road users is consistent.

Safe movement for works traffic must be provided such that vehicles can safely manoeuvre:

- **To and from the site** – ensuring safe access and egress from or into live lanes of traffic; and
- **Around the site** – ensuring safe travel paths are available for access to ancillary and off site work areas such as depots, stockpile sites, quarries and gravel pits.

The impact of works traffic or any associated TTM measures on general road users must be considered.

All vehicles on a work site must have the ability to warn workers of the potential hazard of reversing vehicles. This includes clearly audible and working reversing alarm and lights.

5.2.2 Vehicle movement plans

A vehicle movement plan (VMP) is a document or diagram showing the safe travel paths for works traffic and may include details of safe entry, drive through, exit or navigation around the site. A VMP must be developed if any of the following applies:

- Vehicles will be entering or leaving from a high speed road with an existing permanent speed limit of greater than 75 km/h;

- Vehicles will be entering or leaving from a divided carriageway;
- Where site access constraints require complex manoeuvres on-site;
- Where site access constraints require off-site location movements to turn around;
- Where repetitive movements of plant (greater than 20 per day) are present;
- Where plant on-site may travel faster than 35 km/h;
- Where plant may reverse long distances (greater than 100 m) without the protection of temporary safety barriers; or
- Where specific hazards to vulnerable road users are identified from plant carrying out the work on-site.

A VMP should be developed using the considerations provided in [Table 5-1](#).

Table 5-1. Key considerations for the movement of vehicles

Key consideration	Details
Existing traffic conditions	Hourly traffic flows (including peak hour), traffic speed (85th percentile) and traffic composition (i.e. light vs heavy) will influence the road capacity to accommodate works vehicle movements. Possible source of data is the Transport Traffic Volume Viewer .
Site access	The site environment and constraints will influence the type of access that can be provided and the vehicle movements that can be accommodated.
Expected works traffic movements	The number of works vehicle movements, class of works vehicles (light vs heavy vehicles) and the origin/ destination of works vehicles.
Speed zone	The roadwork speed zone will influence the type and design of the site access and whether additional control measures are required to operate the access. Consideration needs to be given to balancing site access requirements for works traffic with the need to minimise the delay for to general traffic.

When developing works traffic VMP, the project manager or other person as nominated by divisional requirements must:

- Ensure that satisfactory arrangements are planned and implemented for vehicles associated with works;
- Ensure a VMP is developed that provides for safe access to, from and around the worksite; and
- Approve and make available a VMP before work begins.

Where a VMP identifies the installation of additional TTM on the road network, a PWZTMP qualified person must approve the relevant VMP and supporting TGSs.

For long-term, high risk or complex works, multiple VMPs may be needed for a project, and should reflect project staging and work areas. Preparation of a VMP must consider the whole of route for significant suppliers and haul requirements (i.e. from quarries, or through town sites for girder delivery, for example). Subject matter experts in road design or traffic engineering should be consulted to determine the safest and most appropriate treatments.

All drivers of vehicles at work sites are responsible for driving safely and in accordance with the road rules, exercising care and working in accordance with VMPs. Care must be taken by all drivers at work sites when entering and leaving traffic streams and turning at work sites. Contractors and agents of Transport should adopt the principles contained in Transport's [Safe Driving Policy](#).

Although this Section of the Technical Manual details the requirements for VMPs associated with access *to* and *around* a work site, it does not specify the requirement for the development of a VMP *through* a work site to ensure the separation of workers and plant. For information relating to the management of the risk of workers on foot and separation of plant and workers see the Transport WHS procedures for [Working with Mobile Plant](#).

5.2.3 Access to and from the work site

5.2.3.1 General

When developing a TMP, site accesses must be identified and designed in accordance with Austroads [Guide to Road Design](#) and Austroads [Guide to Traffic Management](#). Work sites may be accessed through the options provided in [Table 5-2](#), depending on the location of the works.

Table 5-2. Site access options

Access type	Description
Access via local roads	The use of the existing road network to access construction site off local roads, where generally the safety and traffic risks are lower. A safety assessment of the proposed route should be undertaken as upgrade work and stakeholder consultation may be required.
Left in/left out only	Utilising existing road treatments (roundabout, U-turn bay, alternate route) or designated acceleration and deceleration lanes to provide left only movements to and from site. This requires planning for works vehicles to turn around, when on the road network. This option minimises impact to existing traffic flow while also minimising the safety risks for turning construction vehicles. This may require widening of existing pavement, utilising existing shoulders, narrowing of the existing travel lane, or a combination of these. The design should be prepared in accordance with normal design practice and subject to a Road Safety Audit.
Restricted / separated access	To maintain safe and efficient access, turn movements may be separated or restricted (e.g. no right turn out movements).
Full access	All movements in and out of the site are provided at one intersection. The safety and efficiency impacts should be considered.
Dedicated access	Dedicated work site accesses, such as auxiliary lanes, must be designed by a road designer. Auxiliary lanes are also used to eliminate the need to stop traffic for work site vehicles to enter or exit the site, thus removing the need for traffic controllers and reducing the risk of rear end collisions. Dedicated work site accesses should always be located and designed using the same principles and technical guidance as a permanent treatment.

When planning and designing a work site access, [Table 5-3](#) must be referenced as part of the traffic management planning and risk assessment process.

Table 5-3. Planning and designing for work site access

Consideration	Detail
Existing traffic conditions	Review TMP data or collect recent data to understand existing traffic conditions and to support evidence based decision making. Data may include: • Traffic volumes (daily and hourly peak) including the number of heavy vehicles; • Speed data; • Crash history; • Vulnerable road user volumes and desire lines; and • Public transport (i.e. bus routes and frequency).
Site environment/constraints	Consider site specific constraints such as: • Existing lane widths; • Shoulder widths; • Horizontal and vertical alignment; • Existing road intersections and private property accesses; • Pavement type and condition; and • Road furniture (e.g. barriers, poles, gantries).
Construction traffic	Estimated construction traffic movements into and out of the work site (light and heavy vehicles). Origin and destinations of these vehicles and the turning movements at each access point should also be considered.
Speed zone	The roadwork speed zone will influence the type and design of the site access based on fundamental design considerations. When determining whether a reduced speed zone is required consideration needs to be given to balancing the site access requirements with the need to minimise the delay/increased travel time to non-construction traffic.
Positioning of site access gates	Positioned along the road length where sight distance is adequate and can operate unattended without the need for traffic controllers. See Section 4.3.6 Sight distances
Design vehicles	Appropriate turn paths, lane widths and auxiliary lane lengths must be provided as the size and type of work traffic will influence the access design.
Median crossovers	Median crossovers must only be used by work traffic as a predetermined work site access point when the sight distances detailed in Section 4.3.6 Sight distances are met. Any use of a median crossover for median work site access must be consulted with the relevant traffic management centre and be documented as part of the work site TMP, TGS and VMP.

As part of the risk assessment and design process, additional site specific risks may be identified that warrant the provision of auxiliary lanes beyond the normal design requirements.

5.2.3.2 Designing site access on multi-lane roads or high speed environments

Work site vehicles entering and leaving through traffic lanes from a multi-lane road can represent a significant hazard for both road workers and road users, particularly in a high speed, high traffic volume road environment. To mitigate this potential risk, Transport has developed a standard all work site access treatment to be adopted for work sites on motorways which provides a clear and consistent message for road users driving through roadwork sites in NSW. The treatment may be adapted for access in the near or offside shoulders.

The delineation and sign posting of the standard work site treatment shown in [Figure 5-1](#) must be adopted at the roads and localities given in [Table 5-4](#) and designed in accordance with this Technical Manual, with reference to AS 1742.2, the Transport [Delineation guide](#) and relevant linemarking specifications.

Table 5-4. Road names and route numbers for applicable to auxiliary lane provisions

State road name	Route number	Locality
Pacific Motorway	M1	Wahroonga to Beresfield
Princes Motorway	M1	Waterfall to Yallah
Eastern Distributor	M1	Entire
Western Distributor	A4	Entire
Hume Motorway	M31	Casula to Federal Highway Interchange
Hunter Expressway	M15	Entire
Westlink	M7	Entire
Western Motorway	M4	Entire
The Hills Motorway	M2	Entire
South Western Motorway	M5	Entire
NorthConnex	No route number	Entire
WestConnex	M8, M5, M4	Entire
Lane Cove Tunnel	A1, M2	Entire
Sydney Harbour Tunnel	M1	Entire
Sydney Harbour Bridge	No route number	Entire

The work site access treatment shown in [Figure 5-1](#) should also be adopted on high speed, high volume roads. The treatment may also be used in other situations where the road environment and/or other site-specific risks require a higher level of site access control.

When designing these accesses, clear and uninterrupted travel lanes should be maintained for motorists travelling past the work site. As such, retroreflective markers should not be used in the shoulders for the specific use of guiding work vehicles into or out of a work site. All work site acceleration and deceleration lanes must be designed by a road designer in accordance with Austroads *Guide to Road Design*, unless specific requirements or distances are specified in this Technical Manual.

When used, the following provisions apply:

- All yellow linemarking and diagonal chevron markings must:
 - Be Golden Yellow colour Y14 in accordance with QA Specification R145 Pavement Marking (Performance Based) and AS 2700;
 - Not have any retroreflective pavement markers; and
 - Have dimensions, spacing and angles in accordance with [Figure 5-2](#).
- For the ingress, a Yellow C1 (broken) linemarking should be installed on the outside of an edge line to delineate the diverge area of the travel lane and where vehicles enter the work site (i.e. work site deceleration lane). Refer to [Table 5-5](#) for the length of the C1 line;
- The Golden Yellow (Y14) diagonal markings are an alternative delineation method and a higher control to a Yellow C1 line. The diagonal markings should be used when:

- The pavement surface in this location is not the permanent pavement surface after the works are completed; and
- An unacceptable risk is present for public vehicles to enter the work site, or a high risk of vehicles stopping or in the shoulder. Increased risk occurs in poor vertical or horizontal alignment, high speed road, road furniture and vegetation, etc.

Note: In such cases, Golden Yellow (Y14) diagonal markings in the shoulder should be installed for a length of at least the diverge length and may extend further. Refer to [Table 5-5](#) for minimum length of diagonal markings.

- For the egress, a white continuity C1 (broken) line must be installed where the merge taper is located on the work site acceleration lane, to delineate for vehicles exiting the lane to merge and give way to through traffic. Refer to [Table 5-5](#) for length of C1 line;
- A minimum of three (3) yellow angled pavement arrow markings (ARU5), spaced in accordance with [Table 5-5](#), must be placed in merge taper of the work site access acceleration lane, to indicate to road users of merging works traffic. The middle arrow must be located at the commencement of the C1 line;
- A NO ENTRY (R2-4n) sign with a CONSTRUCTION VEHICLES EXCEPTED (R9-203-1) supplementary plate must be installed at the commencement of the work site deceleration lane diverge taper;
- A Trucks symbolic (W5-22 or T2-25) sign must be installed at D and 2D in advance of the commencement of the work site deceleration lane diverge taper. A supplementary plate '___m ON RIGHT (or LEFT)' (W8-207n) must accompany the first sign at 2D; and
- A Trucks symbolic (W5-22 or T2-25) sign must be installed with relevant supplementary plates 'MERGING' at D and '___m ON RIGHT (or LEFT)' (W8-207n) at 2D in advance of the commencement of the work site diverge area (white C1 line). The spacing of these may be lengthened to suit any existing or temporary barriers or road furniture.

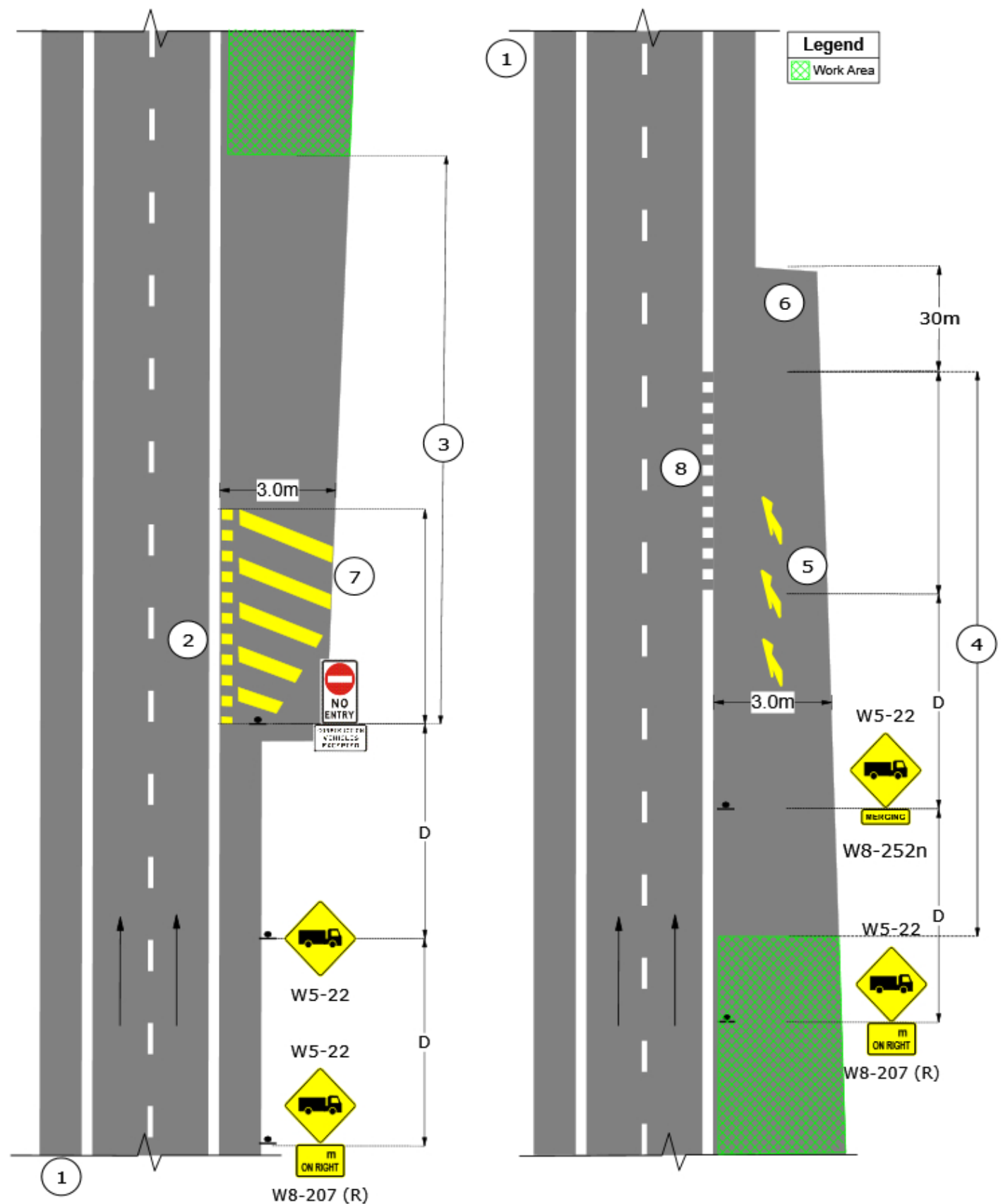


Figure 5-1. Concept layout for a deceleration and acceleration lane into/from a work area located in an 80 km/h zone in the median of a dual carriageway

Notes to Figure 5-1:

1. Additional temporary traffic management signage may be required subject to the risk assessment;
2. Length of yellow C1 (broken) line is as per Table 5-5;
3. Deceleration length (or diverge length) of work site access deceleration lane to be designed in accordance with Austroads Guide to Road Design Part 4a by a road designer;
4. Length of acceleration lane for work site access to be designed in accordance with Austroads Guide to Road Design Part 4a by road designer;
5. Yellow angled pavement arrow markings (ARU5) spaced in accordance with as per Figure 5-2 with middle arrow placed at start of C1 line.
6. Run off area of 30 m x 3 m at end of work site acceleration merge must be included;
7. Yellow (Y14) diagonal markings are an alternate treatment option and higher order control to the yellow C1 (broken line). The dimensions, spacing and width of the diagonal are shown in Figure 5-2 and Table 5-5;
8. White C1 (broken) line is as per Table 5-5

Table 5-5. Lengths of C1 line and spacing of angled arrow pavement markings

D	Length of Yellow C1 line and diagonal markings (min.)	Length of white C1 line	Spacing of angled arrow pavement markings
60	40 m	138 m	30 m
80	50 m	175 m	40 m
100	62 m	212 m	60 m

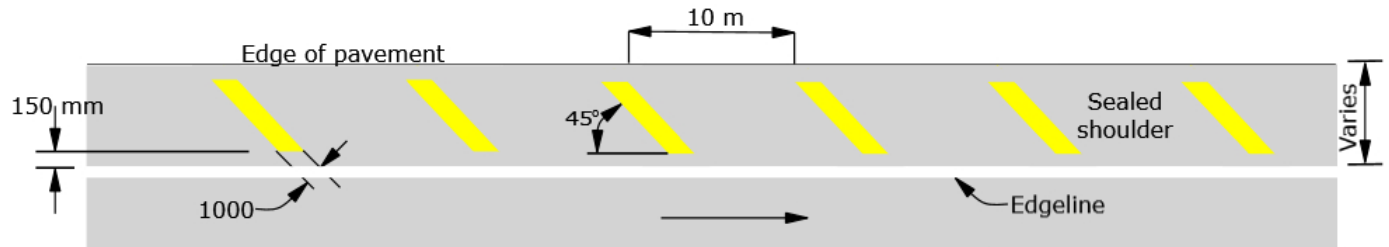


Figure 5-2. Required yellow linemarking and diagonal marking dimensions, spacing and angles

5.2.3.3 Providing for truck movements

Providing for truck movements includes any instance where trucks will be crossing or using a road to access or leave site.

Care must be exercised when designing traffic control for haul roads that cross roads. Traffic control for haul roads must be one of the following:

- PTCB;
- Traffic controllers; or
- STOP (R1-1) signs;

Where sight distance is less than 2D, Table 5-6 must be used for determining controls for managing truck movements where auxiliary lanes are not provided, depending on traffic volumes, sight distance, number of truck movements and traffic speed. Where sight distance is greater than 2D, Table 5-7 must be used.

Table 5-6. Providing for truck movements where sight distance is less than 2D

ADT	300 – 1500		More than 1500	
Number of truck movements per shift	less than or equal to 20	greater than 20	less than or equal to 20	greater than 20
Traffic control required		Yes	Yes	Yes
VMP required	Yes	Yes	Yes	Yes
Warning signs required during shifts		Yes	Yes ^{Note 2*}	Yes

Note 1 to Table 5-6: Where approach speed is greater than 95 km/h every effort should be made to choose turning locations where sight distance exceeds 2D.

Note 2* to Table 5-6: Not required when approach speed is less than 85 km/h.

Table 5-7. Providing for truck movements where sight distance is greater than 2D

ADT	300 – 1500		More than 1500	
Number of truck movements per shift	less than or equal to 20	greater than 20	less than or equal to 20	greater than 20
TGS with traffic control		Yes	Yes ^{Note 1#}	Yes ^{Note 2*}
VMP required		Yes		Yes
Warning signs required during shifts		Yes		Yes

Note 1# to Table 5-7: where approach speed is greater than 95 km/h.

Note 2* to Table 5-7: If acceleration and deceleration cannot occur on shoulders.

In any instance where traffic control is required, then the approach speed of traffic must be reduced to 65 km/h or less as per Section 5.4 Traffic control.

Care must also be taken when vehicles are leaving work areas, to ensure that gravel or mud is not deposited on the through road. The respective environment representative should be consulted for advice regarding management of this risk. Consideration therefore should be given to the use of rumble grids and coarse gravel layers or hosing to clean the vehicles before they enter or cross the through road.

5.2.4 Access around the site

Locations for turning vehicles associated with the work across lanes carrying traffic, must be restricted to predetermined locations. The following relevant factors should be considered when selecting these locations:

- Sight distance;
- Grades;

- Volumes and approach speeds of through traffic; and
- Areas clear of traffic lanes for accelerating and decelerating.

Where roadworks generate truck turning movements remote from work sites, temporary warning signs should be used at the approaches to the points of access on through roads. This is particularly important where there might be little or no other indication of likely vehicle turning movements, for example at roadside stockpile sites or via accesses to private property. Where this occurs, the Trucks symbolic (W5–22 or T2-25) sign should be used, which may be supplemented by the ON LEFT/RIGHT (W8-207) (L) or (R) signs.

If installed, all signs must be displayed before the haulage operations begin, and removed or covered at the end of each shift.

When planning for movement of work traffic, the movements described in [Table 5-8](#) are restricted.

Table 5-8. Work site vehicle restricted movements

Movement	Conditions
U-turns	Work vehicles must not perform single movement U-turn across lanes carrying traffic, unless documented in the TMP, site specific risk assessment and detailed on the relevant TGS and: • Safe intersection sight distance is achieved. The safe intersection sight distance must be determined in consultation with a technical specialist from Traffic Engineering or Road Design and based on the principles of the Austroads <i>Guide to Road Design</i>; or • Traffic is being controlled. The above permissions do not apply for illegal manoeuvres, such as U-turns across unbroken centre-lines.
Median crossovers	Median crossovers are designed for the purpose of providing access from one carriageway to another under specific controlled conditions*. For Transport work site related vehicles, median crossovers must not be used as a point of U-turn unless the following conditions are met: • A risk assessment has been completed for each cross over; • Sight distances detailed in Section 4.3.6 Sight distances are achieved; and • Use has been consulted with the relevant traffic management centre and is documented as part of the work site TMP, TGS and VMP. Where it is determined that crossovers provide a safe alternative to work site manoeuvres for a particular corridor, this must be risk assessed and documented in accordance with Section 2.8 Departures from this Technical Manual.
Reversing movements	Work vehicles must not reverse (including 3-point U-turns) in lanes carrying traffic, unless: • Traffic is being controlled by a PTCO or traffic controllers; or • The approach sight distance exceeds 6D i.e. 20 seconds of travel time.

Note* to Table 5-8: Emergency vehicles such as police service, ambulance service and fire brigade as well as NRMA and nominated Transport emergency response vehicles are permitted to use the median crossover to make a U-turn, provided it is performed in accordance with the relevant organisational safety requirements.

5.2.5 Access to adjoining properties

Pedestrian and vehicle access to adjoining properties must be maintained, unless clear agreement has been reached with the occupants for a temporary restriction. Effective communication with residents must be conducted for the work to be carried out safely and efficiently.

Temporary access to adjacent side streets and properties must be trafficable under all expected weather conditions and must be constructed to reasonable standards matching the side streets and property driveways. If conditions of access cannot be maintained other arrangements must be made for those forms of movement (i.e. wheelchair, bicycle, etc.).

5.3 Work site specific requirements

5.3.1 Traffic incident management

A TMP must include a traffic incident plan. The following is a list of recommended inclusions in a traffic incident plan:

- Names and contact details of nominated personnel responsible for dealing with traffic incidents occurring at the work site;
- Contact details of the person responsible for the works, TMC operation representative, police, and emergency services representatives (where appropriate);
- Procedure to be followed in the event of a traffic incident at the site;
- List of plant that will be available for moving portable concrete safety barriers (if in use on-site);
- Inventory of safety barriers, signs etc. and their storage location(s) that will be available to replace damaged barriers in event of a traffic accident (if in use on-site);
- Procedure for carrying out investigations of traffic incidents involving members of the public or workers. This should include:
 - Checking that the traffic control measures in place are in accordance with the TMP and its component plans, and ROL conditions;
 - Carrying out a “drive through” and video recording of the roadway, including the location where the incident has taken place;
- Information required for initial notification to the person responsible for the works, and where necessary, other relevant authorities; and
- Format for reporting and communication of the results of traffic incident investigations, and lessons learned.

5.3.2 Working at intersections

Intersections vary greatly in type, number of legs, number of traffic lanes and forms of traffic control. Due to the complex nature of working at intersections, the following principles should be considered when developing a TMP that requires work at these locations:

- Sufficient capacity should be maintained, and delays should be kept to acceptable levels, for example, excess capacity should be avoided which encourages high speeds;
- Blockages at the intersection from downstream queuing should be avoided;
- Provision for alternative routes should be made if right hand turns or other movements are temporarily prohibited at the intersection;
- Work periods should be restricted to outside peak traffic flows if the intersection is a critical one within the road network;
- Work areas should be clearly defined so that road users can easily identify paths available for traffic within the intersection;

- Guide signs should be clearly visible and not obstructed by road plant or temporary signs. If necessary they should be relocated; and
- Effective control using signs, traffic signal or controllers, linemarking and delineation should be maintained.

Where distance between intersections is sufficient, standard treatments for the signs and tapers on the approaches must be used. Within the intersection, the work usually needs to be staged so that sufficient lanes are kept open to handle the traffic flows. These stages of the work must be clearly defined using closely spaced traffic cones or temporary barriers. An audit should be undertaken on all temporary works arrangements, prior to a traffic switch or similar being implemented at intersections. If the works are to be in place during night hours, the audit should also be done at night to ensure appropriate delineation is in place.

5.3.3 Working at or close to traffic signals

A need to work at or near traffic signals may arise for a variety of reasons including operation and maintenance of equipment, emergency fault repairs, pavement resurfacing or linemarking.

When planning works that involve or impact traffic signals the following should be considered:

- Consultation with police where major or complex traffic signals will need to be flashed yellow or blacked out;
- Where permanent traffic signals are still in operation, alternative traffic control arrangements, such as a PTCD or a traffic controller with a manual STOP/SLOW bat must not be used to control traffic;
- Determine if turn restrictions or detours are necessary and:
 - Check that alternative routes are available and satisfactory for all classes of vehicle being detoured;
 - Consider the impact that additional traffic will have on the operation (phasing) of any traffic signals on the alternative route;
 - Consider the impact that any lane closures, particularly turn lanes will have on signal operation (phasing) and intersection efficiency.
- Where there are likely to be significant impacts on the operation of traffic signals or coordination between signals engage with Transport Network Operations for advice and mitigation measures;
- Notification to road users in advance of proposed works of impact to the signals. This may be by way of advertising, for example using VMS roadside noticeboard advertising, letter drops or press releases; and
- Ensure that adequate provision is made for pedestrians and mobility aid users if it is necessary to turn off pedestrian signals or close a pedestrian crossing at intersection traffic signals.

This variety of works means that the scale and complexity of the TTM required to facilitate the work activities will also vary significantly. When determining TGS requirements, [Section 7.7.4.2 Working at traffic signals](#) must be followed.

5.3.4 Working in the vicinity of railway lines

The safety procedures provided in this Section must be applied whenever work is carried out in or adjacent to the rail corridor (refer to the [Rail Safety National Law National Regulations](#) and the [Work Health and Safety Act 2011](#) for additional requirements).

For the purposes of this Section, the terms and definitions of [Table 5-9](#) apply.

Table 5-9. Terms and definitions applicable to working in the vicinity of railway lines

Term	Definition
danger zone	everywhere within 3 m horizontally from the nearest rail and any distance above or below these 3 m
protection officer	authorised qualified worker responsible for managing the rail safety component of work site protection
rail corridor	area within rail boundary fence lines, or if there are no fences, everywhere within 15 m of the outer most rails and includes rail tracks, rail junctions, level crossings, station buildings, platforms, signal boxes, tunnels, bridges and other associated structures
rail infrastructure manager	owner and maintainer of the rail network
rail traffic	trains and all other on-track vehicles

When working in or adjacent to the rail corridor, the rail owner and the relevant rail infrastructure manager should be consulted early during development of the TMP and risk assessment, to ensure all constraints, risks and requirements are identified and mitigated. This includes the completion of an Access Application Form (and requires at least 20 working days for processing).

If the situation is such that any person, plant or equipment are required to be within the rail corridor, then the manager authorising the work must ensure that the rail infrastructure manager is advised and arrangements are made for the provision of a protection officer to ensure the safe interaction between the work and the rail network. The rail infrastructure manager must also be notified if the work affects or has the potential to impact the railway line or rail structures. This includes travel to and from the work site within the rail corridor ensuring vehicle and personnel remain outside the danger zone.

When performing work where a protection officer is required, all workers must follow the instructions of the protection officer and work must not be carried out without the presence of the protection officer.

The supervisor in charge of the site must ensure that:

- All workers understand the requirement to follow instructions of the protection officer;
- All workers must wear approved rail standard safety vests and other personal protective equipment as required by the rail infrastructure manager;
- Where required, arrangements are made with the rail infrastructure manager to provide persons with applicable rail safety awareness training; and
- Work is not carried out within the rail corridor, including travel to and from the work site when the protection officer is off-site.

Notwithstanding the fact that the manager authorising the work may be satisfied that there is a physical barrier of solid construction that prevents persons or their equipment from coming into contact with rail traffic or overhead wiring, the manager must still notify the rail infrastructure manager of the work and comply with the its requirements.

In the event that the physical barrier becomes dismantled or damaged in such a way as to allow the possibility of persons coming into contact with rail traffic or overhead wiring, then the supervisor must stop all work on-site and notify the rail infrastructure manager so that the barrier can be re-established or the situation reviewed.

Persons on-site must not damage or attempt in any way to overcome a physical barrier.

5.3.5 Working at night or in low visibility

When working in any poorly lit situations, such as during periods of fog or working in tunnels, a TGS must specify the requirements and details provided in [Table 5-10](#). For night works in particular, it should be noted that;

- Obstacles are less conspicuous and peripheral vision is reduced; and
- There is a higher risk of motorists being fatigued, alcohol-impaired or driving at higher speeds.

Table 5-10. TGS requirements for sites with poor visibility

Requirement	Details
The work site has appropriate flood lighting	PTCDs or traffic controllers with STOP/SLOW bats at the approaches to a night time work area must be clearly visible to road users. This may require additional lighting. The ITCP, works supervisor or team leader must check floodlighting at work sites to ensure that floodlights do not adversely affect road users, adjacent dwellings or businesses. These checks must be made by driving around, past and through the work site in all directions of travel. On divided carriageway roads, these checks must be carried out from all carriageways, even if the work area is only on one carriageway.
Additional signs and devices	On high speed, high volume roads and on busy roads in built-up areas, flashing arrow signs must be used at night. Care must be taken to ensure that the dimming facility is operating correctly. It is recommended that back-up units are made available on critical works. Only type-approved flashing arrow signs must be used in NSW. For information and approval requirements, see Section 6.9.2 Illuminated flashing arrow signs .
Delineation in accordance with AS 1742.3.	The standard ROADWORK AHEAD (T1-1), Worker symbolic (T1-5) and Traffic Controller symbolic (T1-34) signs must be used for night works. All signs must be manufactured with Class 400 or Class 400T (retroreflective) yellow sheeting. As of 30 June 2020 the NIGHT ROADWORK AHEAD (T1- 223), Worker symbolic (T1-224 and T1-5-2) and Traffic Controller symbolic (T1-200-2 and T1-200-3) must not be used.
All personnel on the site wear approved high visibility clothing	In addition to traffic controllers, all personnel working at night must wear approved high-visibility external clothing, including wet weather clothing, in accordance with Transport's current Personal Protective Equipment Procedure . PPE must be clean and bright. Additional PPE such as hearing, eye and foot protection must also be worn as required by the relevant SWMS and in accordance with Transport's current Personal Protective Equipment Procedure .

5.4 Traffic control

5.4.1 General

Traffic control must be used if road users will be directed to disobey a traffic regulation, such as crossing a barrier line (traffic control does not override requirements for road users to obey traffic signals).

In instances of inclement weather where visibility and skid resistance is compromised, traffic control should not be used. However, if this is not practical additional controls must be implemented to ensure the safety of road workers and road users. Prior to installation of traffic control, a contingency plan should be developed to determine the actions and controls that are required in the instance of inclement weather, such as rain.

At any point of traffic control, four cones must be placed at 4 m spacing on the centre line, edge line or both, prior to the traffic control point. Traffic cones are used to highlight the traffic control position, where vehicles are to stop and for traffic management purposes.

5.4.2 Traffic control types

For all work sites requiring traffic control, a PTCD must be used when the existing permanent speed limit is above 45 km/h. This requirement is not applicable to instances and environments of emergency response.

A manual traffic controller may be used provided all of the following conditions are met:

- The use of a PTCD is demonstrated to not achieve the safest outcome;
- The decision to use a manual traffic controller instead of a PTCD is documented in the TMP or supporting risk assessment; and
- Approval is granted by the one-up manager of the PWZTMP qualified person responsible for the works relevant TMP.

Under the Road Transport Act 2013 a person must not install or display a prescribed traffic control device on, above or near a road without appropriate authority. All traffic control devices, including road markings, traffic signals, or any other device used to direct or warn traffic at a roadwork site must be accepted for use in accordance with this Technical Manual.

Devices detailed in this Technical Manual and the relevant specifications where required, provide the authorisation and conditions for use by which those devices can be installed on a Transport work site. Only approved PTCDs detailed in [Section 6.6 Portable traffic control devices \(PTCDs\)](#) must be used.

Any device that is not accepted for use in this Technical Manual must not be used unless the conditions in [Section 2.8.3 Use of unaccepted devices](#) have been met or approval has been obtained by a person with appropriate authority under Part 11 of the Transport Delegations.

5.4.3 Requirements for traffic controllers

[Table 5-11](#) provides the general requirements that must be applied when a traffic controller is used, including the use of a PTCD.

Table 5-11. General requirements for the use of a traffic controller

Aspect	Requirements
Training	Traffic controllers must be trained in their duties and verified as competent. At a minimum, traffic controllers must have the 'Traffic Controller' qualification and be deemed competent in the use of the relevant PTCD or STOP/SLOW bat.
Identification	All traffic controllers performing traffic control work must be identified as such. This must be by wearing a badge or other distinguishing mark clearly stating "traffic controller". Reference to Transport (or RMS/RTA) is not permitted unless the traffic controller is an employee of Transport.
Fitness for duty	Traffic controllers must be fit for duty when reporting for work and remain so during working hours (including breaks and travel time). Traffic controllers must comply with the drug and alcohol procedure in place at their work site. See Section 2.4.3 for further information.
Equipment	Traffic controllers must only control traffic using an accepted traffic control device as per this Technical Manual.

Aspect	Requirements
Approach speed	Where traffic control is used, the speed limit applied to approaching traffic must be reduced to less than 65 km/h. Additionally, a speed zone of less than 65 km/h must commence more than 100 m from the traffic control position, in accordance with Section 4.5.5 Implementation .
Visibility	Traffic controllers must be located in a position where the sight distance between them and oncoming traffic is a minimum of 1.5D, unless: • A site specific risk assessment has been undertaken; • Additional control measures identified in the site specific risk assessment are in place; and • It has been documented in the relevant TMP and TGS.
Positioning	A traffic controller must be positioned: • Outside the travel path of traffic; • Facing the traffic; • Where a clear and safe escape path is available; • So their body is adequately illuminated by installed lighting when working at night or in low visibility; and • Such that they do not obstruct motorists' view of other signs and devices, or be hidden by them. A traffic controller must not leave their position (control point) unless directed by the ITCP qualified person, equivalent person responsible for site, or upon relief by another traffic controller.
Communication	Traffic controllers must ensure that they are able to communicate effectively to other nominated members of the work crew, such as other traffic controllers or plant operators. Communication may be via: • Direct verbal communication; • Verbal communication via a two way radio; or • Through an intermediate person. The use of verbal communication either directly or via a two way radio is the preferred means of communication between traffic controllers. The use of a two way radio is essential if clear sight between operators is not available, are located an extended distance apart, or they are working at night.
Signage	A PTCD sign relevant to the device used, such as Boom Barrier symbolic (T1-272n) or Signals symbolic sign (T1-30), or a Traffic Controller symbolic sign (T1-34) must be used to give advance warning of the presence of traffic control. A PREPARE TO STOP (T1-18) sign must also be used when traffic is required to stop at the traffic control location. The above signs must only be used when the traffic control is in operation and must be removed or covered up when traffic control is discontinued or during breaks, such as lunch.
Period of duty	Traffic controllers must be relieved every two hours of work. They may return to traffic control duties after a minimum of 15 minutes of rest or alternative duties. This also applies to traffic controllers operating a PTCD.
PPE	Traffic controllers must wear all relevant PPE for their works as required by their employer. At a minimum, these requirements must meet the standard set by the Transport Personal Protective Equipment Procedure PN066P19

Aspect	Requirements
Performing duties	When performing their duties traffic controllers must: • Check that the appropriate signage has been installed or uncovered by the ITCP qualified person, prior to undertaking traffic control duties; • Remain focussed on their traffic control duties and be aware of the roadworks; • Report adverse driving behaviour from motorists in accordance with Section 8.3 Reporting work site incidents; and • Remain courteous at all times in dealing with the public. Traffic controllers must not: • Use any electronic device, including a mobile phone, unless required for communication with other traffic controllers or their team supervisor; or • Allow persons to stand near or gather around the PTCD or STOP/SLOW bat, which can cause distraction or create confusion for the travelling public.

5.4.4 Additional requirements for the use of a STOP/SLOW bat

If a manual traffic controller has been justified and approved in the TMP, the manual traffic controller must only be in place when traffic speeds have been reduced to less than 65 km/h in accordance with [Section 5.4 Traffic control](#). A traffic controller must:

- Only use approved hand signals shown in [Table 5-12](#) to direct traffic when using a STOP/SLOW bat;
- Only control one lane of traffic, and usually only in one direction;
- Be clearly visible to traffic at all times. This may require the traffic controller to reposition after the first vehicle has stopped in order to be visible to queuing traffic;
- Stay at the head of the traffic queue at a safe distance from the opposing traffic lane; and
- Stand clear of traffic when allowing it to proceed.

An illuminated wand may be used to supplement the STOP/SLOW bat during night works or when visibility is limited.

Table 5-12. Traffic controller hand signals

Traffic direction	Action	Illustration
From Slow to Stop: stop traffic	1. Choose a gap in the traffic flow and/or most appropriate vehicle to stop; 2. Turn the STOP/SLOW bat to STOP; 3. Raise the free arm into the stop signal position with the palm of the hand towards the traffic.	
From Stop to Slow: allow traffic to proceed	1. Check that all traffic from the other end of the work area has passed; 2. Turn the STOP/SLOW bat to SLOW; 3. Give the proceed signal with a sweeping motion.	
Slow traffic	1. When the STOP/SLOW bat is showing SLOW extend the free arm to the side and wave hand up and down.	

5.4.5 Traffic control locations

The traffic control locations must be considered as part of the development of the TMP. When determining the traffic control locations needed for a work site, the following must be considered:

- Type of temporary traffic management;
- Number of approaches to be controlled;
- The way in which traffic is being controlled, including the use of a PTCD;

- Sight distances and factors that impact on this including road geometry and vegetation;
- Expected duration of traffic control;
- Infrastructure such as transit stops, assessed queue lengths for planned stopping duration; and
- Weather conditions (i.e. adverse weather such as extreme heat).

Once the traffic control locations have been determined, the number of traffic controllers required to operate the traffic control devices, or STOP/SLOW bats can be determined.

To determine the number of traffic controllers for operating a traffic control device, see [Section 6.6 Portable traffic control devices \(PTCDs\)](#).

When a traffic controller is utilised for operation of a PTCD or a STOP/SLOW bat, a sight distance of 1.5D must be allowed for.

A traffic controller is not required where all of the following conditions are met, otherwise at least one traffic controller must be in place (see [Figure 5-3](#)):

- ADT is less than 100 vehicles per day;
- Each entry to the work site is visible from the other end; and
- The work area is shorter than 100 m.

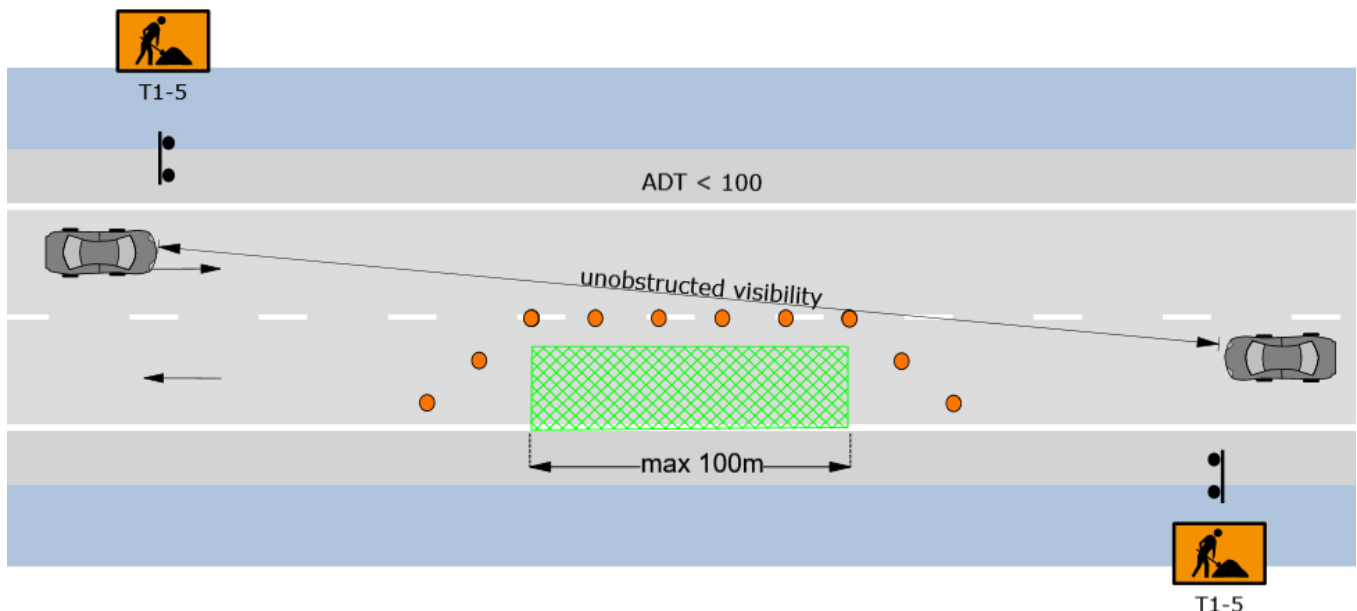


Figure 5-3. No traffic controller required

One traffic controller may be in place where:

- ADT is 500 vehicles or less; and
- Sight distance from the traffic controller exceeds the minimum in [Table 5-13](#).

Two traffic controllers must be in place where:

- ADT is greater than 500 vehicles per day; or
- The work area is longer than 250 m; or
- Sight distance from the traffic controller in the obstructed lane to the other approach is less than the minimum provided in [Table 5-13](#).

Table 5-13. Traffic controller minimum sight distances

Existing permanent speed km/h	Length of Work Area (L)	Minimum clear sight distance to oncoming traffic
less than 105	less than 60 m	300 m
less than 105	greater than or equal to 60 m	L + 250 m
greater than 105	less than 60 m	400 m
greater than 105	greater than or equal to 60 m	L + 350 m

When traffic control is being used for a multi-lane carriageway, a merge or merges must be introduced upstream of the traffic control position so that only one lane is managed at a time as per [Figure 5-4](#).

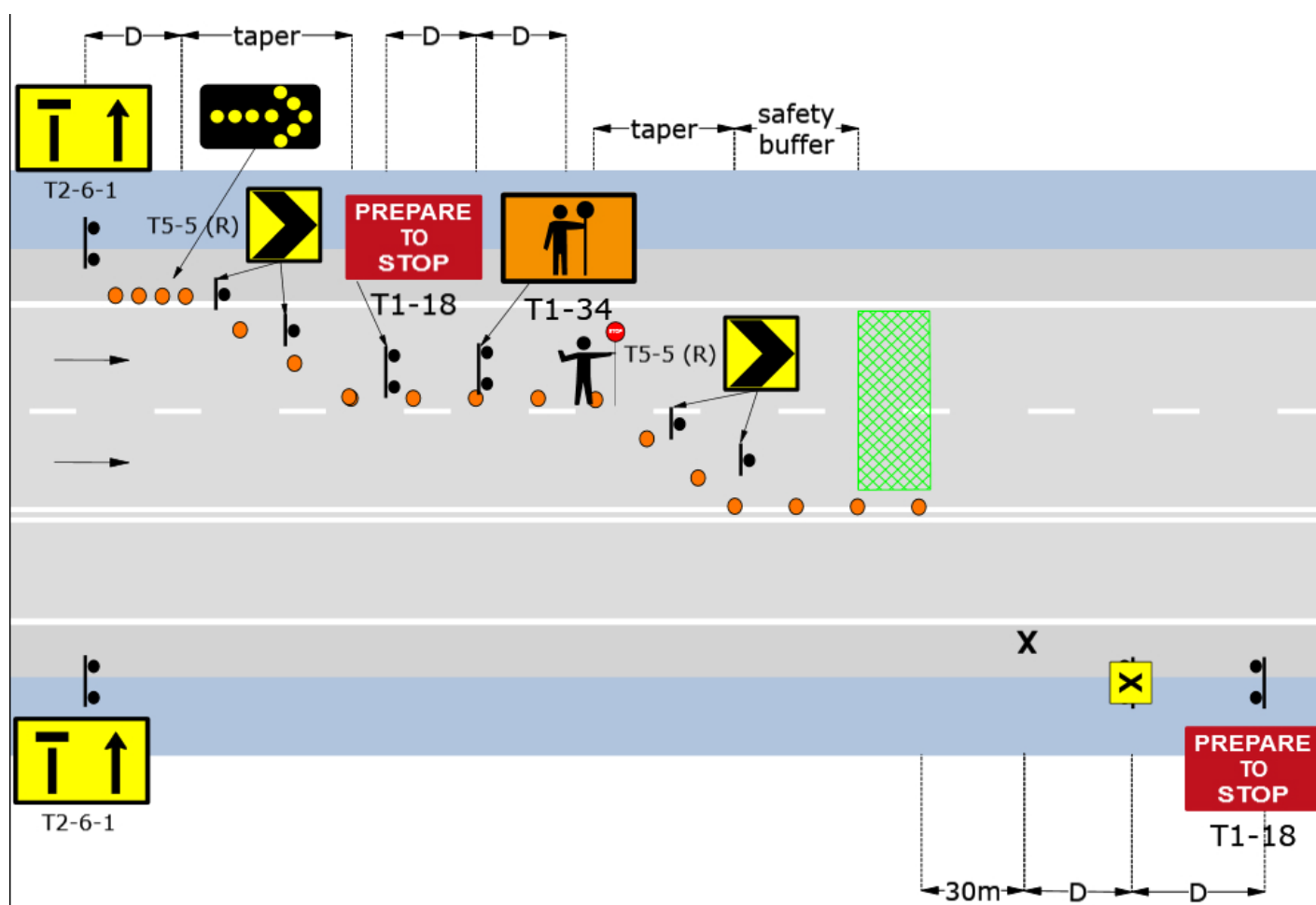


Figure 5-4. Manual traffic controller for multi-lane carriageway

6 Signs and devices

6.1 General

This Section must be used to determine the use and placement of TTM signs and devices at Transport work sites by persons qualified with PWZTMP or ITCP as applicable. The Section includes defining the clearances and spacing for signs and devices as well as the sequence of installation and removal.

The devices in this Section include:

- Portable traffic control devices (PTCDs);
- Safety barriers;
- Traffic guidance and delineation devices; and
- Other safety devices.

In relation to this Section the following must not be used:

- Isolated or non-continuous safety barrier units;
- Barrier boards parallel to the direction of traffic flow;
- Any other traffic control device that is not authorised for use.

For authorising the use of an unaccepted traffic control device see [Section 2.8 Departures from this Technical Manual](#).

6.2 Clearances and spacing of signs and devices

6.2.1 Edge clearances for delineating devices and temporary safety barriers

Clearances between the edge of traffic lane and delineating devices or a road safety barrier system must be in accordance with in [Table 6-1](#), unless alternatives have been determined based on a documented risk assessment and detailed in the TMP as per [Section 2.8 Departures from this Technical Manual](#).

Clearances must be measured to the traffic side edge of delineating devices or barrier. This edge must also be the line from which clearances to the work area are measured for the purpose of determining treatments.

Where barrier shape includes feet, the clearance must be measured from the edge of the foot on the traffic side. Containment fences marking the limit of work area must be placed as specified in [Section 6.8.3 Plastic containment tapes and fences](#).

See [Figure 6-1](#) and [Figure 6-2](#) for further information.

Table 6-1. Edge clearances

Edge of traffic lane to:	Edge clearances
Line of traffic cones or bollards	• 0.5 m for traffic speeds less than 65 km/h • 1.0 m for traffic speeds greater than 65 km/h

Edge of traffic lane to:	Edge clearances
Barrier boards, temporary guide posts or temporary hazard markers	1.0 m
Road safety barrier system	0.3 m for traffic speeds less than 45 km/h 0.5 m for traffic speeds 45 to 65 km/h 1.0 m for traffic speeds 65 to 85 km/h 2.0 m for traffic speeds greater than 85 km/h

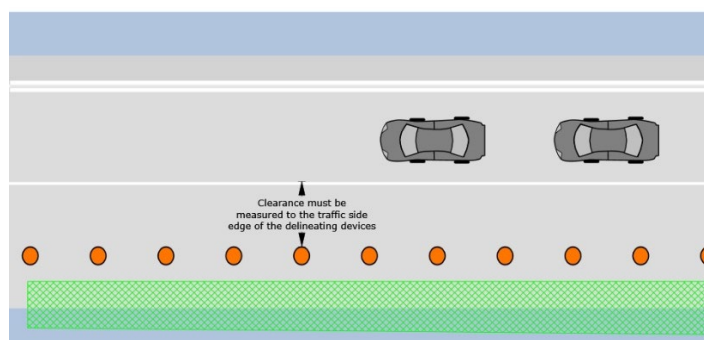


Figure 6-1. Clearance measurement from delineating device (cones)

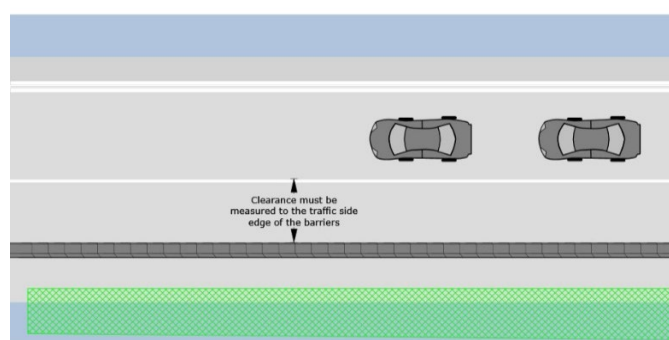


Figure 6-2. Clearance measurement from delineating device (barrier)

6.2.2 Use of delineating devices on kerbs

If the edge of the traffic lane is kerbed, delineation devices must be placed 0.3 m to 0.5 m clear behind the face of kerb.

Temporary kerbing used for delineation of the roadway on detours and side-tracks must be clearly delineated with red delineators on the left side and white on the right (two-way roadway) or yellow on the right (one-way roadway). Additional information relating to the use of temporary kerbing can be found in [Section 6.8.5 Temporary kerbs](#).

6.2.3 Use of temporary safety barriers on kerbs

When an errant vehicle traverses a kerb directly prior to impact with a temporary safety barrier, roll and pitch are developed which affect the interaction of the vehicle with the barrier, increasing the risk of the vehicle mounting the barrier.

Where installation of a temporary safety barrier is required behind a kerb the barrier must be installed either:

- Greater than 1.5 m behind the kerb, to allow an errant vehicle to stabilise before striking the barrier; or
- Within 0.2 m of the kerb so that an errant vehicle has not had adequate time to develop significant pitch and/or roll.

Where temporary safety barriers are required to be installed behind kerbs on roads with a posted speed limit of greater than 75 km/h, a risk assessment must be performed to determine the most appropriate barrier position and detailed within the TMP.

6.2.4 Placement and clearances of registered plant or devices

This Section applies to registered plant or trailer mounted devices that are left unattended for a period of time on the road-side during roadworks, such as lighting towers and VMS. The plant and devices captured within this section generally have a physical size and mass such that they are not frangible and if positioned inappropriately can pose a potential road safety hazard.

Prior to placement of the plant or device, a site-specific risk assessment must be undertaken by the persons responsible for the work site to identify, assess and manage the risks associated with deployment. The risk assessment should consider:

- Location of specific hazards such as traffic, utilities, culverts, medians or embankments;
- Risks introduced as a result of the placement or operation of the device;
- Parking and removal of plant or device; and
- If the items are to be unloaded and loaded onto floats or rigid trucks.

When determining the placement and clearances, the plant or device must be located:

- In a lawful parking location i.e. not on a footpath;
- Behind a Transport accepted safety barrier;
- No less than the appropriate sight distance from an intersection, merge point, exit ramp, traffic control signal or sharp curves;
- In a position where road user sight lines are not obstructed;
- To provide for the safe movements of all road users, including cyclists, pedestrians (inclusive of mobility and vision impaired pedestrians) and heavy vehicles including wide loads;
- Where safe stopping sight distance to pedestrians at crossing points (e.g. refuges, pedestrian crossings, signalised intersections) are not obstructed; and
- Where adequate delineation can be provided such as bollards, traffic cones or temporary linemarking so that road users can perceive the location of the device at night or in low visibility.

When a device cannot be protected by a barrier, the devices or vehicle must be positioned as far away from the edge of the traffic lane as is practical in a position determined suitable based on a documented risk assessment and detailed in the TMP. For more information refer to Austroads [Guide to Road Design Part 6: Roadside Design, Safety and Barriers](#).

Additional issues that should be considered when locating plant or devices include:

- Positioning relative to curves – avoid locating the device, plant item or other vehicle on the outside a curve;
- Avoid locating where vehicles are laterally moving or weaving across traffic lanes (i.e. at the end of merge or in the merge run out area); and
- The deflection zone of the relevant barrier systems
- When a device, plant item or other vehicle is being located behind a crash barrier. Adequate lateral separation should be provided between the back of the barrier and the device, plant item or vehicle to ensure that the performance of the system is not compromised.

Additional information specific to the placement of Variable Message Signs is contained in [Section 6.9.1 Variable message signs \(portable\)](#).

6.2.5 Spacing of cones and bollards

Cones and bollards are used to define the traffic path past or through work areas (refer to [Section 6.8 Traffic guidance and delineation devices](#) for further information on types, sizes and uses of cones and bollards).

Maximum spacing of cones and bollards must be in accordance with [Table 6-2](#). Where traffic volumes are high or other conditions warrant it, consideration should be given to reducing the spacing of cones to as close as 1 m to prevent traffic taking a wrong turn through cones or bollards.

Table 6-2. Required maximum spacing of cones and bollards

Purpose and usage	Speed zone of device location km/h	Maximum spacing m
On approach to a traffic control position (centreline or edge line)	All cases	4
Merge tapers	55 to 75 greater than 75	9 12
Lateral shift tapers	55 to 75 greater than 75	12 18
Protecting freshly painted lines	55 to 75 greater than 75	24 60*
All other purposes	less than 55 55 to 75 greater than 75	4 12 18

Note* to Table 6-2: This spacing should be reduced on curves or crests or if the row of cones is not clearly defined at night.

6.2.6 Spacing of signs

Signs must be spaced in accordance with [Table 6-3](#). The value of 'Dimension D' is used to determine the placement of signs, see [Section 7.3 Dimension D](#) for determining 'Dimension D'.

Table 6-3. Sign spacing requirements

Number of signs	Approach speed	
	less than 65 km/h	65 km/h or greater
One advanced sign	D	2D
Multiple advanced signs	D	D

Where there is more than one advance sign position, the advance sign nearest the work area must be placed D from the beginning of the taper area or diversion and other advance sign positions at successive spacing of D further in advance of the work area.

Tolerances for positioning on signs and devices is detailed in Section 7.10.3 Tolerances on positioning of signs and devices.

6.3 Maintaining temporary signs and devices

Work site temporary signs and devices need to be consistently maintained with particular attention given to the following:

- Signs and devices displayed must remain appropriate for changing circumstances during the work;
- Signs which are not required between shifts must be covered;
- Sign placement, including covers must be checked after weather events;
- Signs and devices must be in good condition;
- Damaged or disfigured signs in the work environment must be replaced as soon as possible, especially if the warnings displayed are not clear;
- Signs and devices erected before they are required must be covered by a suitable, opaque material in accordance with AS 1742.3. A suitable material may be WF 200 woven polypropylene material. The cover must be removed immediately prior to the commencement of work;

Note: *Covering signs with hessian material does not sufficiently inhibit the sign's retroreflective performance and should not be used. Additionally, dark coloured and plastic materials may cause overheating or excessive moisture build-up and therefore damage to the sign.*

- If used at night, signs and devices must be inspected at night (see Appendix E – Inspection checklists and tools).
- Signs must not display conflicting messages; and
- Temporary signs displays must not be altered or changed by adhering non-approved materials to the sign face. For example, roadwork speed zones must not be altered by using tape or any other similar material.

The condition of signs must be checked prior to installation and as a part of regular TTM inspections. See Section 8.1 Work site inspections, reviews and audits.

6.4 Sequence for installation and removal of signs and devices

6.4.1 General

The sequence for installation and removal of signs and devices must be considered in the TMP and documented on the TGS or another site document such as a SWMS. The installation and removal of signs and devices must:

- Be undertaken in accordance with the procedures shown on the TGS or other document;
- Be planned to be in the direction of normal traffic flow;
- Not require workers to cross roads or carriageways on foot; and
- Be undertaken with a work vehicle showing a flashing arrow or rotating/flashing light(s) positioned between the workers and approaching traffic.

Special consideration must be given for the removal of signs on central medians and barriers on multi-lane divided carriageways, i.e. a site-specific TGS or use of a work convoy etc.

6.4.2 Installation

Before work commences, signs and devices at the work site must be installed in a sequence that is safe and efficient. After the work area has been located, via the use of a GPS, survey, landmarks, side streets or chainage, setting up a site to install signs and devices should be in accordance with the general procedures described below:

Two-lane, two-way roads

For two-lane, two-way roads, installation should occur in the following order:

1. Install termination signs (if no side roads).
2. Install on side streets.
3. Install in the non-working lane (unaffected direction).
4. Install in the working lane (affected direction).

Figure 6-3 provides an example sign installation sequence for a two-lane, two-way road.

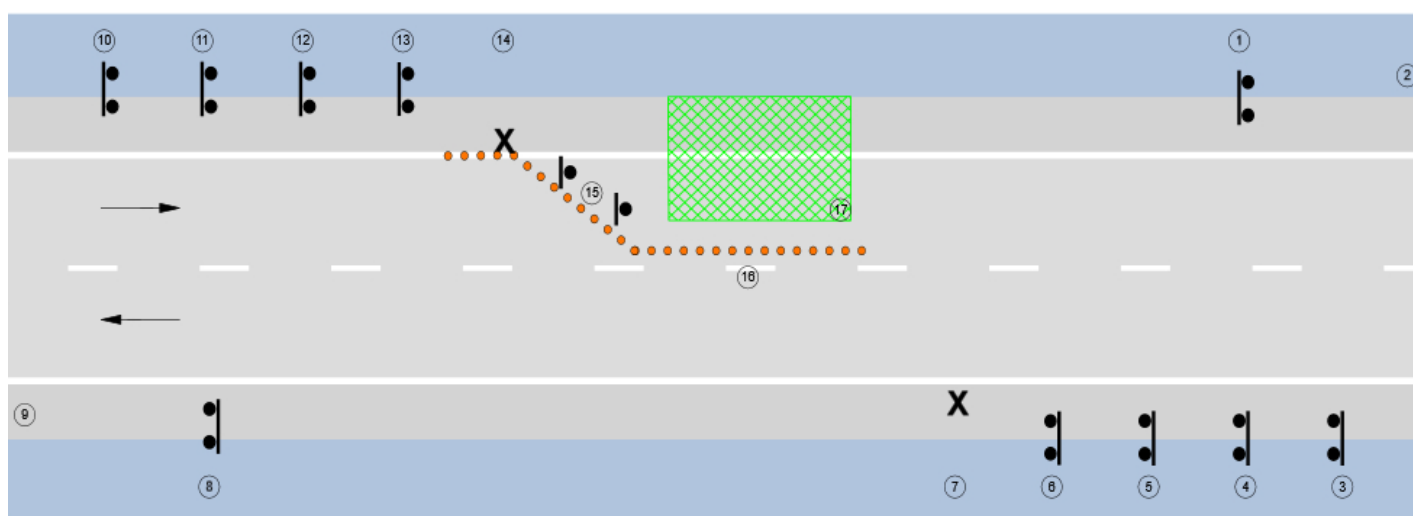


Figure 6-3. Example sign installation sequence for a two-lane, two-way road

Multi-lane roads

For multi-lane roads, installation should occur in the following order:

1. Install signs and devices for the non-working lane (un-affected direction).
2. Install signs and devices for the working lane (affected direction).

Special consideration must be given to the installation of signs on central medians and barriers on multi-lane divided carriageways. In such cases, a site-specific TGS or use of a convoy may be required.

Figure 6-4 provides an example sign installation sequence for a multi-lane road.

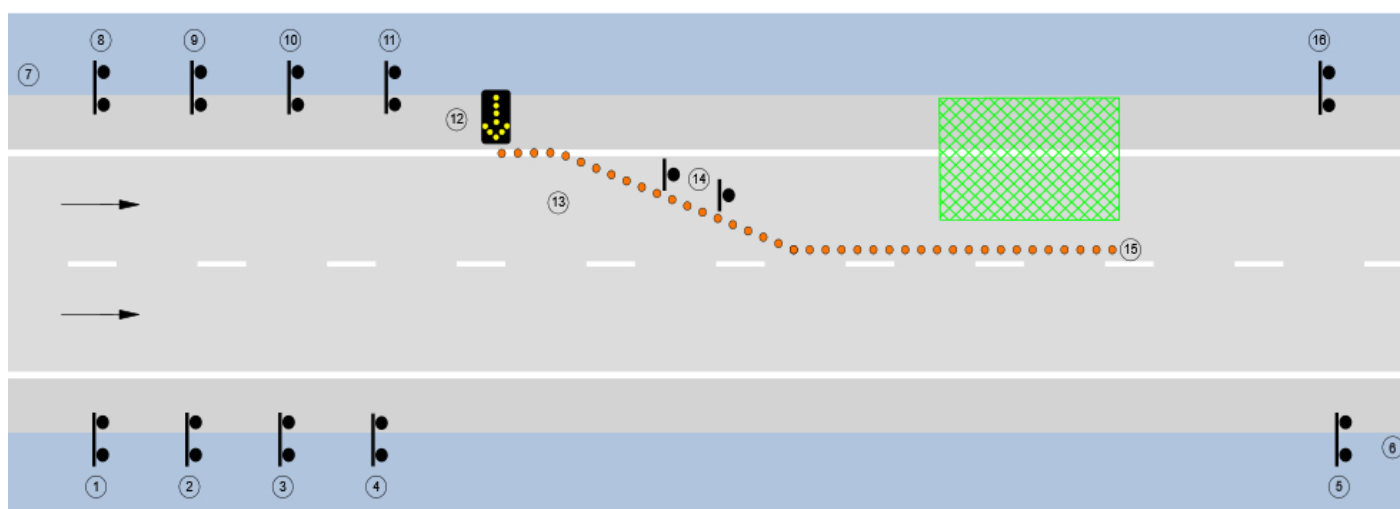


Figure 6-4. Example sign installation sequence for a multi-lane road

A different implementation sequence may be needed for site specific circumstances, e.g. install End Roadwork and reinstate the existing permanent speed limit first.

Where a work area is moving progressively along the road, relocation of the signs ahead should take place in accordance with the sequence described above. Those behind should be relocated in the reverse sequence.

For long-term or recurring short-term sites, consideration should be given to marking the desired location of each sign or device on the road for easy placement.

6.4.3 Removal

Removal of traffic control signs and devices should be undertaken in the reverse order of installation, progressing from the work area out toward the approaches. On motorway type carriageways, the removal of signs can be difficult in this sequence, in which case, signs should be removed in the same order that they were installed. The work vehicle should be positioned between the workers and approaching traffic when removing signs in this manner.

When removing delineation devices, such as cones, bollards or barrier boards used to close a lane, an advanced warning vehicle should be used to warn road users of workers on foot and a work vehicle must also be positioned between the workers and approaching traffic.

A work vehicle must only proceed in a forward direction towards approaching traffic along the closed roadway if it is determined by the PWZTMP or ITCP qualified person that it is safe to do so. This should not occur at night time where it may create motorist confusion or distraction, such as headlight glare.

6.4.4 High risk sites

At sites where it is deemed too difficult or unsafe to install and remove control signs and devices in accordance with the above general principles, special arrangements to complement the above general principles must be adopted to maintain worker safety. These arrangements must be documented as part of the TMP and the relevant TGS or SWMS, and mitigation measures incorporated into the work practice that is adopted. This may mean that signs are removed in the same order that they were erected to allow the work vehicle with a rotating or flashing light(s) to move in the direction of normal traffic flow when use of the travel lane is the only alternative.

6.5 Traffic control signs

6.5.1 General

Standard TTM signs must be used wherever one suitable for the purpose exists. All existing signs can be found in the [Traffic Signs Register](#). Where a suitable standard traffic sign does not exist, the process detailed in [Section 2.8.2 Use of unapproved signs](#) must be followed.

Signs referred to in this Technical Manual follow the naming and numbering conventions of AS 1742.1, which is summarised in [Table 6-4](#).

Table 6-4. Sign naming and numbering convention

Class	Function in TTM	TTM Sign example
Regulatory sign (R)	To regulate the movement of traffic by indicating when or where a legal requirement applies, failure to comply constitutes an offence.	STOP HERE ON RED SIGNAL (R6-6)
Warning sign (W)	To warn road users of unexpected or hazardous conditions on or adjacent to the road.	Traffic Lights symbolic (W3-3)
Direction sign (G)	To inform and advise road users of directions or non-regulatory traffic instructions.	Speed limit AHEAD sign (G9-79)
Temporary signs (T)	To control, warn and guide road users safely through, around or past work sites on roads and footpaths and to warn and advise of other temporary hazardous conditions that could endanger road users.	ROADWORK AHEAD (T1-1)

Where, due to regulatory requirements, NSW adopt a variation to the design of an Australian Standard sign, an 'n' following the sign number is included. Examples include:

- Speed Limit ROADWORK (R4-212n) sign;
- SHOULDER CLOSED (T2-19n) sign; and
- ONE LANE (R9-9n) sign.

6.5.2 Types of signs

Examples of traffic control signs grouped into broad categories, are provided in [Table 6-5](#).

Table 6-5. Examples of signs used for a typical category

Category	Example signs	Sign number
Work site approaches and departures	ROADWORK AHEAD GRADER AHEAD END ROADWORK	T1-1 T1-4 T2-16
Regulatory control of traffic	Speed limit ROADWORK PREPARE TO STOP STOP HERE ON RED SIGNAL	R4-212n T1-18 R6-6

Category	Example signs	Sign number
Detours	DETOUR One lane each way symbolic Arrow marker symbolic	T5-1 T2-24 T5-6
To indicate road conditions	GRAVEL ROAD Loose stones symbolic NEW WORK NO LINES MARKED	T3-13 T3-9 T3-11
Lane and road closures	Lane closed symbolic ROAD CLOSED	T2-6-1 T2-4
Pedestrian control signs	PEDESTRIANS USE OTHER FOOTPATH LOOK BOTH WAYS TWO WAY TRAFFIC	T8-2 T8-3 T8-5
Vehicle height and mass restriction signs	LOW CLEARANCE X.Xm BRIDGE LOAD LIMIT Xt GROSS	R6-11 R6-3

Signs must be designed and manufactured in accordance with AS 1743. Details of each letter must be as shown in AS 1742.2. The retroreflective material used on signs must be Class 400 or Class 400T material complying with AS 1906.1.

6.5.3 Sign sizes

Signs are generally available in four increasing size classes: A, B, C and D. A size and B size are the most commonly used. The dimensions of the sign sizes vary depending on the sign.

Conditions for use of the different sign sizes are provided in [Table 6-6](#). The TGS designer is responsible for selecting sign sizes and therefore must ensure that they are shown on the TGS.

Table 6-6. Sign size conditions of use

Sign Size	Requirement
A size	Must be used when any of the following conditions are met: - Directed at pedestrians or cyclists; - The lateral offset of the sign from the travel path is less than 4.5 m; or - The lateral offset of the sign from the travel path is between 4.5 m and 8 m and the posted speed is less than 95 km/h.
B size	Must be used when any of the following conditions are met: - The conditions for A size signs are exceeded; - The sign is a roadwork speed zone sign used on roads where the existing permanent speed limit is greater than 55 km/h; - The relevant A size sign is less than 1 m² in area and traffic speeds are greater than 65 km/h; - On motorway type roads for added emphasis of the onset of works, detours or closures; or - For display of any other critical safety messages

Sign Size	Requirement
C size and D size	Must be used when any of the following conditions are met: - The sign is a roadwork speed zone sign used on motorway type roads, multi-lane roads, or higher speed roads; - There is considered to be a need to emphasise the message; or - There is excessive lateral offset of the sign.

Note: Details of standard sign sizes are contained in Transport electronic [Traffic Signs Register](#) and AS 1742.1.

6.5.4 Sign mounting

The method of mounting signs must consider the duration of display, the placement location and frequency of removal or covering. Signs may be mounted in frames or on posts; however at all times the mounting method must:

- Provide secure sign attachment;
- Be stable in windy conditions and from the effects of passing traffic;
- Be suitable for both gravel and bitumen surfaces;
- Be able to accommodate various sign sizes; and
- Not be a hazard if struck in their normal upright position or after being knocked over.

For ease of installation and removal, signs mounted in frames are the preference for works taking less than two weeks. Signs required for works which will be in progress for less than two weeks may be erected in a permanent manner if it is considered to be justified.

Signs continuously required for works, which will be in progress for longer than two weeks, should be installed on posts in a permanent manner.

Table 6-7 provides the requirements for frame and post mounted signs.

Table 6-7. Frame and post-mounted sign requirements

Sign mount	Requirement
Signs mounted in frames	When using frames to mount signs they must be quick and easy to install as well as easy to handle, transport and store. When in a frame, the minimum height to the lower edge of the sign should be 200 mm. For requirements specifically associated with frame mountings of multi-message signage, see Section 6.5.8 Multi-message signs.
Signs mounted on posts	When signs are mounted on posts, the locations of any underground utilities must be determined prior to installation. The ability to install signs on posts may be restricted: - In narrow cuttings; - Where underground utilities are located; - Behind safety barriers where off-sets might be too great; or - Where it is undesirable to damage the asset for installing. When installed in open road situations, the underside of the sign must be at least 1.5 m above the level of the nearest edge of the travelled path. When installed on a kerb or footpath, the underside of the sign must be at least 2.2 m above the level of the nearest edge of the travelled path to reduce impact on vulnerable road users or interference from parked vehicles.

6.5.5 Sign placement

Placement of signs must be arranged so that they are prominently displayed to traffic and will command attention. Signs must be properly displayed at all times and within the line of sight of the intended road user. Regulatory and detour signs must be located nearest to the travel edge of the lane. Signs must not:

- Be obscured from view such as vegetation or parked cars;
- Obscure other devices from the line of sight of road users;
- Create a hazard to road workers and road users (see Section 4.4 Providing for specific road users for additional provisions for specific road users);
- Be a hazard that deflects traffic into an undesirable path; or
- Restrict sight distance for drivers entering from side roads or streets, or private driveways.

The visibility of a sign can be affected by shade, the direction of the sunlight, and background conditions including lighting and oncoming headlights.

Where installed, signs must be oriented to ensure adequate line of sight for approaching road users (see Figure 6-5):

- On the outside of a curve, the sign face must be at 0 degrees, or “normal to traffic”;
- On a straight, the sign face must be angled at approximately 5 degrees normal to oncoming traffic; and
- On the inside of a curve, the sign face must be angled at approximately 5 degrees normal to oncoming traffic at 200 m preceding the sign.

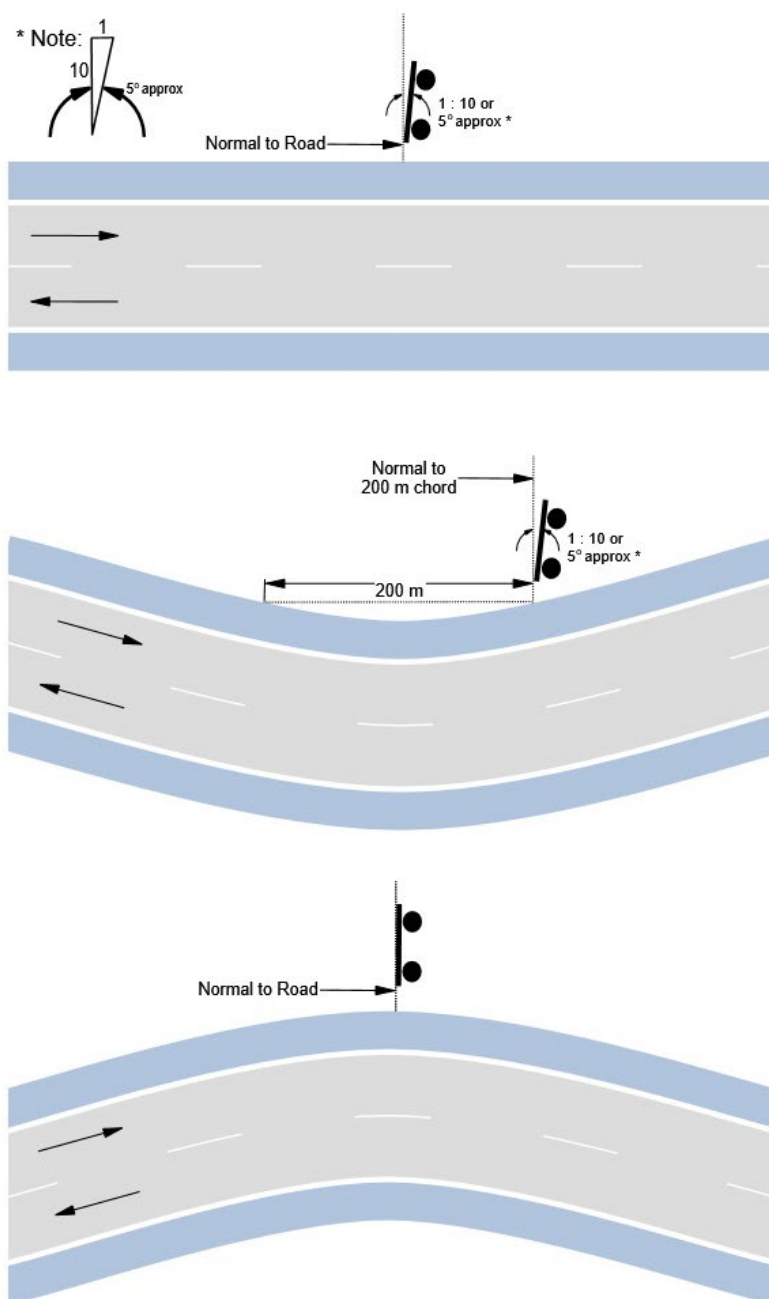


Figure 6-5. Orientations of signs

Signs must be placed in accordance with [Table 6-8](#).

Table 6-8. Sign placement provisions

Placement of signs	Provision
For short term works	Signs mounted on portable supports used for short-term operation must be located as follows: In open road areas – on the road shoulder a minimum of 1 m clear of, and not more than 5 m from the travelled way. In built up areas – behind the kerb if visible to oncoming traffic and not obstructing vulnerable road users or on the pavement, as near as practicable to the kerb, provided the sign is not obscured by parked vehicles and without obstructing moving traffic.

Placement of signs	Provision
For long term works	Signs mounted on posts used for long-term operation must be located as follows: In open road areas or roads without kerbing – signs should be placed clear of the outer edge of the shoulder and, where possible, at least 2 m clear of the travelled way, whichever is the greater clearance. In built up areas or roads with kerbing – signs should be placed at a minimum of 300 mm clear distance behind the kerb. All signs mounted on posts should be mounted at heights specified in Section 6.5.4 Sign mounting.

6.5.6 Duplication of signs

Consideration should be given to duplicating signs as a measure to improve worker and road user safety when developing a TMP and TGS. Where practical, signs should be located on both sides of the roadway for undivided roads, or on the left hand side and on the median for divided roads where there is sufficient median width.

Signs should be duplicated:

- On multilane roads with volumes of 10,000 vpd or greater;
- For lane status signs (T2-6-1, T2-6-2 etc.) regardless of vpd;
- On the outside of left hand curves where the sign is seen on approach to the work area;
- On medians of dual carriageways where parked vehicles or other objects obscure kerb side/footpath signs; and
- At other locations where conditions are such that duplicate signs improve safety and guidance as identified in the TMP or risk assessment.

Duplication of signs should not be used, where:

- It will introduce potential safety risks during installation;
- It will be necessary to cross the road on foot carrying signs;
- The shoulder is too narrow to position the signs or to park the work vehicle; or
- The duplicated signs have too large a lateral offset as to not be obvious to motorists e.g. on a six lane, two-way undivided road.

The decision to not duplicate signs in accordance with the above provisions must be documented in the TMP.

On multi-lane roads where there is no room for duplicate signs on medians, consideration should be given to placing supplementary signs on the left hand side.

Details of duplication of speed zone signs are contained within the speed zones [Section 4.5.5 Implementation](#).

6.5.7 Dual sign arrangements

Dual sign arrangements are two independent signs displayed together at one position either side by side, as shown in [Figure 6-6](#) or 'stacked', as shown in [Figure 6-7](#). Dual sign arrangements may be used, provided all of the following conditions are met:

- The size of both signs, including the legend, size of symbol or area occupied by the legend is unchanged from the standard sign;

- The lateral offset meets the requirements of [Section 6.2.6 Spacing of signs](#); and
- Where used in a dual sign arrangement, regulatory or detour signs must be located nearest to the travel edge of the lane.



Figure 6-6. Side-by-side dual sign arrangements

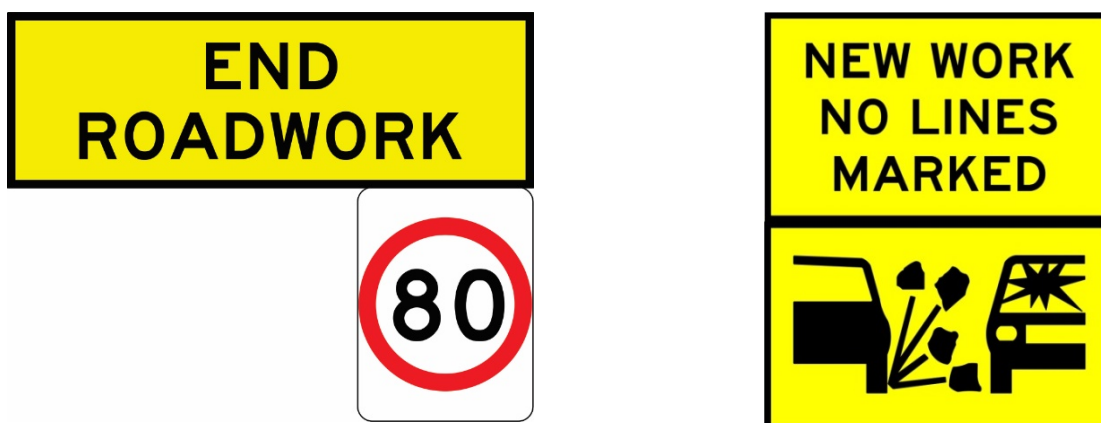


Figure 6-7. Stacked dual sign arrangements

6.5.8 Multi-message signs

6.5.8.1 General

Multi-message signs (MMS) are two or more logically related signs displayed within one mounting frame. For Transport roadwork, MMS may be used, provided both of the following conditions are met:

- The existing permanent speed limit is less than 65 km/h; and
- The MMS frame has the dimensions shown in either [Figure 6-10](#) or [Figure 6-11](#).

Additionally, an MMS must not:

- Be used on multi-lane carriageways;
- Contain messages that are not relevant to the works being performed; or
- Contain empty panels (empty voids).

MMS should not be used where the lateral offset of the sign is close to maximum allowable limits, due to insufficient time for drivers to fully read and comprehend the message being displayed. Examples of typical MMS use and layouts are contained in [Figure 6-8](#) and [Figure 6-9](#).

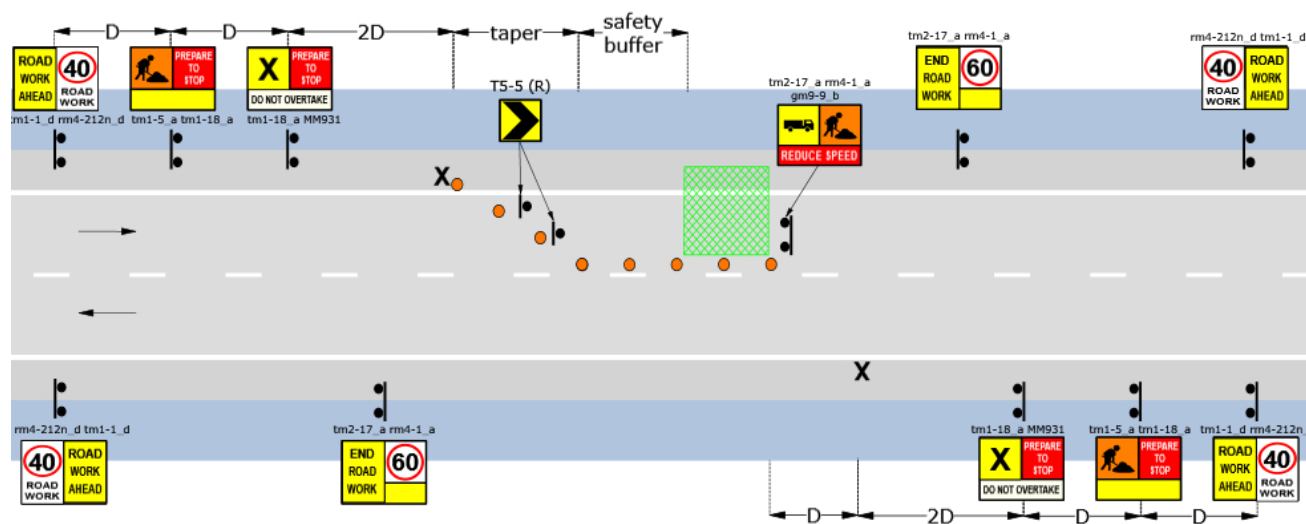


Figure 6-8. Typical MMS layout – carriageway

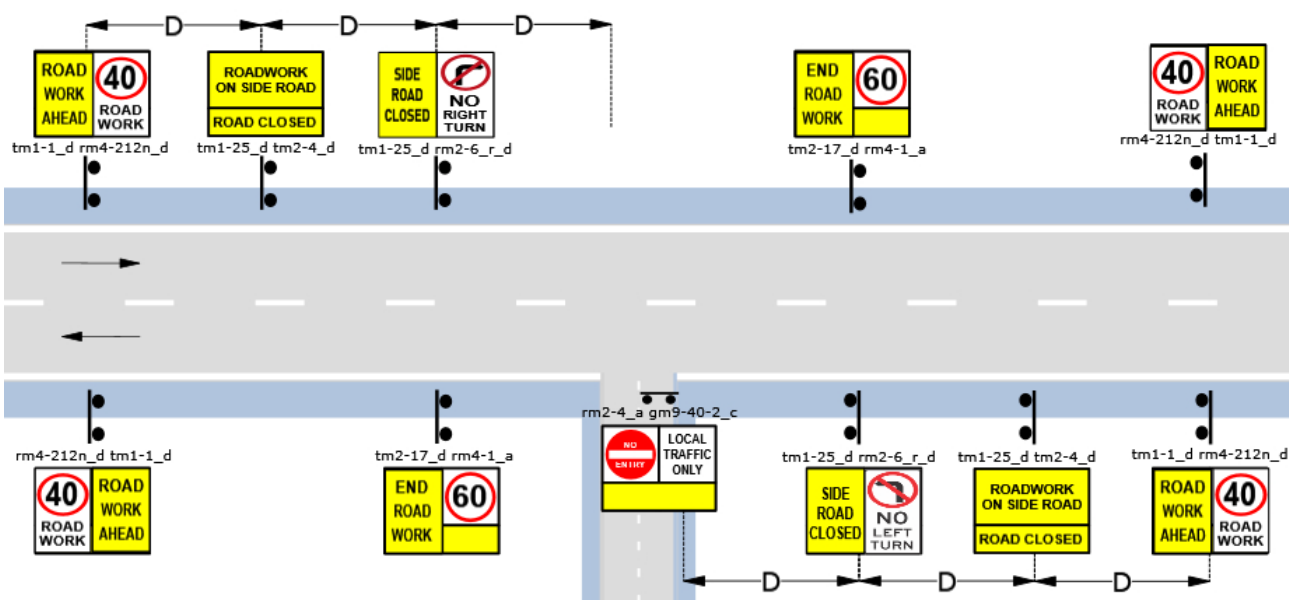


Figure 6-9. Typical MMS layout – side road closed

6.5.8.2 Frame and sign sizes

Figure 6-10 and *Figure 6-11* show the two permitted sizes of multi message sign frames.

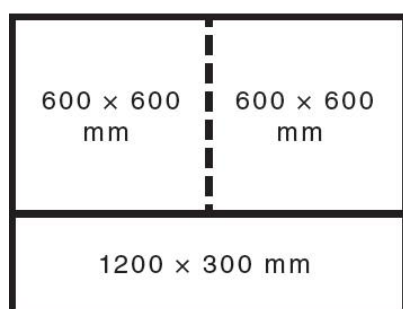


Figure 6-10. Three sign or stacked dual sign MMS frame

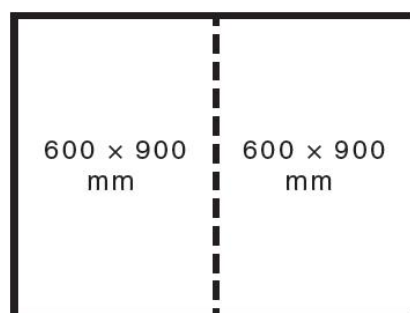


Figure 6-11. Side-by-side dual sign MMS frame

Table 6-9 shows the permitted MMS dimensions and sizes.

Table 6-9. Permitted MMS sign dimensions and sizes

Sign dimension (mm)	Sign Size
600 x 600	M-A
1200 x 300	M-B
1200 x 600 (used in a stacked dual sign MMS frame)	M-C
600 x 900	M-D

6.5.8.3 Combinations

When determining the combination of signs to be used in an MMS frame, the following conditions must be met:

- No more than one regulatory sign is used per MMS frame;
- The regulatory sign must be located in the top position of the frame, closest to the traffic;
- There must not be more than two signs consisting of words only;
- When using two 600 mm x 600 mm signs, at least one 600 mm x 600 mm panel must contain a symbolic sign;
- All signs are placed horizontally;
- A blank retroreflective yellow panel must be placed within any unused panel of the frame; and
- Signs that are not published in the [Traffic Signs Register](#) for MMS use must not be used.

Figure 6-12 to **Figure 6-26** are examples of possible MMS combinations.



Figure 6-12. Sign numbers: TM1 and GM9-79



Figure 6-13. Sign numbers: TM1-1 and RM4-212n



Figure 6-14. Sign numbers: TM1-32 and RM2-66n-L



Figure 6-15. Sign numbers: RM2-66n-R and TM1-32

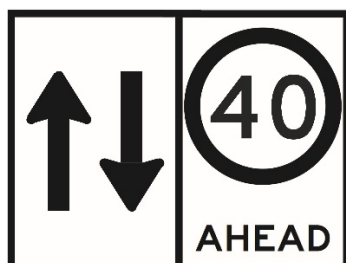


Figure 6-16. Sign numbers: RM2-11 and GM9-79

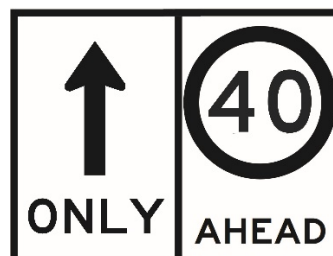


Figure 6-17. Sign numbers: RM2-7 and GM9-79



Figure 6-18. Sign numbers: TM1-3-1and TM1-28



Figure 6-19. Sign numbers: TM2-17 and RM4-1



Figure 6-20. Sign numbers: TM1-5, TM1-18 and TM1-100



Figure 6-21. Sign numbers: TM1-5, TM1-3-1 and TM1-100



Figure 6-22. Sign numbers: TM1-5, TM2-25 and TM1-1



Figure 6-23. Sign numbers: TM1-47 , TM1-46 and TM1-18

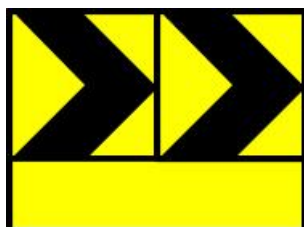


Figure 6-24. Sign numbers: TM5-5 and TM1-100



Figure 6-25. Sign numbers: TM1-100, TM1-30 and TM1-18



Figure 6-26. Sign numbers: TM1-30, TM1-18 and TM1-100

6.5.9 Requirements for specific signs

Table 6-10 provides the requirements and conditions of use of specific TTM signs. During aftercare arrangements, when not in use, or the relevant work is not being performed, the signs must be covered or removed.

Table 6-10. Requirements and conditions of use for specific TTM signs

Sign	Conditions	Notes
Workers symbolic (T1-5)	Must be used where worker on foot will be visibly working adjacent to traffic.	The sign must be: - Covered or removed where there are no workers on foot; and - Used with the NEXT 2km (T1-28) sign at frequently changing work areas.
PREPARE TO STOP (T1-18)	Must be used where traffic is required to stop at a PTCD or traffic controller.	The sign must be used with the relevant PTCD warning sign or traffic control sign.
Trucks symbolic (W5-22) Trucks symbolic (T2-25)	Must be used where roadworks generate greater than 20 truck turning movements per day.	The sign (T2-25) is restricted to short term work only. These signs may be used in conjunction with ___ m ON LEFT/RIGHT (W8-207).

Sign	Conditions	Notes
ROAD PLANT AHEAD (T1-3-1) GRADER AHEAD (T1-4)	Must be used to give warning of frequently changing work areas.	The sign must be used with the NEXT 2km (T1-28) sign for frequently changing work areas. 2 km may be increased to 10 km for shoulder grading and mowing in open road areas and for maintenance grading on unsealed roads.

Table 6-11 provides the requirements and conditions of use of specific TTM signs that may remain uncovered during aftercare arrangements. The signs must be used where the stated conditions apply.

Table 6-11. Requirements and conditions of use for specific aftercare TTM signs

Sign	Conditions	Notes
ROADWORK AHEAD (T1-1) or (T1-31) BRIDGEWORK AHEAD (T1-2)	Must be used at: - Long-term road or bridge work sites; - A diversion of traffic along a side track or detour; or - Unexpected conditions, such as loose stones or the absence of linemarking.	The signs may be used: - With the NEXT 2km (T1-28) sign for frequently changing work areas; and - Short-term works where additional advance warning is warranted.
ROADWORK X km AHEAD (T1-16) BRIDGEWORK X km AHEAD (T1-29)	Must be used where: - The approach speed is greater than 85 km/h; or - Relevant sight distance is less than 150 m (to work area/end of queue etc)	This sign to be positioned X km from: - The start of the taper area; - The traffic diversion; or - The traffic control position.
ROADWORK ON SIDE ROAD (T1-25)	Must be used in advance of an intersection to warn of the relevant activities on the side road where there is insufficient distance on that road to provide the required warning.	
SIDE ROAD CLOSED (T1-32)	Must be used in advance of an intersection where the side road is closed to all traffic.	
TRAFFIC HAZARD (T1-10)	Must be used only for emergencies and for a maximum of 24 hours.	
STOP HERE ON RED SIGNAL (R6-6)	Must be used: - Where traffic is required to stop in compliance with a PTS (placed 6 m in advance of the PTS); and - With the SIGNALS AHEAD (T1-30 or W3-3) sign.	

Sign	Conditions	Notes
END ROADWORK (T2-16) or (T2-17)	Must be: - Placed at a distance D from the work site to indicate that normal traffic conditions have resumed when ROADWORK AHEAD (T1-1) or ROADWORK X KM AHEAD (T1-16) signs are used; and - Placed adjacent to or after any signs indicating the reinstatement of an existing permanent speed limit.	The T2-16 sign is preferred wherever space is available as the site allows it to be used
Slippery symbolic (T3-3) SOFT EDGES (T3-6) ROUGH SURFACE (T3-7) GRAVEL ROAD (T3-13) Loose stones symbolic (T3-9) LOOSE SURFACE (T3-14)	Must be installed to warn motorists of conditions which make a roadway surface temporarily hazardous.	On long work sites these signs should be repeated at intervals of not more than 500 m
NO LINES DO NOT OVERTAKE UNLESS SAFE (T3-12) NO LINES DO NOT OVERTAKE (TM3-12-1n)	Must be used in a two-lane, two-way road when: - Lines have been removed; or - A new seal has been installed.	- T3-12 may be used where overtaking would normally be permitted in an oncoming traffic lane. - TM3-12-1n must be used where barrier lines would normally be installed and overtaking is not permitted.
NEW WORK NO LINES MARKED (T3-11)	Must be used on a multilane road when: - Lines have been removed or - A new seal has been installed.	Must not be used where there is an unacceptable risk of collision due to oncoming traffic. In such cases, T3-12 or TM3-12n as applicable must be used.

6.5.10 Signposting of roadwork speed zones

In accordance with [Section 4.5 Speed zones](#), roadwork speed zones must be implemented to assist in managing the risk to road users or road workers where there is a hazard on or surrounding the road. When designing a TGS that includes a roadworks speed zone the following requirements apply:

- Roadworks speed zone signs used on roads where the existing permanent speed limit is greater than 55 km/h, must be at least 'B' size. On motorway type roads, multi-lane roads, or higher speed roads, 'C' size signs may be erected;

- When a roadworks speed zone is introduced, speed limit signs or markings (see requirements below) and advisory speed limit signs in the zone which show conflicting speeds, must be covered or removed;
- When erected, roadwork speed limit restriction signs must be:
 - Located within 5 m of the edge of the outer travel lane; and
 - Clearly visible to traffic. In urban areas the location of the sign may be adjusted to avoid parked vehicles, other signs or obstructions.
- Where there are multiple work crews within a job site or over a length of road, separate zones may be needed for each work area in order to avoid an excessively long and restrictive speed zone. This will be relevant where 30 km/h speed zones have been installed to support the safety of workers on foot in a particular location within a work site. The minimum lengths stated in [Section 4.5.6 Minimum length zones](#) may be used as a guide when designing the TGS in this instance; and
- For long-term work, in place for longer than four weeks, the existing speed limit pavement numerals should be removed in accordance with [Section 6.8.8 Removal of pavement markings and markers](#). Pavement markings for roadworks speed zones are typically not installed.

[Table 6-12](#) provides the requirements and conditions for use of specific roadwork speed limit signs that must be used in accordance with the stated conditions.

Table 6-12. Requirements and conditions of use of specific roadwork signs

Sign	Conditions	Notes
Speed limit AHEAD (G9-79) sign	Speed Limit AHEAD signs must be erected where the speed of traffic on the approach to the temporary speed zone is 30 km/h or more than the temporary limit. The Speed Limit AHEAD signs must be located 2D in advance of the initial roadworks speed zone (R4-212n) signs.	Speed limit ahead signs should only be used when it is essential to provide motorists with information that is not otherwise evident, or where the reduction in speed is significant. Speed Limit AHEAD signs should also be considered for use: • Where there is insufficient sight distance; • On downhill approaches; • Where, under normal driving expectations, the change in speed zone is not apparent to road users; and • In complex roadside environments where there is competing signage or factors such that the speed zone change might not be apparent to road users
Speed Limit ROADWORK (R4-212n) sign	At the start of a roadworks speed zone, Speed Limit ROADWORK signs must be erected on both sides of the carriageway. Repeater signs must be erected on the left side of the carriageway at a maximum spacing of 500 m. They must also be erected where traffic enters from a side road within a roadworks speed zone.	Where this is not possible a second sign must be erected 0.5D from the start of the zone.

Sign	Conditions	Notes
ROADWORK SPEED LIMITS ENFORCED (T4-216)	ROADWORK SPEED LIMITS ENFORCED signs should be erected on long-term roadwork sites where a speed restriction is used for an extended period, for example in excess of four weeks, and compliance with the roadwork speed restriction is expected to be challenging. If used, these signs must be located D metres after the first roadwork speed limit sign. Additional ROADWORK SPEED LIMITS ENFORCED signs may be used at 500 m intervals thereafter or where the designer considers most appropriate for speed compliance.	These signs may also be used if compliance becomes difficult during shorter periods of work These signs do not need to be applied when utilising buffer roadwork speed limits.

6.6 Portable traffic control devices

6.6.1 General

A portable traffic control device (PTCD) is a device designed to manually control traffic. A PTCD is designed to reduce risk to traffic control personnel by enabling use and control of the device remotely so that the operator can be located outside of the live lane of traffic. PTCDs may include but are not limited to PTS and boom barriers.

In accordance with [Section 5.4 Traffic control](#), a PTCD must be used instead of a manual traffic controller for all work sites under traffic control when the existing permanent speed limit is above 45km/h, unless the decision to not use a PTCD has been documented in the TMP and considered in the associated risk assessment.

When developing a TMP, or selecting or designing a TGS for the use of a PTCD, the relevant qualified person must consider:

- Queue length estimates;
- Expected traffic flows;
- Operational efficiency of the device and the expected delay and queue lengths; and
- Any lost time associated with use for e.g. lowering of boom.

[Appendix B – Device use requirements](#) provides additional requirements in relation to the use, installation, operation and removal of accepted PTCD types. The requirements provided for these devices must be read in conjunction with relevant specifications and manufacturer's instructions.

6.6.2 Delegation and approval

6.6.2.1 Type 2 (automatic) portable traffic signal (PTS) system

Prior to the use of a Type 2 (automatic) portable traffic signal (PTS) system, the following approvals must be sought and granted:

- Type approval of the equipment as per the relevant Usage Procedure for the PTS device; and
- Approval from the relevant delegated representative.

Within Transport, a person at or above the level of section manager within Transport has the delegated authority to approve the installation of the PTS.

Other Road Authorities in NSW (i.e. Councils) have delegated authority to use PTS in specific situations.

No other delegated authority has been given to other agencies or persons. This means that public utility authorities and contractors working on public streets have no delegated authority to use PTS and will need to apply to Transport for authorisation to do so.

6.6.2.2 Type 1 (manual) portable traffic signal (PTS) systems

No delegation is required for the installation of a Type 1 (manual) portable traffic signal (PTS) system, however before use, two approvals are required:

- A type approval as per the relevant Usage Procedure for the device; and
- Project approval for use on each job.

6.6.2.3 Portable boom barriers

No delegation is required for the installation of a portable boom barrier, however before using a portable boom barrier, two approvals are required:

- A Transport type approval as per the relevant Usage Procedure for the device; and
- Project approval for use on each job.

Note: The 'Portaboom' device was accepted for use under the Transport Infrastructure Product Evaluation Scheme (TIPES) certificate number 16/0001, which is deemed to be equivalent to a Transport type approval. No other devices are accepted under TIPES.

6.6.3 Type 2 (automatic) portable traffic signal (PTS) systems

The type 2 PTS, also known as automatic PTS can be controlled under vehicle actuated or fixed time operational modes. A type 2 (automatic) PTS can also be controlled manually by a traffic controller using a remote control. An example of a type 2 PTS setup on-site is shown in [Figure 6-27](#).



Figure 6-27. Type 2 PTS

Type 2 PTS may be used for traffic control applications lasting up to three months. For sites where work will continue for longer periods, without the location of the work site changing, a risk assessment and feasibility analysis must be carried out to determine whether the extended use of portable signals is acceptable or whether a temporary signal installation should be provided. This risk assessment and feasibility analysis must be submitted to Traffic Engineering for review and the concurrence of the Director, Roads and Traffic Engineering obtained for the use of the portable signals for an extended period.

A type 2 PTS must be used in accordance with [Appendix B.2 Usage procedure: Type 2 \(automatic\) portable traffic signal systems](#).

6.6.4 Type 1 (manual) portable traffic signal (PTS) systems

The Type 1 PTS is a compact portable system that is manually controlled by a traffic controller using a remote control. A Type 1 PTS may be used with or without the addition of a boom arm. Where used:

- Type 1 (manual) PTS devices must have type approval in accordance with [Transport Specification TSI-SP-059 Type 1 Portable Traffic Signals](#).
- Type 1 (manual) PTS devices with a boom arm (Type 1 PTS-B) must have type approval in accordance with [Transport Specification TSI-SP-081 Type 1 Portable Traffic Signal with Boom Barrier](#).

Type 1 PTS or Type 1 PTS-B may be used when controlling a single-lane of traffic where either shuttle flow or plant crossing control is required.

Type 1 PTS and Type 1 PTS-B must not be installed on multi-lane carriageways unless a merge or merges have been installed upstream of the traffic control position such that the device is managing only one lane.

Type 1 PTS and Type 1 PTS-B must not be installed for work sites where the traffic speed in the work zone is greater than 65 km/h.

An example of a type 1 PTS setup on-site is shown in [Figure 6-28](#).



Figure 6-28. Type 1 PTS

All Type 1 PTS and Type 1 PTS-B must be used in accordance with [Appendix B.3 Usage procedure: Type 1 \(manual\) portable traffic signal systems](#).

6.6.5 Portable boom barriers

A portable boom barrier is controlled by a traffic controller using a remote control. A portable boom barrier is an alternative to other devices such as PTS. Portable boom barriers must have a Transport type approval and project approval in accordance with [Section 6.6.2.3 Portable boom barriers](#). At the time of publication of this Technical Manual, a Transport specification for portable boom barriers is not available.

Portable boom barriers are not suitable in all traffic environments, but may be suitable for locations where there are sufficient gaps in traffic to safely lower the boom.

Portable boom barriers may be installed when controlling a single-lane of traffic in a single direction.

Note: *Two portable boom barriers may be used when controlling a single-lane in opposite directions for either shuttle flow or plant crossing control situations.*

Portable boom barriers must not:

- Be installed on multi-lane carriageways; or
- Be installed for work sites where the traffic speed in the work zone is greater than 65 km/h.

A portable boom barrier must be used in accordance with [Appendix B.4 Usage procedure: portable boom barriers](#).

6.7 Temporary safety barriers

6.7.1 General

Work zone temporary safety barriers are used at work sites to prevent vehicles encroaching on work areas and to ensure the safety of the workers from errant vehicles.

Work zone temporary safety barrier products must be accepted for use by Transport. Accepted safety barriers have been assessed and conform to the requirements of AS 3845. Unless the barrier, including attachment, has been tested and accepted, devices such as signage, lighting posts, screens or work platforms must not be attached to temporary safety barriers.

The use of safety barriers during temporary works should be considered as part of the project TMP and risk assessment. The Austroads [Guide to Road Design Part 6: Roadside Design, Safety and Barriers](#) and Transport *Supplement to AGRD Part 6* provides detailed guidelines for analysing risks, severity and design of safety barriers on temporary roads and detours as well as for the protection of workers in defined work areas adjacent to traffic.

It should be noted that barriers are an introduced obstruction. When the reason for their inclusion in the road side environment no longer exists, the barriers should be removed as soon as practicable. Barriers must only be used for their intended purpose, since they can present a hazard in the work area or work site if used otherwise. If temporary safety barriers are proposed to be used such that they are not placed parallel to traffic, Road Design must be consulted to determine minimum lengths and placement angle in relation to passing traffic so that the risk of injury is minimised if the barrier is inadvertently struck.

For long-term, complex or high risk projects, advice should be sought from Road Design for barrier selection and design.

6.7.2 Use of temporary safety barriers

Work zone temporary safety barriers must be installed in accordance with their Transport acceptance conditions and manufacturer's requirements. This ensures in-field performance during impact is as expected based on crash testing undertaken. The list of Transport-accepted work zone temporary safety barriers systems and their acceptance conditions can be found on the Transport [Safety Barrier Products website](#).

Additional information is provided in [Section 6.2.1 Edge clearances for delineating devices and temporary safety barriers](#) for clearance requirements and restrictions on installing safety barriers on kerbs.

Temporary safety barriers may be used:

- To separate opposing traffic streams where there are potentially hazardous conflicts, such as the risk of head-on collisions;
- Where there are excavations or hazardous fixed objects close to the travelled way;
- Where there is inadequate separation from temporary foot paths or bicycle paths; or
- Where there are embankments within the vicinity of works.

Temporary safety barriers are not delineation devices and therefore must not be used alone for the purposes of delineation. However, they may be used to assist delineation, provided other signs and devices, such as cones or bollards are also in place.

When determining the appropriate temporary safety barrier for use, the dynamic deflection and working width relevant to the barrier type must be considered, and an exclusion zone delineated to ensure workers do not access these areas.

Dynamic deflection is the largest transverse deflection of a road safety barrier system during an actual crash or during a full-scale impact test (i.e. the amount the road safety barrier deflects from its initial position during impact (see [Figure 6-29](#))).

An exclusion zone is the area behind the barriers that must be maintained as being clear of materials, workers or plant considerate of the dynamic deflection.

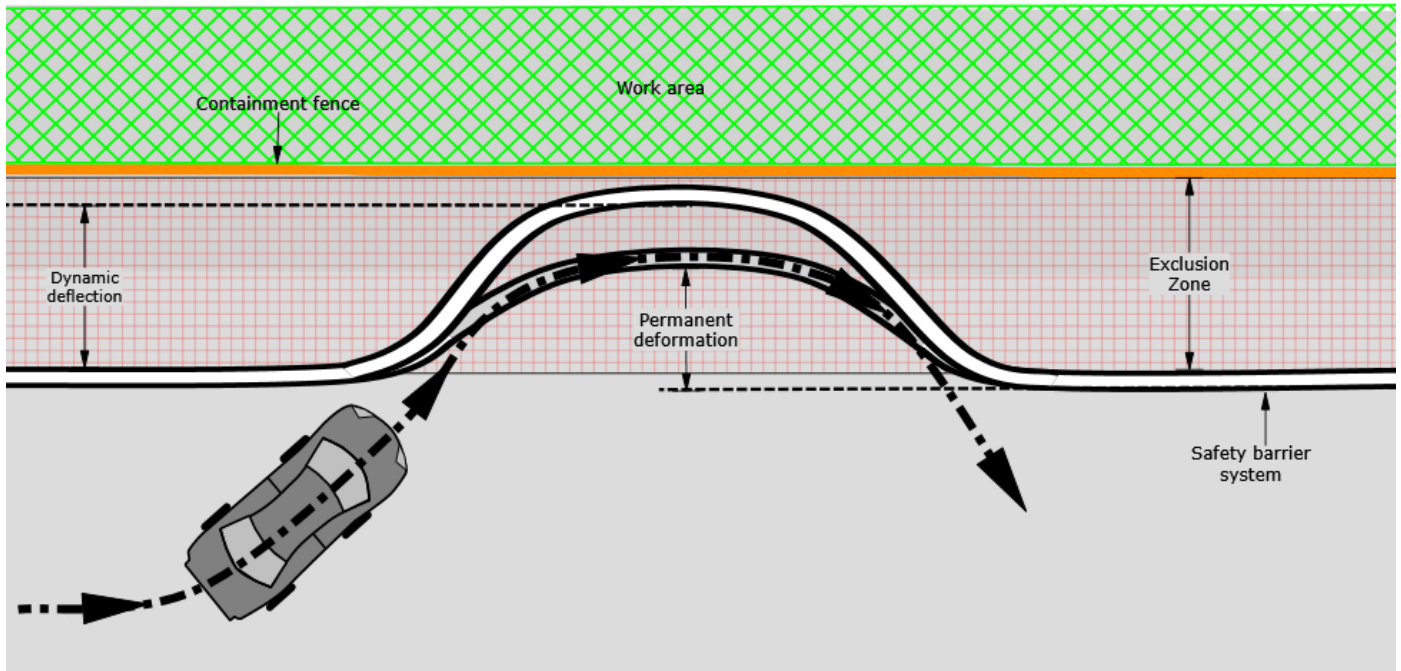


Figure 6-29. Dynamic deflection and exclusion zone

See [Section 6.2.1 Edge clearances for delineating devices and temporary safety barriers](#) for details on clearances, placement and location of barriers.

Where work zone temporary safety barriers are designed for use as part of a work site, a Temporary Safety Barrier Design Statement should be completed and included in the TMP. A Temporary Safety Barrier Design Statement should include:

- Location and direction barriers;
- Barrier type and specific design requirements;
- Terminals required;
- Transitions, including overlap requirements of different systems;
- Installation sequence; and
- Any relevant diagrams to assist the installation of the design.

The Temporary Barrier Design Statement should also include the ability for persons installing or inspecting to sign off that the barrier system is in accordance with the Design Statement. A Temporary Barrier Design Statement template is provided in [Appendix A.2.4 Temporary barrier design statement](#) for this purpose.

6.7.3 Mobile safety barriers

Mobile safety barriers are a portable barrier system attached to a prime mover and are designed to provide protection to workers in both static and dynamic work environments.

A barrier trailer, when deployed correctly, aims to absorb the impact and deflect a vehicle which has left the intended path and entered the work site. The purpose of the mobile barrier is to protect the road workers behind it whilst maintaining the safety of the road user.

In a static environment, mobile safety barriers provide an additional form of work site protection.

When used for dynamic work, the portable temporary barrier system is designed to take the place of the work vehicle and the shadow vehicle within the work convoy (see [Figure 6-30](#)).

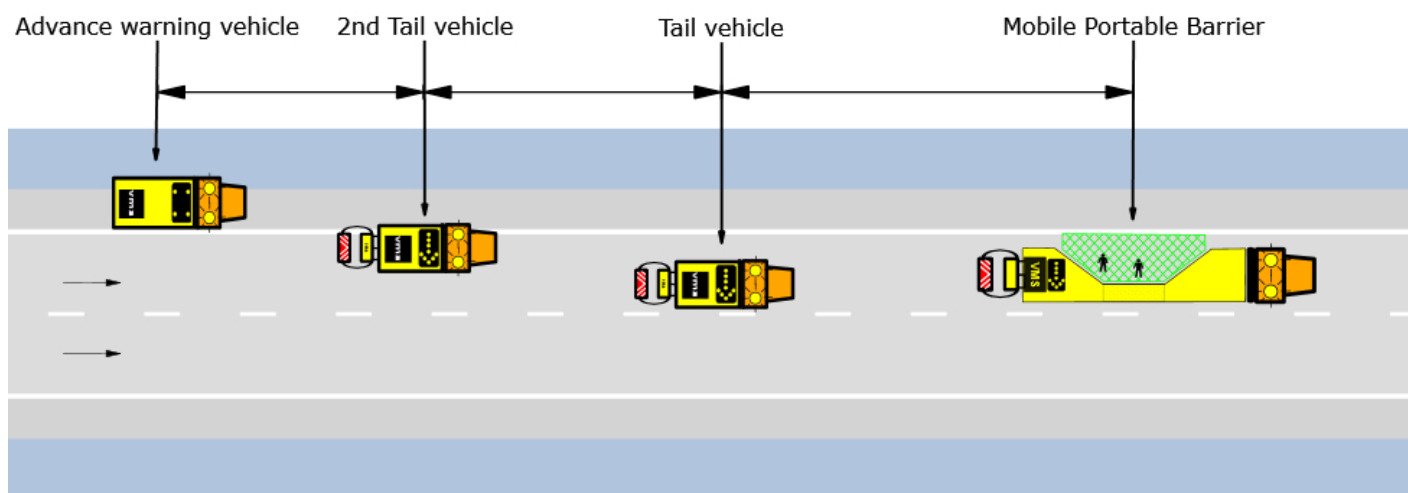


Figure 6-30. Example of a mobile safety barrier

Mobile safety barriers must only be used when all of the following conditions are met:

- The mobile safety barrier is accepted for use by Transport;
- **Note:** At the time of publication of this Technical Manual, the only accepted mobile safety barrier is “Mobile Barrier Truck – MBT-1”.
- The temporary speed limit is less than 85 km/h;
- The exclusion zone behind the barrier is a minimum of 0.4 m;
- When deployed parallel to direction of traffic; and
- When a crash tested truck-mounted attenuator (TMA) is used as a tail vehicle.

Mobile safety barriers must not be used:

- As a work platform or with attachments such as lifting equipment or working platform; or
- With any additional attachments for the purpose of undertaking work on a work site.

Mobile safety barriers must comply with Transport Specification R132 and with Transport’s [Acceptance Conditions – Mobile Barriers MBT-1](#).

Example of work site layout for use in static and dynamic work sites are provided in [Appendix D – Work type layout examples](#).

6.8 Traffic guidance and delineation devices

6.8.1 General

Delineators used at or near works on roads must meet the requirements of AS/NZS 1906.2 for either the sheeting or discrete device type. Delineators made from orientation-sensitive material must be made and installed in accordance with the manufacturer’s recommended orientation for optimum performance. Delineators must be used in accordance with AS 1742.2 and [Delineation guide](#).

6.8.2 Barrier boards

For use as traffic guidance devices, barrier boards must:

- Comply with Transport QA Specification 3385;
- Alternate diagonal stripes of black and retroreflective yellow, terminating in yellow at each end as illustrated in [Figure 6-31](#);
- Have diagonal stripes aligned to face down (see [Figure 6-31](#));
- Be placed at right angles to traffic flow;
- Be placed at a maximum spacing of 100 m; and
- Be secured so that they are not moved or blown over by winds or pressure from heavy vehicles.

Where barrier boards are placed facing traffic, the bars need to be consistently pointing in the same direction. The bars on the barrier board must point down toward the side that vehicles are required to pass. For example, traffic is required to pass to the right of the barrier board as shown in [Figure 6-31](#).

Barrier boards must not:

- Be used as delineation devices;
- Be placed parallel to the direction of traffic flow. This ensures maximum visibility of the barrier board and prevents it from becoming a spearing hazard if struck by an out-of-control vehicle; and
- Be used adjacent to a pedestrian or cyclist path of travel so as to not become a tripping or falling hazard.

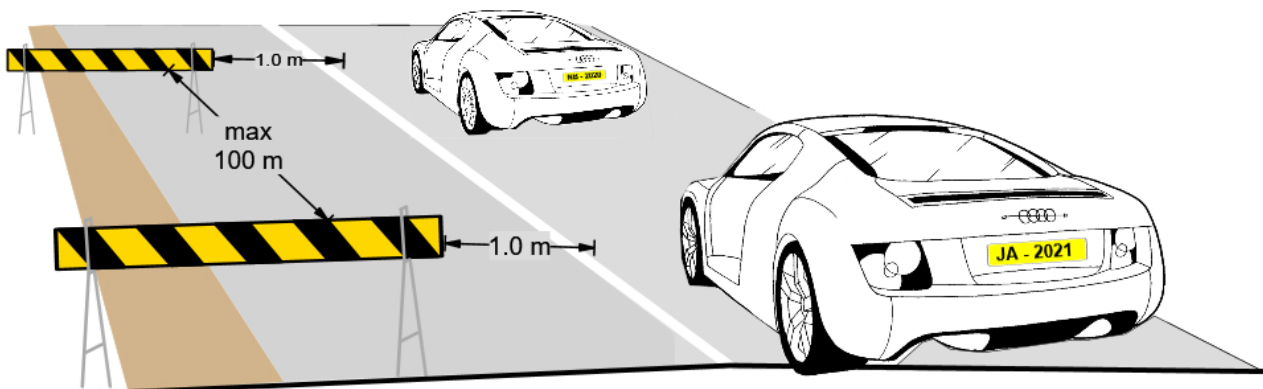


Figure 6-31. Typical example of a barrier board

6.8.3 Plastic containment tapes and fences

Containment tapes and fences may be used to provide visible separation between a travelled way and the work area, where physical protection by use of a safety barrier is not warranted.

When installed, containment tape and fencing must:

- Be supported by posts;
- Have posts installed at maximum spacing of 5 m;
- Ensure horizontal deflection of the tape or fence material does not exceed 0.5 m from the nominal line of the tape or fencing as a result of wind, air turbulence from passing traffic or minor impacts; and
- Be monitored and maintained as they can become a hazard to traffic if poorly maintained, especially in windy conditions.

Plastic containment tapes and fences must not:

- Be used as temporary safety barriers; or
- Act as delineation devices when used adjacent to traffic.

Containment fences or delineation should be used behind safety barriers to delineate and prevent workers entering the exclusion zone behind the temporary safety barrier. See [Section 6.7 Temporary safety barriers](#) for further information.

Additional conditions of use of containment tapes and fences are provided in [Table 6-13](#).

Table 6-13. Conditions of use of containment tapes and fences

Containment type	Conditions of use
Containment tapes	Containment tapes may be used to contain workers on foot and plant within the safe workplace boundary established at the particular work site. Where used, tapes must be 100 mm wide with alternate stripes of contrasting colour and supported on posts approximately 1 m high such that the height of the tape above the ground is never less than 800 mm. Tapes must not be used for pedestrian containment adjacent to traffic.
Plastic mesh fencing	Plastic mesh fencing may be used for excluding pedestrians from a work area and for the containment of workers on foot and plant. When in use, plastic mesh fencing must comprise a flexible orange mesh approximately 1 m high. When in use, the top of the fence must not be less than 800 mm above the ground. When used for containment from the travelled way, the clearance requirements of Section 4.3.4 Minimum clearances of workers to traffic must be met.

6.8.4 Traffic cones and temporary bollards

Traffic cones and temporary bollards may be used to define the traffic path *past* or *through* the work area. Cones and temporary bollards must not be used as a substitute for barrier boards and signs at either end of the work.

Traffic cones and temporary bollards must comply with Transport QA Specification 3352 Fluorescent Plastic Traffic Cones. Traffic cones and temporary bollards must have a white horizontal retroreflective band of Class 400 material that are:

- 150 mm wide on 450 mm height traffic cones; or
- 250 mm wide on the traffic cones and temporary bollards higher than 450 mm.

Traffic cones and temporary bollards must be used on works in accordance with the conditions provided in [Table 6-14](#).

Table 6-14. Traffic cone and temporary bollard conditions of use

Cone or bollard size (height)	Conditions of use
General	Cones and bollards must only be used while work is in progress (day or night) where workers are in attendance to reinstate any of the cones and bollards dislodged by traffic or wind. Cones or bollards may be used on long-term unattended works provided they are securely fixed to the pavement or weighted to provide adequate stability. The requirements for spacing of cones and bollards are provided in Table 6-2.
300 mm cones	300 mm traffic cones must not be used.
Small (450 mm) cones	450 mm traffic cones may be used for most built-up and open road applications including footpaths, shared paths and bicycle paths provided traffic speeds are less than 75 km/h.
Standard (700 mm) cones	700 mm traffic cones must be used in locations where traffic speeds are greater than 75 km/h. Note: 700 mm traffic cones may also be used on lower speed roads.
Large (900 mm) cones	900 mm traffic cones may be used: - On high speed, high volume roads instead of standard size cones, e.g. expressway type roads; or - On any work site where increased visibility is required or as a means to provide additional delineation. Note: 900 mm traffic cones may also be used on lower speed roads.
Temporary bollards (min 900 mm)	Temporary bollards must be: - At least 900 mm high and 100 mm in diameter; - Made from fluorescent red or orange material; and - Resilient to impact. Note: Temporary bollards may also be used on any speed roads.

6.8.5 Temporary kerbs

Temporary kerbing may be used to form temporary medians, traffic islands, pavement edges, or as a temporary lane divider in appropriate situations during long term works.

When used temporary kerbing must be:

- Yellow in colour;
- Not greater than 150 mm in height;
- Secured to the pavement;
- A continuous line at least 150 mm wide as seen by approaching traffic; and
- Clearly delineated in accordance with [Section 6.2.2 Use of delineating devices on kerbs](#).

6.8.6 Roadworks temporary guideposts

Roadworks temporary guideposts with delineators may be installed to provide a single continuous line defining the travelled path.

Where used for delineation and at a minimum, roadworks temporary guideposts must be used as follows:

- For delineation of the travel path through or past the work site as an alternative to traffic cones or bollards, guideposts must have yellow delineators and must be placed on both sides of the roadway; or
- For delineation of the roadway on detours and side-tracks, guideposts must have red delineators on the left side and white on the right for a two-way roadway, or yellow on the right (one-way roadway).

Roadworks temporary guideposts with delineators must be installed in accordance with AS 1742.2.

Delineators used at or near works on roads must meet the requirements of AS/NZS 1906.2 for either the sheeting or discrete device type. Delineators made from orientation-sensitive material must be made and installed at the manufacturer's recommended orientation for optimum performance.

6.8.7 Temporary pavement markings and markers

Pavement markings on temporary roadways and detours must be of a similar standard to that in use at either end of the adjoining sections of road. Where the adjoining road is delineated with edge lines, temporary roadworks must be similarly marked with an edge line.

Pavement markings and markers used at temporary work sites generally comprise:

- Barrier, lane and edge lines;
- Turning arrows; and
- Raised retroreflective pavement markers (temporary or permanent).

Where, during or at the conclusion of pavement-surfacing works, a section of roadway is to be left for a period of time without linemarking, temporary raised pavement markers should be used to provide delineation at the dividing or lane lines. Application of the pavement markings must take place as soon as practicable.

Where temporary linemarking, i.e. linemarking not in its final location, is required on the final wearing surface, or adjacent pavement, pavement marking tape should be used. Where used pavement marking tape must meet the performance requirements of [Transport QA Specification R145 Pavement Marking \(Performance Based\)](#).

Where it is determined that any temporary pavement marking or marker has become ineffective, remarking and/or replacement must be undertaken as soon as practicable.

Where a single carriageway is opened adjacent to, or used in lieu of, an existing dual carriageway length, pavement arrows (in tape if they are required to be removed from a final wearing surface) indicating the direction of flow of traffic must be placed as directed with the maximum allowable spacing being 500 m. The arrows must be removed if the section is then reincorporated as dual carriageway.

Where existing pavement markings are required after the temporary works, these may be masked over with suitable black tapes where they exhibit similar characteristics to the existing pavement colour and surface finish, during the works period. This may be a suitable consideration where limited traffic impact is expected, to limit the chance of dislodgement by traffic. Otherwise, permanent removal must be undertaken and agreement must be received from the asset owner in accordance with [Section 6.8.8 Removal of pavement markings and markers](#).

On long-term works raised retroreflective pavement markers complying with [Transport QA Specification R142 Retroreflective Raised Pavement Markers](#), may be used in conjunction with temporary pavement markings. The spacing and application must be as specified in [Transport QA Specification R142 Retroreflective Raised Pavement Markers](#).

Pavement markings and retroreflective markers should be considered in conjunction with the placement of other delineation devices, and must be used where temporary safety barriers are used, to ensure road users are safely directed through the site without conflicting messages.

All pavement markings must be retroreflective in accordance with [Transport QA Specification R145 Pavement Marking \(Performance Based\)](#).

6.8.8 Removal of pavement markings and markers

All redundant pavement markings and raised pavement markers that will not be required must be immediately removed in such a way as to leave a clean, undamaged pavement with a surface texture, reflectivity characteristics and colour comparable to the adjacent pavement surface.

Blacking out of redundant pavement markings with paint is not permitted and must not occur.

Where existing pavement markings are to be removed and replaced by other pavement markings, removal must not begin until adequate provisions have been made to complete the installation of the replacement markings. Pavement markings must be removed so that the markings that remain in place at any time will not be in a pattern that will mislead or misdirect road users. Reinstatement of the pavement markings must take place as soon as possible.

The markings must be removed so that the surface is in proper condition for adequate bonding of the new markings. Any material deposited on the pavement as a result of removing pavement markings must be promptly removed as the work progresses by acceptable methods.

When these operations are completed, the pavement surface must be clear of any residue or debris. A minimum of 90 percent of the total area of the existing pavement markings must be removed to uniformly expose the existing pavement surface.

Any damage to the pavement, pavement joint materials or the pavement surface caused by the removal of pavement markings must be repaired. The pavement surface must be left in a condition that will not mislead or misdirect road users.

6.9 Illuminated warning devices

6.9.1 Variable message signs (portable)

6.9.1.1 General

A portable variable message sign (VMS) is an electronically powered on-road sign used for traffic management or driver information applications. All portable VMS used on a Transport work site must meet the requirements of AS 4852.2 *Variable Message Signs, Part 2 Portable Signs*. Where portable VMS are to be used at or near a work site to carry warning or other messages relating to the works, this Section must be applied. If a conflict exists between requirements in this Section and AS 4852.2, then AS 4852.2 must be applied.

Where a VMS is used, the rationale for use must be documented in the TMP and/or risk assessment. The risk assessment should:

- Include identification, assessment and management of risks introduced to road users and the workers who install and maintain the VMS; and
- Consider the risks associated with the proposed VMS location such as proximity to traffic, overhead electricity lines, and positioning them near culverts, medians or steep embankments.

The inclusion of VMS at a work site must be shown on the approved TGS.

Table 6-15 provides the conditions of use that apply in relation to the display of portable VMSs.

Table 6-15. Conditions of use of portable VMS displays

Display type	Conditions of use
General	When a VMS is used, it must meet the following conditions: - The VMS display must be levelled; - The visual lighting intensity of the VMS display must not generate a distraction or blinding risk for road users, particularly at night; - Complementary messages must only be placed on a VMS when a flashing arrow sign is utilised; - When on-site, the VMS display must always be displaying a sign or message. Where there is no relevant sign or message to be displayed, the messages shown “<i>Report Traffic Incident</i>” with “131 700” as alternating messages as shown in <i>Figure 6-32</i> should be used as the default messages. Otherwise, the VMS must be switched off; and - If there is a malfunction with the VMS the display must be rotated at 90 degrees or folded down until such time as it can be removed. In these circumstances, consideration should be given to placing additional delineation to highlight the presence of the out of service VMS.
Displaying messages	- The content of new and revised messages must be approved in accordance with standard procedures for each road authority. Refer to Austroads <i>Guide to Traffic Management (AGTM) Part 10: Traffic Control and Communication Devices</i> and Transport Supplement to AGTM Part 10; - Messages must comprise not more than four words or numbers on any one screen; - Letter forms and legend height must be adequate to be comfortably read by drivers at the prevailing approach speed of traffic; - There must not be more than two separate screens in any alternating series of screens; - Where there are alternating screens, the ‘on’ time of each screen must be 0.6 + 0.1 second per word or number, and the total time required to read the message on both screens must be taken into account when determining message length and letter height; Note: A procedure for determining letter sizes for signs is provided in AS 1742.2. The letter series which most nearly match the on-screen fonts should be used in the calculations. It is recommended that the calculated letter height be doubled for this purpose. - Symbols must not be used unless they replicate an existing static sign or have been tested for comprehension in their on-screen format, i.e. taking into account distortions due to pixel size limitations; - Messages must be relevant to the nature and phase of the work in progress and must be changed or switched off when they are not relevant; Note: Other variable message signs near roadworks sites displaying unrelated messages must be switched off. - Messages must be complementary to other signs, or warning or delineating devices required by this Technical Manual. The nature and positioning of the messages must not detract from those signs or devices.

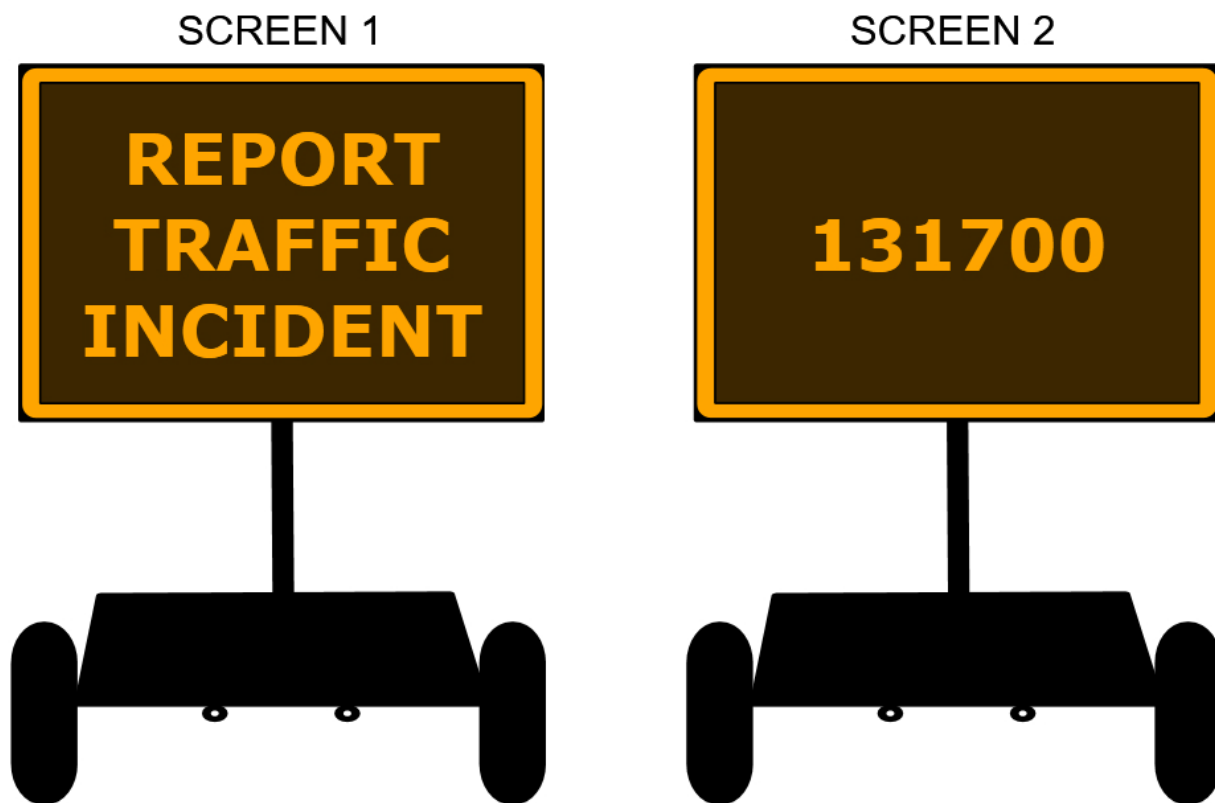


Figure 6-32. VMS default messages

All VMS must be located and installed in accordance with Section 6.2.4 Placement and clearances of registered plant or devices.

In addition to the requirements of Section 6.2.4 Placement and clearances of registered plant or devices, portable VMS must be:

- Placed on a road where there is adequate time for the driver to view and comprehend the message. A greater distance is required for multiple message screens and higher traffic speeds;
- Driven into position in the same direction as traffic, so that the trailer lights and reflectors are seen by on-coming vehicles;
- Placed at least 300 m from the nearest permanent VMS;
- Positioned so that the display is clear of the envelope of passing vehicles (particularly heavy vehicles);
- Be clearly visible at all times, with reflective material delineating the outer extremities;
- Anchored to prevent it moving under wind loading; and
- Risk assessed to determine if additional delineation is required, taking into account factors such as lateral offset from the edgeline, road alignment, horizontal and vertical, speed zone, road environment and the adequacy of the delineation on the trailer body/sign face.

The positioning of VMS must always be verified by the project team or delivery partner as being appropriate. Appendix E – Inspection checklists and tools and forms contains a checklist that is a broad summary of the major items for consideration in locating and placing of portable VMSs.

6.9.1.2 Displaying signs

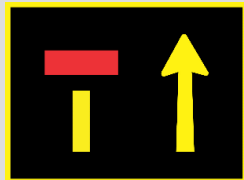
A VMS may be used to display a static sign under the following conditions:

- Where a VMS will display a sign detailed in the [Traffic Signs Register](#), the VMS must:
 - Display a reasonable likeness to the static sign in accordance with the examples provided in the Traffic Signs Register; and
 - Be, at a minimum, displayed at the required size in accordance with this Technical Manual and the relevant Sign Design Plan provided in the [Traffic Signs Register](#).
- Where a VMS is used to replace a regulatory sign or a lane status sign, the VMS must display the message continuously and not alternating with another message or flashing.

Table 6-16 provides a list of the static signs which are, at the time of publication, permitted for display on a VMS.

Table 6-16 Static signs for display on VMS

Sign code	Sign name	Pictorial example
R4-1n-VMS	Speed limit	
T1-16n-VMS	Roadwork X km ahead	
R4-212n-VMS	Speed limit and roadwork	
T1-34n-VMS	Traffic controller	
T1-23n-VMS	Changed traffic conditions	
T1-31n-VMS	Roadwork ahead	

Sign code	Sign name	Pictorial example
T1-5n-VMS	Worker symbolic	
T2-6-1n-VMS	Left lane closed	
T2-6-2n-VMS	Left lane closed	
T2-17n-VMS	End roadwork	
T2-25n-VMS	Truck crossing/entering	
G9-9n-VMS	Reduce speed	

6.9.2 Variable speed limit signs (portable)

6.9.2.1 Description and general

This Section describes portable variable speed limit signs and additionally provides the requirements for their use to regulate speed as part of TTM.

A portable variable speed limit sign (VSLS) is an illuminated sign used for advising road users of a temporary change to the speed limit. The portable VSLS display is mounted on a re-locatable trailer and displays signs in the format and colours specified for speed limit signs in the NSW Road Rules. The displayed speed limits on the VSLS are capable of being changed by electronic means. Generally, VSLS are used as a series of signs along the length of a roadwork site, rather than a singular replacement for a static sign.

All portable VSLS used on a Transport work site must meet the requirements of Intelligent Transport Systems (ITS) Specification no. TSI-SP-082 *Portable Variable Speed Limit Sign*.

The rationale for use of VSLS must be documented in the TMP and/or the supporting risk assessment. The TMP/risk assessment must:

- Include identification, assessment and management of risks introduced to road users and the workers who install and maintain the VSLS; and
- Consider the risks associated with the proposed VSLS location such as proximity to traffic, overhead electricity lines, culverts, medians or steep embankments.

The placement of VSLS, including the spacing of devices and the offset from the nearest traffic lane must be shown on the approved TGS.

Table 6-17 provides the principle conditions of use for portable VSLS.

Table 6-17. Conditions of use of portable VSLS

	Conditions of use
General	• The VSLS must be a Transport type approved device. • The VSLS display must be levelled. • The VSLS display must be a 'C' size speed limit sign as defined in AS 1743. • When on site, the VSLS should always display a speed limit image. • When the VSLS malfunctions, the display must be rotated 90 degrees until the device is repaired or replaced. As a result of the out-of-service VSLS, additional measures should be installed such as delineation around the device and a static speed sign.
Placement and Positioning	• The VSLS must be placed and positioned in accordance with the provisions of Section 6.2.4 Placement and clearances of registered plant or devices. • The VSLS must be duplicated at the start of the roadwork speed zone. Where this is not possible, a second VSLS must be erected 0.5D from the start of the zone. • As part of a series of signs, the VSLS must be repeated on the left hand side of the carriageway at a spacing of no greater than 750 m. A VSLS must also be placed between 50 m and 100 m both upstream and downstream of a junction to a side street/road to inform road users who have entered a roadwork speed zone from a side street/road. • Regardless of the use of VSLS, the normal speed limit must be reinstated at the extremities of the work site in accordance with the provisions of Section 7.6.2.5 Termination area.
Displaying signs	• The VSLS image must display a reasonable likeness to the static speed limit sign (R4-1) in accordance with the NSW Road Rules. • The VSLS image must be displayed in accordance with the size requirements in this Technical Manual and the relevant sign design plan provided in the Traffic Signs Register. • Roadwork speed zones must be selected in accordance with the provisions of Section 4.5 Speed zones.
Maintenance	• As part of the TMP/TGS development, an inspection and maintenance plan for routine preventive maintenance of VSLS must be prepared and implemented in order to proactively check the units for battery health and optimum working condition at regular intervals. • The inspection and maintenance plan must also detail the procedure for identifying, managing and repairing/replacing any faulty VSLS unit.

6.9.2.2 Additional placement requirements

The following additional requirements apply to the placement of VSLS:

- The VSLS must be positioned so there is adequate time for the driver to view and comprehend the information contained within the sign.
- The VSLS must be orientated in the same direction as the flow of traffic, such that trailer lights and reflectors are visible to on-coming vehicles.
- The VSLS must be positioned so the display is clear of the envelope of passing vehicles (particularly heavy vehicles).
- The VSLS must be clearly visible at all times, with reflective material delineating the outer extremities.
- In strong wind areas, the VSLS must be anchored to prevent it moving under wind loading.

The risk assessment must consider whether additional delineation is required, taking into account factors such as lateral offset from the edge line of the nearest traffic lane, horizontal and vertical road alignment, speed zone, road environment and the adequacy of the delineation on the trailer body/sign face. The positioning of VSLS must always be verified by the project team or delivery partner as appropriate.

Appendix E - Inspection checklists and tools and forms contains a checklist that may be used by project teams when locating and placing portable VMS/VSLS.

6.9.2.3 Accompanying static signs

The static sign IF VARIABLE SPEED LIMIT SIGN IS BLACKED OUT (G6-317n or G6-317-1n) must be installed at the extremities of a work site controlled by a series of portable VSLS to advise entering road users of the speed limit when the portable VSLS is blacked out, due to a malfunction or an operational need. G6-317n is the preferred method of display, however G6-317-1n may be selected where there are width limitations and other locality constraints (see [Figure 6-33](#) and [Table 6-18](#)).

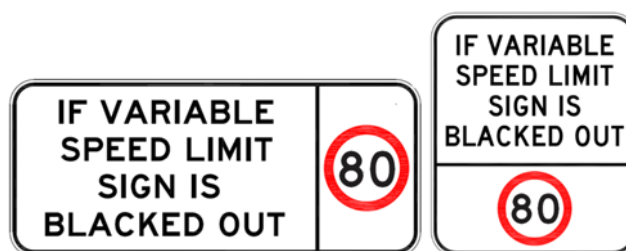


Figure 6-33 G6-317n and G6-317-1n

Table 6-18. Size requirements for G6-317n and G6-317-1n signs.

Road configuration	Approach speed	Sign size
Single carriageway	Less than 95 km/h	A size
	Greater than 95 km/h	B size
Dual carriageway and multilane roads	Less than 95 km/h	A size
	Greater than 95 km/h	B size

The G6-317n or G6-317-1n sign must be installed in the following locations:

- Prior to the first portable VSLS along the length of the work site in each direction (preferably at distance of D);
- 300 m after the first VSLS in each direction; and

- Less than 50 m from a junction with a side road/street, both upstream and downstream, to inform road users who have entered a roadwork speed zone from a side street/road.

These signs should also be repeated at an appropriate spacing along the length of an extended roadwork site.

The default speed shown inside the red annulus of the G6-317n sign should be the speed limit applicable during aftercare arrangements. However, alternative speed limits may be used, provided an appropriate justification is given the TMP and/or supporting risk assessment.

6.9.3 Illuminated flashing arrow signs

Illuminated flashing arrow signs (FAS) are signs which comprise a matrix of lamps or LED aspects in the form of an arrow that is flashed in a cyclic manner to either the left or right, indicating the direction in which approaching vehicles are to pass. Flashing arrow signs are intended to be applied primarily where a lane is closed or a diversion of traffic is required, typically on a multi-lane carriageway.

Illuminated flashing arrow signs must comply with [Transport Specification TSI-SP-060 Illuminated Flashing Arrow Signs](#) and the relevant Australian Standards. General operating instructions are provided in this Section and [Appendix B](#).

Illuminated FAS must only be vehicle-mounted (rear mounted or cab mounted), incorporated into VMS, or a stand-alone sign. [Appendix B.5 Usage procedure: illuminated flashing arrow signs](#) provides additional relevant requirements in relation to their use, installation, operation, signage and removal. The requirements provided for these devices must be read in conjunction with relevant specifications and manufacturer's instructions.

Additional vehicle mounted beacons or lamps that might cause a risk or confusion to road users whilst the flashing arrow sign is operating should be switched off.

6.9.4 Traffic warning (roadwork) lamps – flashing yellow lamps

Vehicle-mounted warning devices must be displayed as follows:

- A single (1) rotating or flashing yellow lamp for emergency or other infrequent use on a vehicle not normally used for roadworks purposes, or for use on a plant item or an inspection vehicle; or
- A pair (2) of rotating or flashing yellow lamps for use on vehicles (e.g. patrol trucks) so that at least one and preferably both lamps are visible from any direction.

The conditions of use for flashing yellow lamps are provided in [Table 6-19](#).

Table 6-19. Conditions of use of flashing yellow lamps

Display type	Conditions of use
Unidirectional flashing yellow lamps	Are used to draw attention to a particular sign. Where used, they must be mounted above the sign. They may be mounted on barrier boards indicating the ends of work areas. They must not be used to delineate a path through a work site because of the confusing light pattern which a series of such lamps can produce. They may also be used to highlight safety barrier end treatments.
Rotating or flashing yellow lamps	Must be used to draw attention to work vehicles and plant to which they are attached or to draw attention to signs in difficult light conditions or in high volume locations.
LED light bars or beacons	May be used if it can be shown that their on-time performance in each flash cycle is equal to or longer than that of the rotating or flashing lamp.

6.10 Truck and trailer-mounted attenuators

6.10.1 Description

A truck and trailer-mounted attenuator (TMA) is a combination of host vehicle and Impact Attenuator Unit (IAU), mounted on or towed by the host vehicle. The IAU is a mobile crash cushion safety device, used as an advance warning device on approach to work sites to provide for increased road user safety if struck. The primary purpose of an IAU is to reduce the impact for errant road users and should not be used as the sole protection method for work sites from unexpected vehicle impacts.

For use on Transport roads, TMAs must be assessed and accepted by [Austroads Safety Barrier Assessment Panel](#) in accordance with AS/NZS 3845.2:2017 and meet the requirements of this document..

6.10.2 Usage policy

TMAs must be used on the roads at the listed localities provided in [Table 6-20](#). If a section of the road listed below, or work type on the section of road listed below does not have a risk profile that justifies the need for a TMA, this must be captured in the relevant TMP as a departure in accordance with [Section 2.8 Departures from this Technical Manual](#).

Table 6-20. Roads at localities where TMAs are required

State road name	Route number	Locality
Pacific Motorway	M1	Wahroonga to Beresfield
Princes Motorway	M1	Waterfall to Yallah
Eastern Distributor	M1	Entire
Western Distributor	A4	Entire
Hume Motorway	M31	Casula to Federal Highway Interchange
Hunter Expressway	M15	Entire
Westlink	M7	Entire
Western Motorway	M4	Entire
The Hills Motorway	M2	Entire
South Western Motorway	M5	Entire
NorthConnex	No route number	Entire
WestConnex	M8, M5, M4	Entire
Lane Cove Tunnel	A1, M2	Entire
Sydney Harbour Tunnel	M1	Entire
Sydney Harbour Bridge	No route number	Entire

TMAs are not required where one or more of the following are in place:

- Road closure;
- Contraflow; or
- Work is behind temporary safety barriers.

TMAs are recommended to be used when the posted speed limit exceeds 85 km/h and traffic volumes exceed 20,000 vpd. Additionally, TMAs may also be used in other instances.

The need for a TMA must be determined in the TMP and the site specific risk assessment. Factors that should be considered when determining if a TMA is required include:

- Work type and activity;
- Speed of approaching traffic;
- Location of work, including sight distances;
- Crash history at location of work;
- Posted and temporary speed limits;
- Number of carriageways;
- If the location is a nominated heavy vehicle route; and
- Volume of heavy vehicle movements.

6.10.3 Operation

The operational requirements of a TMA are as follows:

- The TMA must be set-up and installed, operated and disassembled in accordance with the manufacturer's instructions. The requirements of this document in relation to TMAs are additional to the manufacturer's instructions, and in the case of discrepancy, this document prevails;
- The host vehicle must be staffed by at least one qualified operator (see Section 6.10.5 TMA Operator);
- Original equipment manufacturer (OEM) seatbelts must be worn at all times. Where a harness is fitted to a host vehicle, the operator should be harnessed while the attenuator is lowered;
- The TMA must be located between 100 m and 200 m to the work area during dynamic work, including the roll ahead distance (see Figure 6-34);
- The TMA must not be lowered when people are in the immediate vicinity of the attenuator;
- When used as an advance warning vehicle (motorway application) and when in environments with narrow shoulders and reduced edge clearances (if the advance warning vehicle cannot achieve the required 0.5 m clearance from edge line), the TMA must be located outside the traffic lane;
- When used as a shadow vehicle (protected by the tail vehicle) for the protection of workers on foot or small plant items, must follow no less than 40 m behind in the work lane (see Figure 6-35); and
- Consideration should be given to switching off any vehicle mounted beacons that might cause a risk or confusion to road users whilst flashing arrow sign is operating.

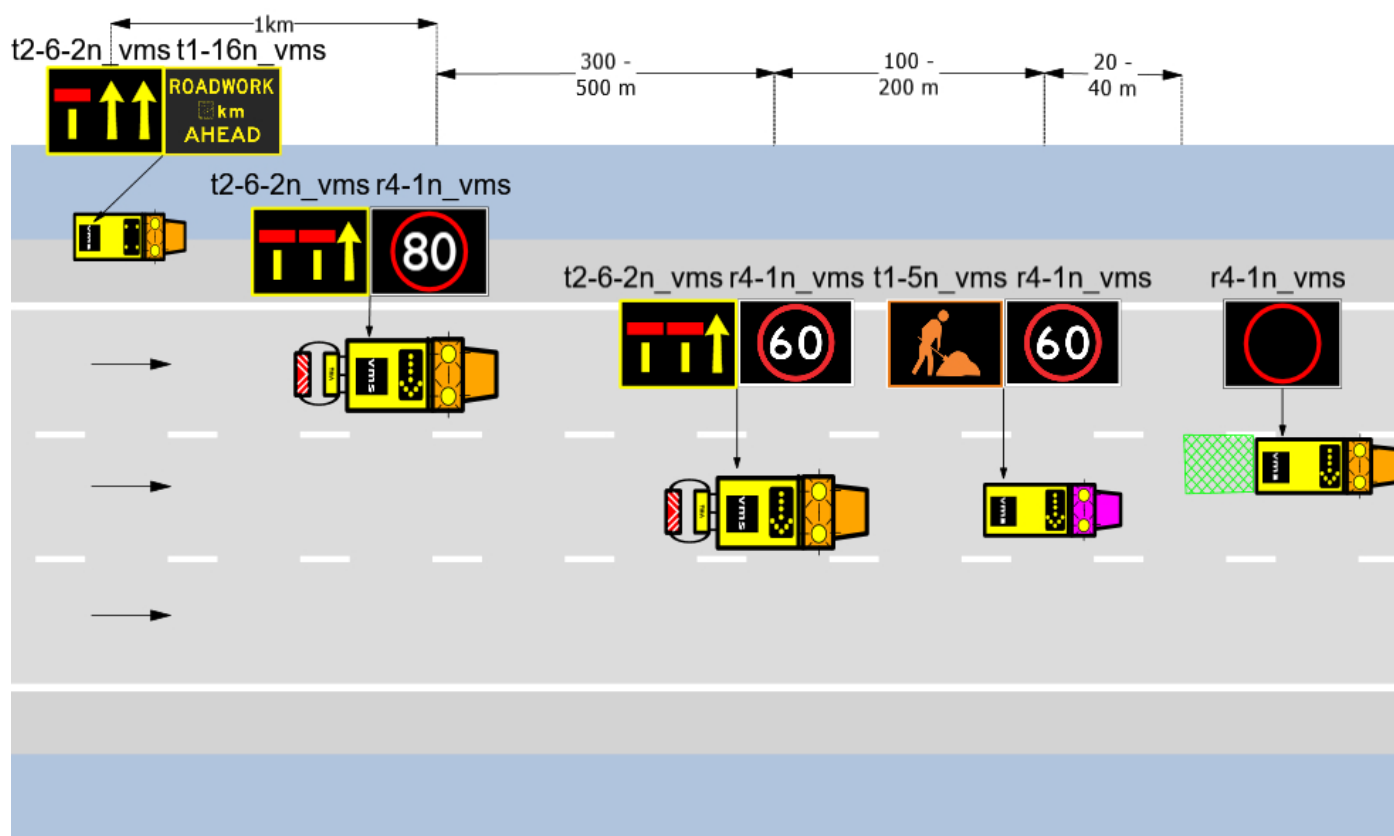


Figure 6-34. TMA layout – dynamic work

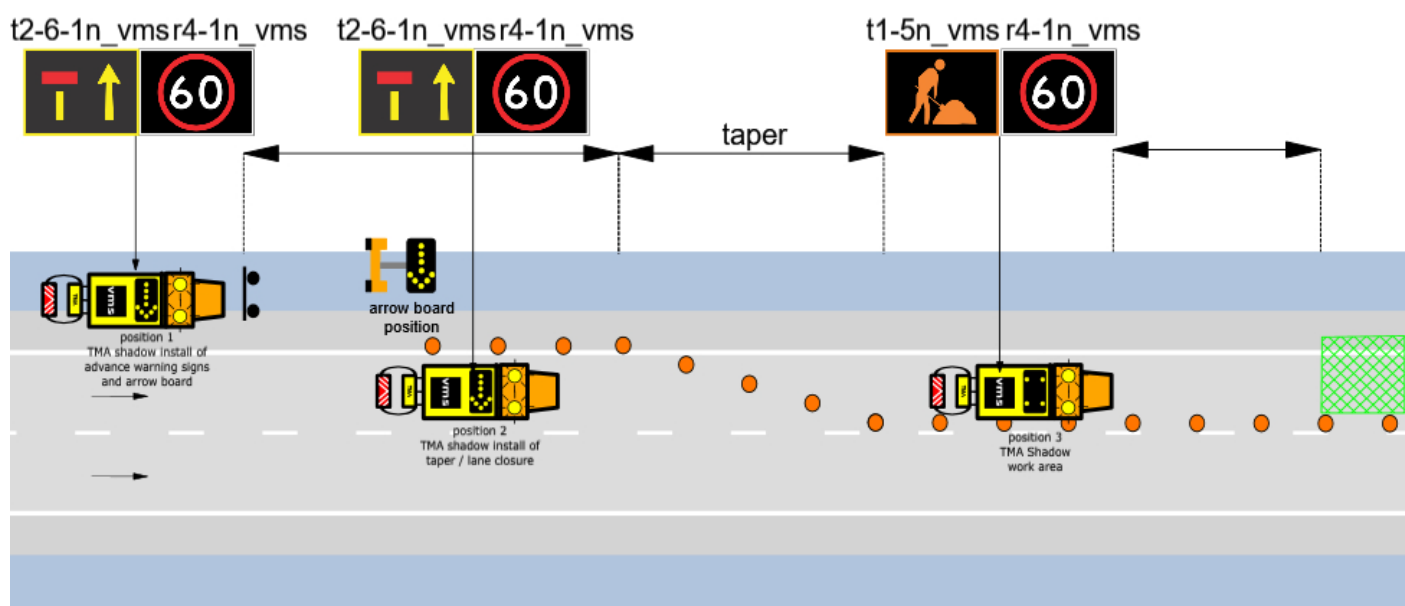


Figure 6-35. TMA layout – static work

6.10.4 Host vehicle features

The host vehicle must:

- Have at least two dedicated flashing warning lights, usually amber in colour;
- Have a rear overhang of 6.5 m maximum;

- Be at least 15 tonnes GVM;
- Be a single cab truck, with automatic transmission;
- Be fitted with automatic impact brakes (AIB);
- Be fitted with a variable message sign (VMS);
- Be fitted with rear facing CCTV cameras; and
- Be fitted with a Type C flashing arrow board.

6.10.5 Signage

Signage on a VMS that is attached to a TMA must be in accordance with Section 6.9.1 Variable message signs (portable).

Advance warning signage should follow the guidance provided in Figure 6-34 and Figure 6-35.

For static work sites, advance warning signage must be installed on all approaches to the work area. For dynamic works, an advance warning vehicle must be in place on all approaches to the work area.

6.10.6 TMA Operator

The operator of the TMA must:

- Hold a current and valid heavy vehicle licence of suitable class to operate the TMA host vehicle;
- Hold a current Implement Traffic Control Plan qualification;
- Have successfully completed the national unit of competency RIIRTM301D (operate a truck or trailer-mounted attenuator) with evidence of completion;
- Have successfully completed training in the relevant business/divisional requirements; and
- Have been inducted into the relevant or site-specific SWMS for operation of TMAs at roadwork sites.

6.11 Temporary portable rumble strips

6.11.1 Description

Temporary portable rumble strips (TPRS) are portable traffic devices placed in a traffic lane perpendicular to the direction of travel. TPRS provide a visual, audible and tactile warning to alert road users of changed conditions.

When used in conjunction with other traffic control signs and devices, TPRS have been found to increase positive road user behaviour in terms of increased compliance at a work site.

The decision to use a TPRS must be considered as part of the risk assessment and documented in the TMP.

TPRS are not suitable in all traffic environments and the decision to use a TPRS must consider:

- Volumes and proportion of heavy vehicles which influence the lateral and rotational movement of the TPRS;
- Maintaining safe travel paths for vulnerable road users;
- Vertical (grade) and horizontal (curves) alignment of road which influence longitudinal, lateral and rotational movement of TPRS;
- Road alignment to ensure clear sight distance of D to the TPRS is maintained; and

- Proximity of residential buildings or other noise sensitive land uses due to noise associated with the TPRS.

A TGS, which includes TPRS, must be designed in accordance with the above considerations.

Only TPRS products accepted by Transport products must be used. At the time of publication, this includes:

“RoadQuake Modular F1791001 or Folding F1791003”

TPRS from the above manufacturer must be yellow in colour.

Other TPRS devices not listed above may be used following approval from the Director, Roads and Traffic Engineering via standards@transport.nsw.gov.au.

6.11.2 Usage policy

TPRS may be installed in locations characterised by all of the following:

- Where the speed zone is less than 65 km/h; and
- On concrete, asphalt or sealed roads.

TPRS must not be installed:

- On multi-lane carriageways, except where all of the above conditions have been met;
- For work sites where the speed zone in the work zone is greater than 65 km/h; or
- On wet, unsealed, freshly sealed, or heavily rutted roads.

To protect oncoming traffic from the risk of swerving behaviour, TPRS must be placed in one of the following methods:

1. Where traffic is reduced to and controlled within a single lane (see [Figure 6-37](#)); or
2. On a two-lane two-way road a temporary median kerb with delineators is installed on the centreline for a minimum distance of D prior to the first strip in each array (see [Figure 6-38](#)).

Loose materials such as gravel and debris on the road surface must be removed prior to installation.

The carriageways near and around the work site that are being controlled by TPRS should remain clean and clear of dirt and construction debris. Debris and other construction waste that have dislodged from passing vehicles onto the carriageway/s should be regularly monitored and removed.

TPRS must be installed in the form of arrays. An array is a series of three rumble strips with 3 m spacing between each rumble strip as shown in [Figure 6-36](#).

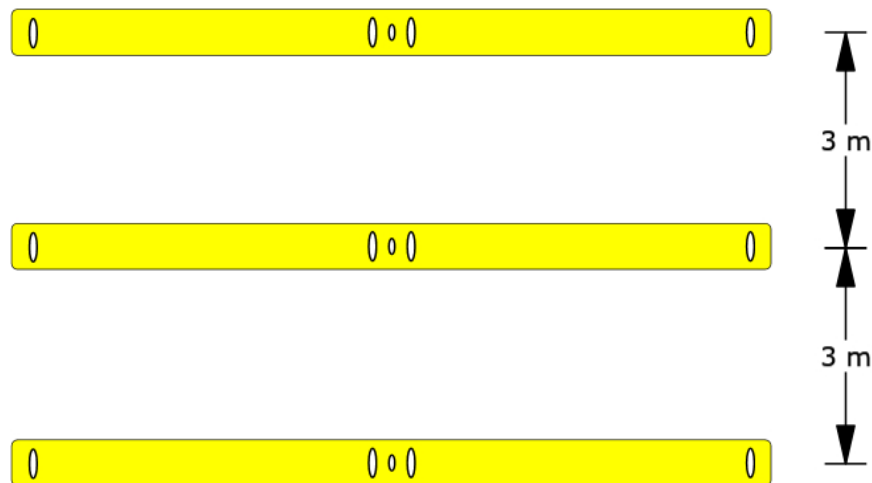


Figure 6-36. Temporary portable rumble strip (TPRS) typical array arrangement

TPRS must be used in accordance with Appendix B.6 Usage procedure: temporary portable rumble strips. The requirements provided for these devices must be read in conjunction with relevant specifications and manufacturer's instructions (see Figure 6-37).

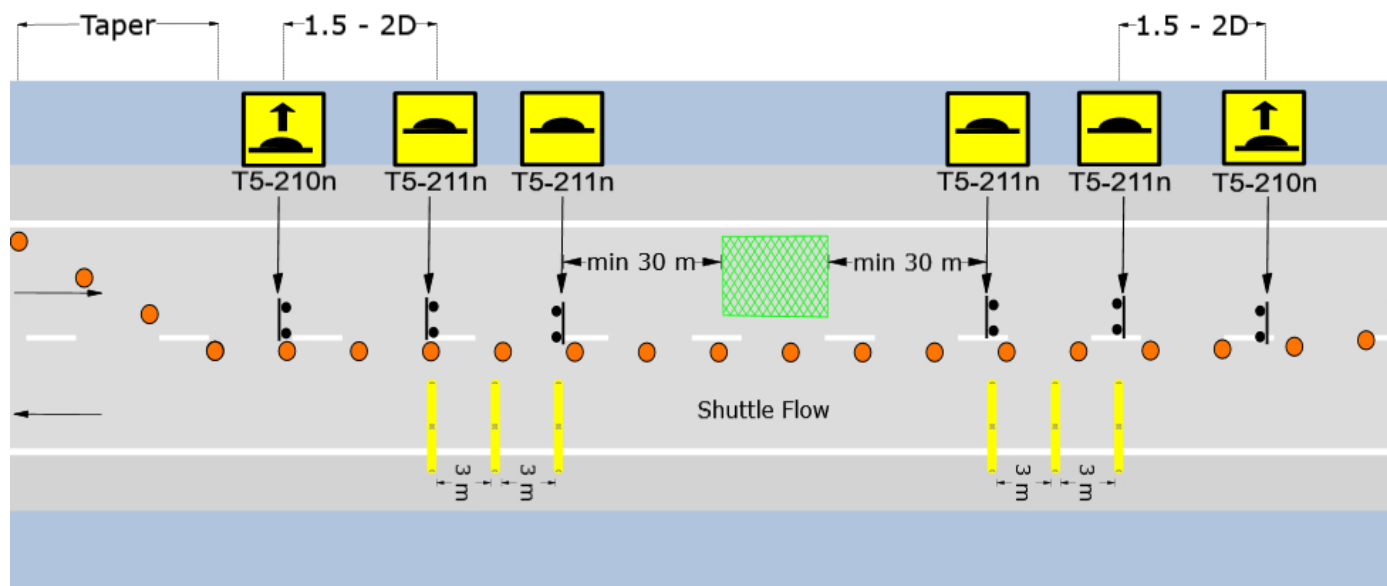


Figure 6-37. TPRS layout – single lane

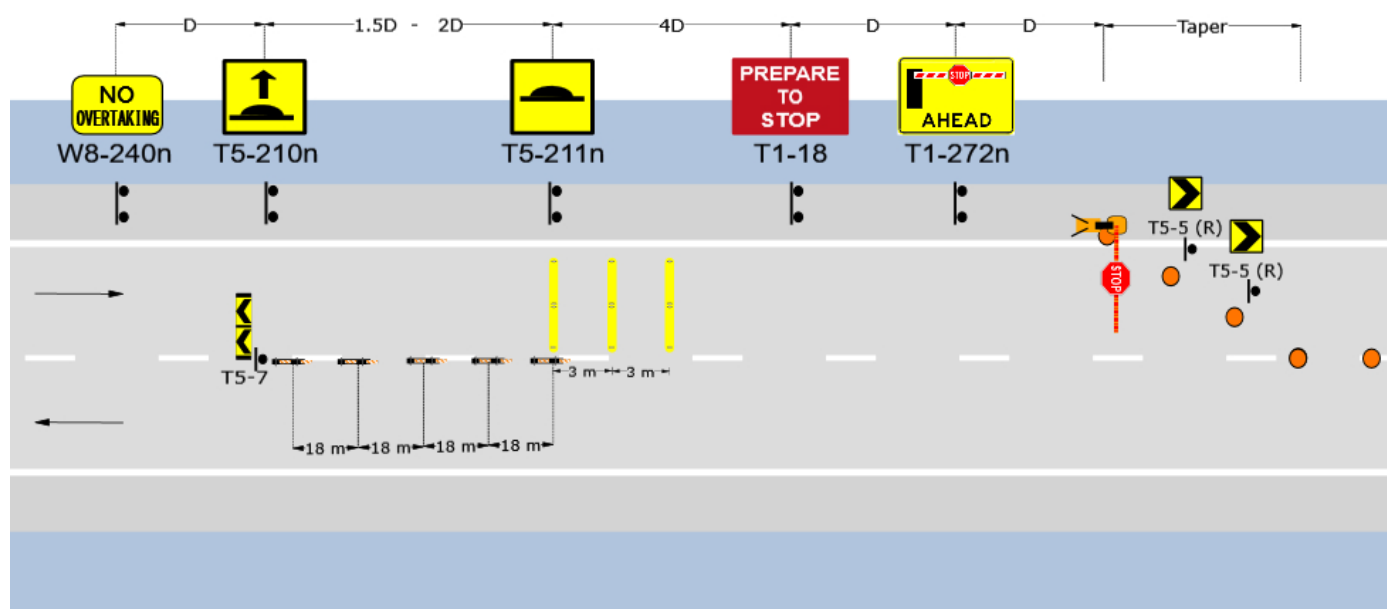


Figure 6-38. TPRS layout – boom barrier

7 Traffic guidance schemes

7.1 General

This Section must be used by those with PWZTMP and ITCP qualifications to develop, select and implement a traffic guidance scheme (TGS). It covers:

- Principles of a TGS, including components of a work site and application of Dimension D;
- Selection, design and implementation of a TGS; and
- Finalisation and record keeping of a TGS.

Prior to the selection or development of a TGS, the person responsible for the works must confirm the type/s of TTM work to be undertaken safely. The two types of work are:

- Static work; and
- Dynamic work.

Often, a combination of work types may be needed. For example, a dynamic work TGS may be used to set up a static work site.

Factors that influence the work type selection include, but are not limited to:

- Duration of works;
- Location of works, e.g. in the traffic lane or shoulder;
- Clearances to traffic;
- Existing posted speed limits;
- Traffic volumes;
- Other nearby or associated works;
- Available sight distance; and
- Workers on foot requirements.

After the TTM work type/s has been established, the controls can be determined and a TGS selected or developed.

As detailed in Section 7.5 Selecting a TGS, a qualified person must determine if a suitable TGS for the TTM work exists in an approved TGS library. If no Generic TGSs can be made Site Suitable, a Site Specific TGS must be designed in accordance with Sections 7.6 Designing a TGS: General principles, 7.7 Designing a TGS: Static work or 7.8 Designing a TGS: Dynamic work.

7.2 Types of TGS

A TGS, previously known as a traffic control plan (TCP), is a detailed layout of temporary signs and devices that communicate the TTM arrangement around, past or through a work site or temporary hazard. Table 7-1 provides a description of the different types of TGS used in this document.

Table 7-1. TGS types and descriptions

TGS type	Description
Generic	A Generic TGS is a TGS that has not been approved for use on-site and has no specific location information. A Generic TGS is a planning tool and should only exist within a TGS library (see Section 7.12 TGS records) held by a work group for their own operating environment and work activities (such as a District Works Depot). A series of generic TGSs may be included in a TGS library. A generic TGS must not be used and implemented unless a copy has been made and the copy has had appropriate details added and verified as Site Suitable or modified to be Site Specific (if required) and approved for use by an appropriately qualified person. See Section 7.5 Selecting a TGS. All Generic TGS must be designed by a PWZTMP qualified person and endorsed prior to being added to the TGS library. When designing a Generic TGS, the principles of this Technical Manual must be followed. The work type layouts provided in Appendix D – Work type layout examples are for information only and are not Generic TGS. They may be used to aid the design of Generic TGSs, but must not be selected and used at a work site without review and amendment.
Site Suitable	A Site Suitable TGS is a copy of a Generic TGS that has been determined suitable for a location and / or works. In order for a TGS to be considered Site Suitable, the following provisions apply: • The Generic TGS must have been selected by a PWZTMP or an ITCP qualified person using the defined selection procedure, see Section 7.5 Selecting a TGS; • A PWZTMP or ITCP qualified person must confirm the selected Generic TGS is appropriate for use for the road environment and the work activity at that site; and • The TGS to be specified does not require modifications beyond the tolerances detailed in Section 7.10.3 Tolerances on positioning of signs and devices. When a Site Suitable TGS is selected, a TGS verification of the site must be conducted by a PWZTMP or ITCP qualified person and mandatory information outlined in Section 7.4 Information required on a TGS must be added prior to the TGS being approved for use, see Section 7.12 TGS records. A Site Suitable TGS is only valid for the time period of works specified on the TGS.
Site Specific	A Site Specific TGS is produced by a PWZTMP qualified person and may be either: • A new TGS designed specifically for a project site location on the road and for a particular work activity; or • A copy of a Generic TGS that has been modified outside the tolerances of Section 7.4 Information required on a TGS to suit the specific location or work requirements. When a Site Specific TGS is designed, a TGS verification of the site must be conducted by a PWZTMP qualified person and mandatory information outlined in Section 7.4 Information required on a TGS must be added prior to the TGS being approved for use, see Section 7.12 TGS records. A Site Specific TGS is only valid for the time period of works specified on the TGS. A Site Specific TGS may be added to the TGS library as per Section 7.12 TGS records, if it is determined appropriate for future re-use activities.

7.3 Dimension D

Dimension D is a measure of distance in metres. It is used to determine taper lengths, the position of signs and devices and for determining sight distances along the road so that road users have sufficient time to absorb the roadwork specific messages, understand the changed traffic conditions and take necessary actions.

Dimension D is calculated by expressing the speed in metres for the zone preceding to where the Dimension D will be applied, this may be either the existing posted speed or a reduced roadwork speed limit.

For example Dimension D in *Figure 7-1* below is:

- 110 m for the yellow shaded area;
- 80 m for the blue shaded area; and
- 60 m for the pink shaded area.

The existing posted speed limit may be used to determine Dimension D throughout the work site, provided the PWZTMP qualified person has determined that there is higher risk of poor driver compliance with speed zones and where space allows.

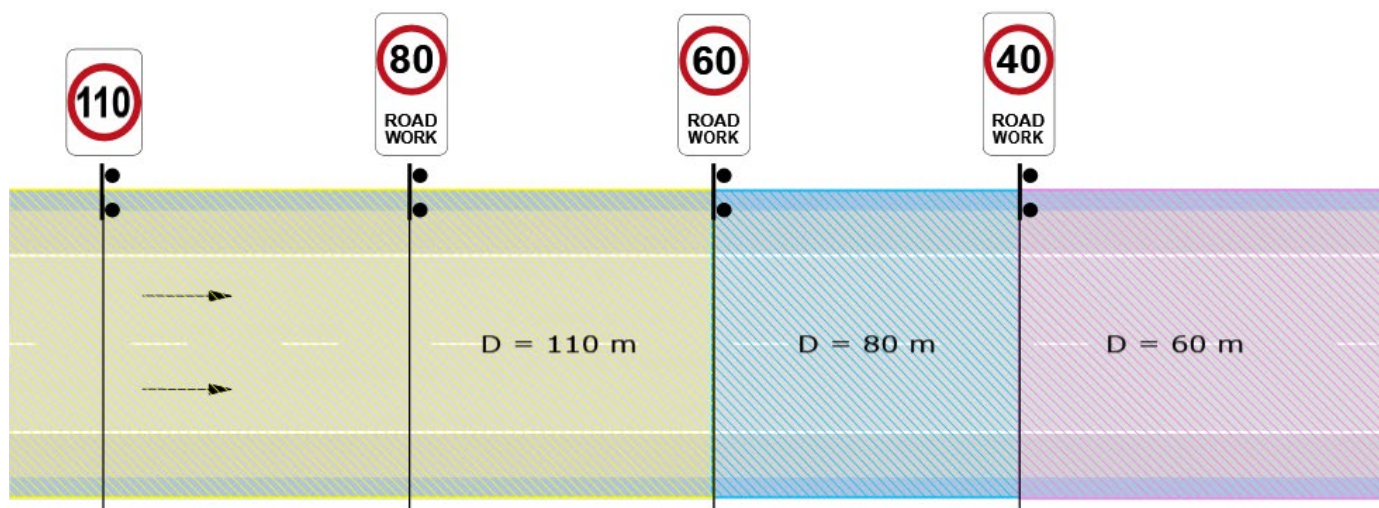


Figure 7-1. Example calculation of Dimension D

The Dimension D to be used on a work site must be determined by the PWZTMP qualified person and must be specified on the relevant TGS.

Where required by site-specific constraints, the application of Dimension D may be varied through the departures process provided in *Section 2.8 Departures from this Technical Manual*.

An example showing application of Dimension D in a 60 km/h roadwork zone with a preceding 80 km/h zone is given in *Table 7-2*.

Table 7-2. Dimension D calculation based on speed zone

Scenario	Dimension D required	Dimension D
Dimension D	Dimension D calculated as	80 m
For determining sight distance to a PTCD or manual traffic controller	Traffic controller must be able to see 1.5 D or greater to the oncoming traffic	80 m x 1.5 1.5D = 120 m
For determining sight distance to end-of-queue	Sight distance to the end-of-queue for approaching traffic must be calculated at 2D for approach speeds greater than 65 km/h and 1.5D for approach speeds of less than 65 km/h	greater than 65 km/h 80 m x 2 2D = 160 m less than 65 km/h 80 m x 1.5 1.5D = 120 m
For determining sign spacing	Distance between signs must be calculated as follows: Single sign: 2D for speeds greater than 65 km/h and 1D for speed zones of less than 65 km Multiple signs (such as dual sign arrangements or multi-message signs): 1D for all permitted speed zones	greater than 65 km/h 80 m x 2 2D = 160 m less than 65 km/h 80 m x 1 D = 80 m
For determining taper lengths	See Section 7.6.2.2 Tapers	
For distance between tapers on multi-lane roads	A distance of 1.5D should be applied	80 m x 1.5 1.5D = 120 m

7.4 Information required on a TGS

7.4.1 General

All TGS must include the following information:

- Project name and the time period that works will be performed;
- Types and location of permanent signs and traffic signals (where applicable);
- Types, locations and clearance requirements of all:
 - Temporary signs; and
 - Traffic control locations and devices including details of determined safe location and escape routes for traffic controllers, where applicable.
- Predicted end-of-queue lengths;
- Location and names of roads and intersecting roads;

- Location reference markers, in order to assist recreating the layout in case of incident. Location reference markers must be recorded for temporary speed zone locations;
- Notes, as per Section 7.4.2 Notes on a TGS;
- Dimension D applied;
- Frequency of shift TTM inspections in accordance with Section 8.1.4 Shift TTM inspections; and
- Sign-off section for the designer and approver including an issue and review date.

Where applicable, the below details should also be included on the TGS:

- Installation and removal order of temporary signs and traffic control devices, (this may also be captured in relevant SWMSs).
- Locations of temporary safety barrier systems including terminals;
- Exclusion zone required for the temporary safety barrier system and details of delineation of zones and maintenance during works;
- Existing footways and cycle ways;
- Location of site access gates;
- Location and access provisions for local properties and/or car parking access;
- Temporary pavement markings and delineation devices required;
- Locations of temporary street lighting;
- Time and date, the designer completed the TGS verification as required by Section 7.9.1 TGS verification;
- Locations of emergency or breakdown lanes; and
- Existing pavement markings including a note of which require covering/removal.

This information must be included on, or attached to the TGS. This may be as information on the front of the TGS, as detail on the rear of the TGS, as an attachment to the TGS or a combination of methods. Appendix C – Example TGS provides an example of a completed TGS with details and notes as per this Section.

7.4.2 Notes on a TGS

Notes must be included on the approved site suitable or site specific TGSs, to detail any specific procedures that need to be communicated to the:

- *Implementer* – for installing or removing the TGS;
- *Person responsible* – for inspecting and monitoring the operation of the TGS;
- *Traffic controller* – for the operation of the TGS; or
- *Workers* – utilising the TGS.

Notes are created by either the PWZTMP or ITCP qualified person and provide additional information to the TGS or reinforcement of key critical safety messages.

The non-exhaustive list below provides some examples of notes that may be included on TGS:

- “Shadow vehicle or TGS No xxx must be used to set out this TGS (installation/dismantle)”;
- “Observer must be utilised when installing this TGS”;
- “Ensure all radios are switched to the appropriate channel and frequency, and are in working order”;

- “Size X signs must be used”;
- “Signs X must be duplicated”;
- “Contradictory signs must be covered”;
- “Where there is not enough lateral clearance, reduce speed to X km/h”;
- “Access and exit points for site/delivery vehicles must be via X”;
- “Containment fence or tape to be used to delineate exclusion zones”;
- “All personnel, plant and equipment to maintain a minimum of X m from traffic”;
- “A variable message sign (truck or trailer mounted) may replace ‘Roadwork Ahead’ or ‘Roadwork Ahead 1 km sign/s (ensure clear of edge line)’; or
- “All signs that are no longer required must be covered or removed when not in use, e.g. between shifts or during meal breaks”.

7.5 Selecting a TGS

7.5.1 General

A Generic TGS that has been developed in accordance with Section 7.6 Designing a TGS: General principles and 7.7 Designing a TGS: Static work, or 7.7.4 Through TTM methods, may be selected by a person who is either PWZTMP or ITCP qualified (see Figure 7-2) to be site suitable. The qualified person must then assess whether the Generic TGS will:

- Be suitable for the road environment and TTM activity and if so, confirm the Generic TGS as a Site Suitable TGS in accordance with Section 7.9 TGS confirmation and approval; and
- Require modifications in excess of the tolerances detailed in Section 7.10.3 Tolerances on positioning of signs and devices.
- If modifications are required, then a PWZTMP qualified person must design a Site Specific TGS.

To appropriately select and assess a generic TGS the PWZTMP or ITCP qualified person must:

- Obtain a copy of the relevant TMP;
- Determine the category of work to be undertaken (short term / long term, static / dynamic);
- Use an approved selection procedure; and
- Identify, assess and mitigate site-specific conditions and risks within the parameters of the selection procedure and the tolerances outlined in Section 7.10.3 Tolerances on positioning of signs and devices.

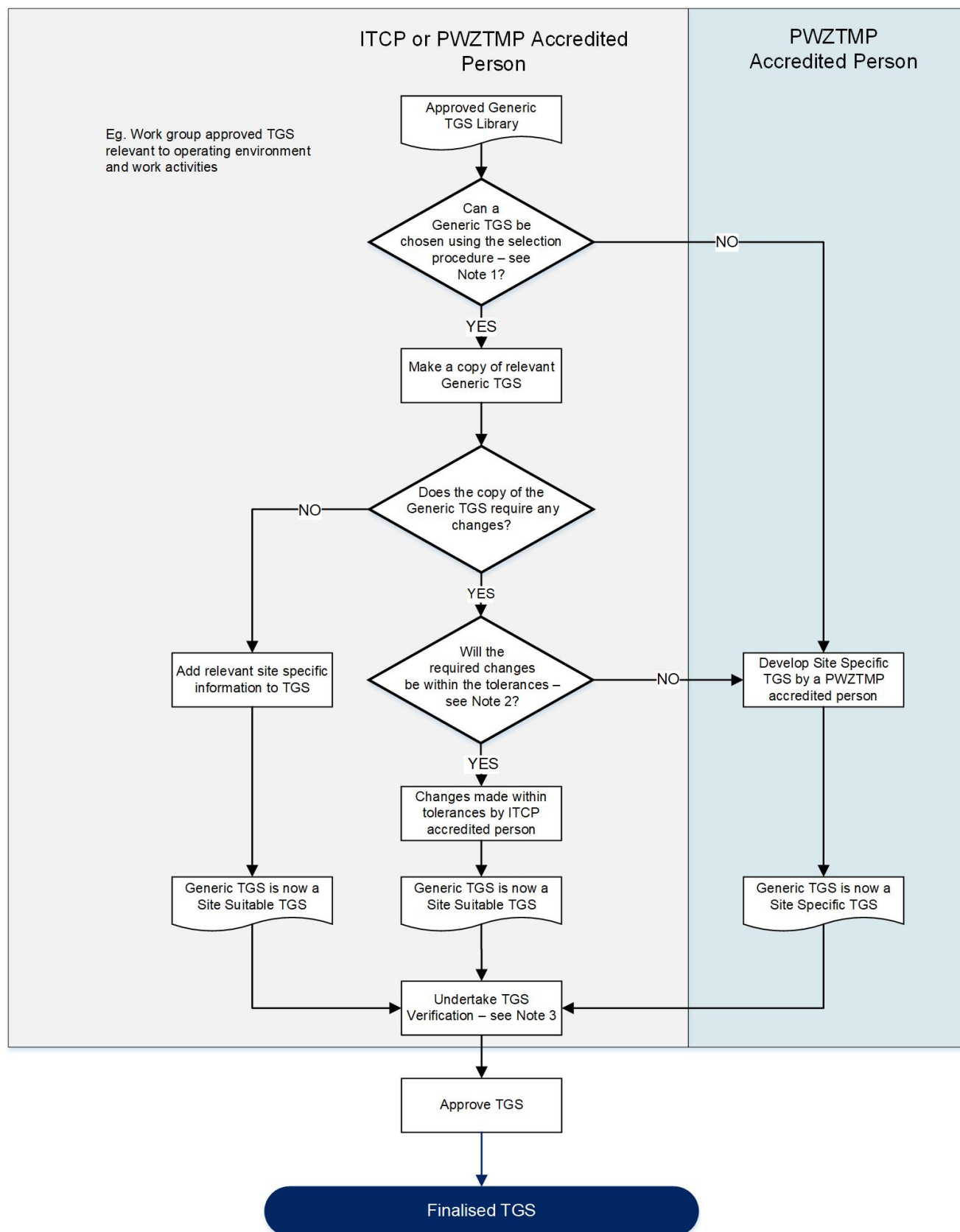


Figure 7-2. TGS selection flow chart

Notes Figure 7-2:

1. Selection procedure must be developed in accordance with Section 7.5.2 TGS Selection Procedure
2. Tolerances must be in accordance with Section 7.10.3 Tolerances on positioning of signs and devices
3. TGS Verification must be completed in accordance with Section 7.9 TGS confirmation and approval

7.5.2 TGS Selection Procedure

A selection procedure must be prepared and endorsed by a PWZTMP qualified person to accompany a generic TGS Library. The selection procedure must define the factors or inputs to be used to determine whether a generic TGS is suitable or not. When using the generic TGS Library to select a site suitable TGS, the PWZTMP or ITCP qualified person must:

- Reference the TMP under which the TGS will be implemented; and
- Ensure that the generic TGS has been reviewed and approved within the previous 12-month period. See Section 7.11.1 Generic TGS review and Section 7.9 TGS confirmation and approval for review and approval of a TGS.

A selection procedure should consider factors such as the:

- Activity to be undertaken;
- Type of work being planned, i.e. static or dynamic work;
- Type of road i.e. a one-way or two-way road and single versus divided carriageway;
- Number of lanes;
- Road environment, for example rural/urban;
- Posted speed limit and approach speeds of traffic;
- Time of day, for example whether the TGS is suitable for day work, night work or both;
- Traffic volumes; and
- Where the work site is located within the road corridor, for example on shoulder or in lane.

If a TGS does not exist in the generic library that can be selected as Site Suitable, a Site Specific TGS must be designed in accordance with Section 7.6 Designing a TGS: General principles and Section 7.7 Designing a TGS: Static work, or Section 7.7.4 Through TTM methods.

7.5.3 Confirming a Site Suitable TGS

After a Generic TGS has been identified as being suitable for the works using the selection procedure, the PWZTMP or ITCP qualified person must make a copy of the relevant Generic TGS. This copy will become the Site Suitable TGS after the below information has been added to it:

- Distance D;
- Details of modifications within the tolerances outlined in Section 7.10.3 Tolerances on positioning of signs and devices;
- Location information; and
- Signature and approval date of the selected TGS.

Once the site suitable TGS has been created, it must be confirmed and approved in accordance with Section 7.9 TGS confirmation and approval before it is implemented in the field.

7.6 Designing a TGS: General principles

7.6.1 General

This Section provides the general principles for designing a TGS.

There are two types of TGS that may be designed by a PWZTMP qualified person. These include:

- Generic TGS; and
- Site Specific TGS.

All TGSs must be designed and approved by a PWZTMP qualified person and prepared in accordance with this manual. Key principles for designing a TGS include:

- Ensure the risks and controls detailed in the TMP are addressed;
- Eliminate risks SFAIRP;
- Provide protection SFAIRP to workers and the work site, including:
 - Separating the work area from the route of vehicles wherever possible; and
 - Prioritising the use of PTCDs rather than a manual traffic controller.
- The needs of all road users are provided for e.g. pedestrians, cyclists, school children etc., including:
 - Avoiding the need to slow and/or stop traffic wherever possible.
 - The use of a pilot vehicle is considered;
- Only accepted signs and devices are used, including condition signs where the road surface at the work area is sufficiently different from the approach roads and may be hazardous to traffic;
- Preparing a separate TGS for each stage of the work; and
- TGSs are drawn to scale to ensure geometric standards are achieved and implemented.

To ensure a TGS is easily understood, all TGSs should be produced electronically with hand drawn TGSs only produced in extenuating circumstances.

7.6.2 Components of the work site

The work site is the length of road, made up of five smaller areas as shown in *Figure 7-3* below, which includes the area where the work is being undertaken and any additional length of road used for traffic control including signs, tapers, traffic lights and other devices.

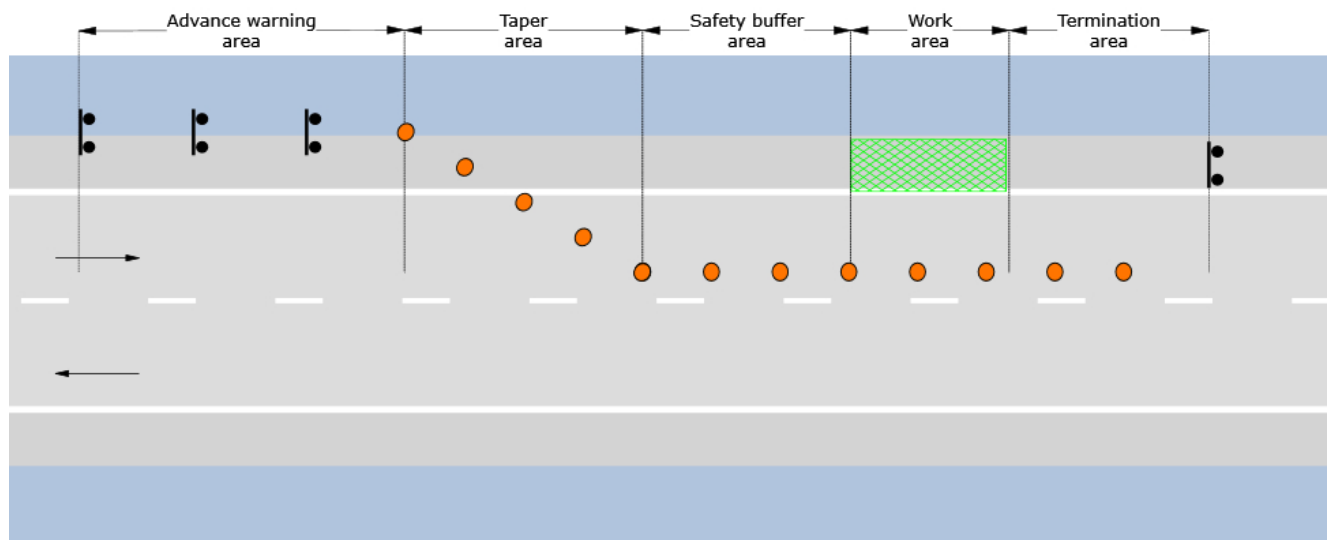


Figure 7-3. Components of a work site

Note to Figure 7-3: the five areas noted above are shown in the direction of an affected lane.

7.6.2.1 Advanced warning

Advanced warning for work sites must be installed to communicate to road users that there are changes to the road conditions ahead. Advanced warning may be required to notify road users of:

- Traffic hazards;
- Speed reductions;
- Workers on foot;
- Stopped traffic; or
- Changes to lane configurations.

Most work sites will require advance warning signage to be included on the TGS with spacing in accordance with [Section 6.2.6 Spacing of signs](#).

Advance warning signs are not required in the following situations:

- Where the work is sufficiently remote from a roadway that no action is required of road users;
- Where the work is undertaken as intermittent work i.e. in gaps in traffic, in either open road or built-up areas or takes less than five minutes;
- If the effects of the works are confined to one direction of travel and no extra vigilance is required of road users travelling in the other direction; or
- On unsealed roads.

7.6.2.2 Tapers

The following types of approach tapers may be used (see [Figure 7-4](#), [Figure 7-5](#) and [Figure 7-6](#)):

- **Traffic control taper** — used immediately after the position of a PTCD or traffic controller, to guide road users past a roadwork site or temporary hazard.
- **Lateral shift taper** — where traffic is required to shift laterally without conflict with another traffic stream.
- **Merge taper** — where one lane of traffic is required to merge into another.

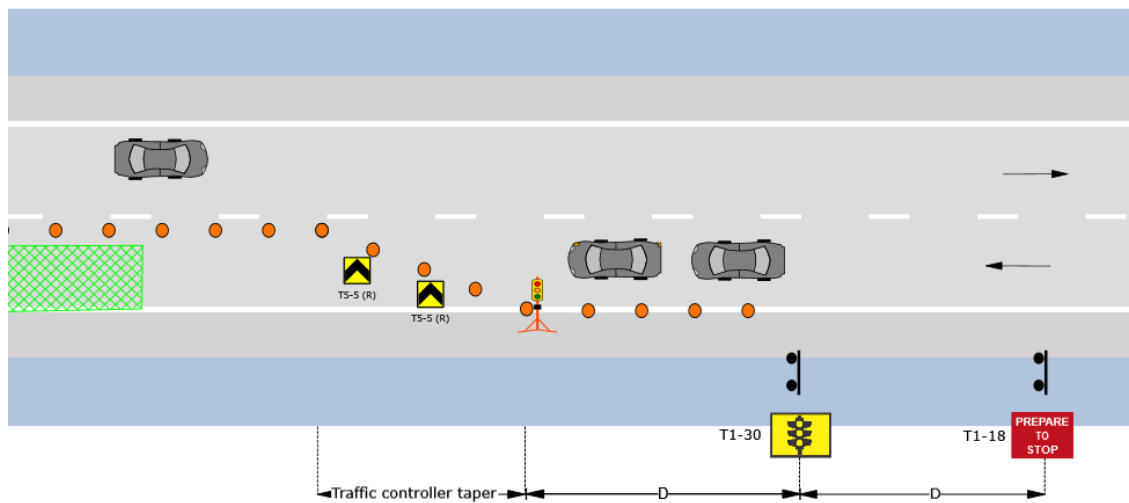


Figure 7-4. Traffic control taper with Type 1 PTS

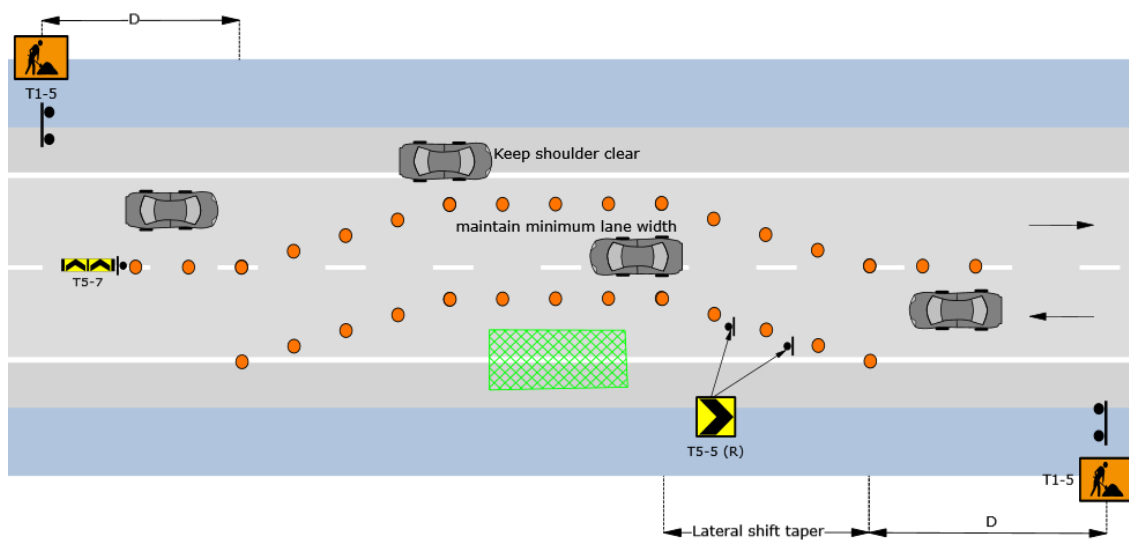


Figure 7-5. Lateral shift taper

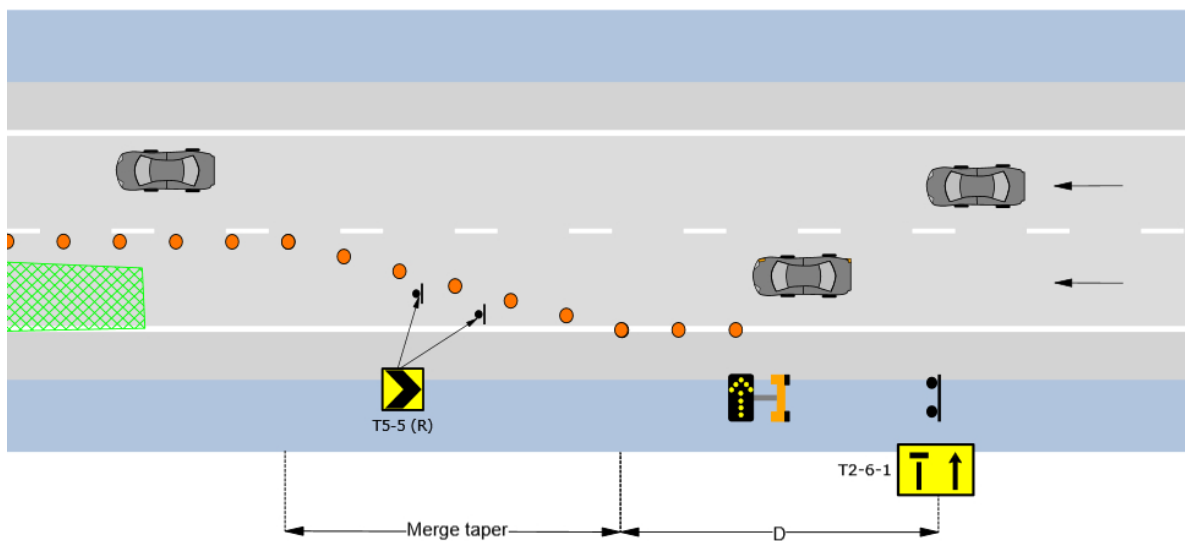


Figure 7-6. Merge taper

Recommended taper lengths are provided in [Table 7-3](#). Taper lengths should be increased at locations with poor sight distances or speed compliance, and supported by a site-specific risk assessment and documented in the TMP.

Table 7-3. Recommended taper lengths

Speed (km/h)	Recommended taper length (m)		
	Traffic control taper	Lateral shift taper	Merge taper
45 or less	15	15	15
46 to 55	15	15	30
56 to 65	30	30	60
66 to 75	N/A	70	115
76 to 85	N/A	80	130
86 to 95	N/A	90	145
96 to 105	N/A	100	160
Greater than 105	N/A	110	180

Note to Table 7-3: Speed is defined as the speed (km/h) of traffic at a position in the TGS where a device is located (e.g. start of a taper). This should be one of the following, in order of preference:

- The measured speed;
- The predicted speed of traffic;
- The preceding roadwork speed zone in accordance with [Section 7.3 Dimension D](#); or
- The existing posted speed limit.

When installing multiple tapers, the distance between each taper must be in accordance with [Table 7-4](#).

Table 7-4. Minimum taper lengths

Speed (km/h)	Distance between tapers (m)
45 or less	10
46 to 55	25
56 to 65	70
greater than 65	1.5 x Speed

Note to Table 7-4: Speed is defined as the speed (km/h) of traffic at a position in the TGS where a device is located (e.g. start of a taper). This should be one of the following, in order of preference:

- *The measured speed;*
- *The predicted speed of traffic;*
- *The preceding roadwork speed zone in accordance with Section 7.3 Dimension D; or*
- *The existing posted speed limit.*

7.6.2.3 Safety buffer

A safety buffer is the unoccupied space between the taper and work areas, and allows for a driver of an errant vehicle to correct, slow or stop before entering the work area. When designing a TGS, a space of no less than 30 m must be provided prior to the work area for the safety buffer.

Safety buffers are not required on departure, however if road users are able approach the work area from either end, a safety buffer must be provided at both ends of the work area.

Roadworks and storage of vehicles, plant, equipment and stockpiled materials must not occur in safety buffers. Road workers must not occupy the safety buffer area except when accessing the work area.

7.6.2.4 Work areas

The work area is the space required for the immediate work to be completed. Work areas must be set up in accordance with this manual and, where possible, work vehicles within a work area must be parked so that:

- They are clear of the minimum clearances described in Section 4.3.4 Minimum clearances of workers to traffic and Section 4.3.5 Protection of work area;
- Workers are not required to enter the minimum clearances to access tools and stores from the side of the vehicle; and
- The work vehicle provides a shield to workers from the path of errant vehicles.

7.6.2.5 Termination area

Provision of appropriate termination signage is critical to ensure road users are aware that the roadwork or changed conditions are no longer present and normal road conditions are reinstated.

Where a ROADWORK AHEAD sign is installed, END ROADWORK (T2-16) or (T2-17) termination sign must be located D from the last point of the work area where people are working and displayed adjacent to or after any signs indicating the reinstatement of the existing permanent speed limit. The SPEED LIMIT (R4-1) symbolic sign must be placed no further than 50 m past the work area.

It is important to note that if the termination signs are placed at a distance too great from the work area, driver compliance may reduce and road user complacency for work sites may increase.

7.6.3 Traffic control types

Traffic control must be designed in accordance with Section 5.4 Traffic control. Portable traffic control devices (PTCDs) must be used when the existing permanent speed limit is above 45 km/h for all work sites under traffic control. This requirement is not applicable to instances and environments of emergency response.

A manual traffic controller may be used instead of a device provided all of the following conditions are met:

- The use of a PTCD does not provide the safest outcome, SFAIRP;
- The decision to use a manual traffic controller instead of a PTCD is documented as part of the TMP and site specific risk assessment; and
- Approval is granted by the one-up manager to the PWZTMP person approving the TMP.

For all traffic control types, four cones placed at 4 m spacing must be located immediately preceding the location of the traffic control (see [Figure 7-4](#)).

7.6.4 End-of-queue

In addition to the TMP requirements provided in [Section 4.6 End-of-queue management](#), if work is planned to stop traffic, the TGS designer must assess the likely maximum queue length, through a consideration of:

- Expected delay in minutes;
- Peak hourly traffic volumes at the time of the work;
- The type or mix of traffic; and

Note: Heavy vehicles are longer than light vehicles and therefore affect the calculation of maximum queue length.

- The gap between stopped vehicles.

Note: In rural environments the gap is generally longer than that of an urban road environment.

If the TGS is planned to stop traffic, the TGS designer must ensure that signs and spacing specified on the TGS are appropriate for all queue lengths up to the maximum design queue length plus thirty per cent (30%) to allow for anomalies, such as greater demand. Road users will generally need 2D sight distance to the end of expected queues in open road areas and 1.5D in built-up areas.

Expected queue lengths must be determined and detailed on the TGS. Procedures should be put in place where the predicted maximum queue length will be exceeded. This may include increasing gap between signs, extra signs or preparing to be able to release traffic to ease the congestion in the queue.

7.6.5 Pilot vehicles

A pilot vehicle is a work vehicle used to guide traffic by controlling the desired path and to manage speed through the work area. A pilot vehicle must:

- Be registered for road use in NSW;
- Only act as a pilot vehicle specifically for the function of the work site;
- Have its function documented in the TMP and the relevant TGS prior to operating on a work site;
- Be operated by a person as either Traffic Controller or Implement Traffic Control Plans and appropriately licenced to operate the relevant vehicle; and
- Be a car, utility or truck; however the vehicle must not be heavily loaded or towing;
- Have two rotating or flashing yellow lamps, an effective two-way radio, headlights and tail lights; and
- Have a PILOT VEHICLE DO NOT OVERTAKE (T6-5) sign instructing traffic to follow it but not pass it.

When in use, a PILOT VEHICLE IN USE (T6-6) sign must also be installed at an appropriate distance in advance of where the pilot vehicle operates.

A pilot vehicle should be used to guide traffic through a work area when:

- Traffic needs to follow a particular path through the site which might not be obvious to road users;
- There is low compliance with the reduced speed limit, for example, switching traffic onto a new carriageway or releasing traffic after a full closure;
- Part of the work area is out of view of the supervisor, work gang and the traffic controller;
- The traffic speed has been reduced to less than 45 km/h due to an increased risk to workers (see [Section 4.5 Speed zones](#)); or